Microorganism Pronunciation Guide

The pronunciation of each name is given in parentheses. The phonetic spelling system is explained at the beginning of the glossary (p. G1)

Bacteria

Acetobacter (ah-se″to-bak′ter)
Acinetobacter (as″ĭ-net″o-bak′ter)
Actinomyces (ak″tĭ-no-mi′sēz)
Agrobacterium (ag″ro-bak-te′re-um)
Alcaligenes (al″kah-lij′ĕ-nēz)
Anabaena (ah-nab′ē-nah)
Arthrobacter (ar″thro-bak′ter)
Bacillus (bah-sil′lus)
Bacteroides (bak″tĕ-roi′dēz)
Bdellovibrio (del″o-vib′re-o)
Beggiatoa (bej″je-ah-to′ah)
Beijerinckia (bi″jer-ink′e-ah)
Bifidobacterium (bi″fid-o-bak-te′re-um)
Bordetella (bor″dĕ-tel′lah)
Borrelia (bō-rel′e ah)
Brucella (broo-sel′lah)
Campylobacter (kam″pĭ-lo-bak′ter)
Caulobacter (kaw″lo-bak′ter)
Chlamydia (klah-mid′e-ah)
Chlorobium (klo-ro′be-um)
Chromatium (kro-ma′te-um)
Citrobacter (sit″ro-bak′ter)
Clostridium (klo-strid′e-um)
Corynebacterium (ko-ri″ne-bak-te′re-um)
Coxiella (kok″se-el′lah)
Cytophaga (si-tof′ah-gah)
Desulfovibrio (de-sul″fo-vib′re-o)
Enterobacter (en″ter-o-bak′ter)
Erwinia (er-win′e-ah)
Escherichia (esh″er-i′ke-ah)
Flexibacter (flek″sĭ-bak′ter)
Francisella (fran-si-sel′ah)
Frankia (frank′e-ah)
Gallionella (gal″le-o-nel′ah)
Haemophilus (he-mof′ĭ-lus)
Halobacterium (hal″o-bak-te′re-um)
Hydrogenomonas (hi-dro″jĕ-no-mo′nas)
Hyphomicrobium (hi″fo-mi-kro′be-um)
Klebsiella (kleb″se-el′lah)
Lactobacillus (lak″to-bah-sil′lus)
Legionella (le″jun-el′ah)
Leptospira (lep″to-spi′rah)
Leptothrix (lep′to-thriks)
Leuconostoc (loo″ko-nos′tok)
Listeria (lis-te′re-ah)
Methanobacterium (meth″ah-no-bak-te′re-um)
Methylococcus (meth″il-o-kok′-us)
Methylomonas (meth″il-o-mo′nas)
Micrococcus (mi″kro-kok′us)
Mycobacterium (mi″ko-bak-te′re-um)
Mycoplasma (mi″ko-plaz′mah)
Neisseria (nīs-se′re-ah)
Nitrobacter (ni″tro-bak′ter)
Nitrosomonas (ni-tro″so-mo′nas)
Nocardia (no-kar′de-ah)
Pasteurella (pas″tĕ-rel′ah)
Photobacterium (fo″to-bak-te′re-um)
Propionibacterium (pro″pe-on″e-bak-te′re-um)
Proteus (pro′te-us)
Pseudomonas (soo″do-mo′nas)
Rhizobium (ri-zo′be-um)
Rhodopseudomonas (ro″do-soo″do-mo′nas)
Rhodospirillum (ro″do-spi-ril′um)
Rickettsia (rĭ-ket′se-ah)
Salmonella (sal″mo-nel′ah)
Sarcina (sar′sĭ-nah)
Serratia (sĕ-ra′she-ah)
Shigella (shi-gel′ah)
Sphaerotilus (sfe-ro′tĭ-lus)
Spirillum (spi-ril′um)
Spirochaeta (spi″ro-ke′tah)
Spiroplasma (spi″ro-plaz′mah)
Staphylococcus (staf″ĭ-lo-kok′us)
Streptococcus (strep″to-kok′us)
Streptomyces (strep″to-mi′sēz)
Sulfolobus (sul″fo-lo′bus)
Thermoactinomyces (ther″mo-ak″tĭ-no-mi′sēz)
Thermoplasma (ther″mo-plaz′mah)
Thiobacillus (thi″o-bah-sil′us)
Thiothrix (thi′o-thriks)
Treponema (trep″o-ne′mah)
Ureaplasma (u-re′ah-plaz″ma)
Veillonella (va″yon-el′ah)
Vibrio (vib′re-o)
Xanthomonas (zan″tho-mo′nas)
Yersinia (yer-sin′e-ah)
Zoogloea (zo″o-gle′ah)

Viruses

Vernacular, nonscientific virus names are not written in italics, and thus the following names are not italicized.

adenovirus (ad″ĕ-no-vi′rus)
arbovirus (ar″bo-vi′rus)
baculovirus (bak″u-lo-vi′rus)
coronavirus (kor″o-nah-vi′rus)
cytomegalovirus (si″to-meg″ah-lo-vi′rus)
Epstein-Barr virus (ep′stīn-bar′)
hepadnavirus (hep-ad″nə-vi′rus)
hepatitis virus (hep″ah-ti′tis)
herpesvirus (her″pēz-vi′rus)
influenza virus (in″flu-en′zah)
measles virus (me′zelz)
mumps virus (mumps)
orthomyxovirus (or″tho-mik″so-vi′rus)
papillomavirus (pap″ĭ-lo″mah-vi′rus)
paramyxovirus (par″ah-mik″so-vi′rus)
parvovirus (par″vo-vi′rus)
picornavirus (pi-kor″nah-vi′rus)
poliovirus (po″le-o-vi′rus)
polyomavirus (pol″e-o-mah-vi′rus)
poxvirus (poks-vi′rus)
rabies virus (ra′bēz)
reovirus (re″o-vi′rus)
retrovirus (re″tro-vi′rus)
rhabdovirus (rab″do-vi′rus)
rhinovirus (ri″no-vi′rus)
rotavirus (ro′tah-vi″rus)
rubella virus (roo-bel′ah)
togavirus (to″gah-vi′rus)
varicella-zoster virus (var″ĭ-sel′ah zos′ter)
variola virus (vah-ri′o-lah)

Fungi

Agaricus (ah-gar′ĭ-kus)
Amanita (am″ah-ni′tah)
Arthrobotrys (ar″thro-bo′tris)
Aspergillus (as″per-jil′us)
Blastomyces (blas″to-mi′sēz)
Candida (kan′dĭ-dah)
Cephalosporium (sef″ah-lo-spo′re-um)
Claviceps (klav′ĭ-seps)
Coccidioides (kok-sid″e-oi′dēz)
Cryptococcus (krip″to-kok′us)
Epidermophyton (ep″ĭ-der-mof′ĭ-ton)
Fusarium (fu-sa′re-um)
Histoplasma (his″to-plaz′mah)
Microsporum (mi-kros′po-rum)
Mucor (mu′kor)
Neurospora (nu-ros′po-rah)
Penicillium (pen″ĭ-sil′e-um)
Phytophthora (fi-tof′tho-rah)
Rhizopus (ri-zo′pus)
Saccharomyces (sak″ah-ro-mi′sēz)
Saprolegnia (sap″ro-leg′ne-ah)
Sporothrix (spo′ro-thriks)
Trichoderma (trik-o-der′mah)
Trichophyton (tri-kof′ĭ-ton)

Protozoa

Acanthamoeba (ah-kan″thah-me′bah)
Amoeba (ah-me′bah)
Balantidium (bal″an-tid′e-um)
Cryptosporidium (krip″to-spo-rid′e-um)
Entamoeba (en″tah-me′bah)
Giardia (je-ar′de-ah)
Leishmania (lēsh-ma′ne-ah)
Naegleria (na-gle′re-ah)
Paramecium (par″ah-me′she-um)
Plasmodium (plaz-mo′de-um)
Pneumocystis (noo″mo-sis′tis)
Tetrahymena (tet″rah-hi′mĕ-nah)
Toxoplasma (toks″o-plaz′mah)
Trichomonas (trik″o-mo′nas)
Trypanosoma (tri″pan-o-so′mah)

Algae

Acetabularia (as″ĕ-tab″u-la′re-ah)
Chlamydomonas (klah-mid″do-mo′nas)
Chlorella (klo-rel′ah)
Euglena (u-gle′nah)
Gonyaulax (gon″e-aw′laks)
Laminaria (lam″ĭ-na′re-ah)
Prototheca (pro″to-the′kah)
Spirogyra (spi″ro-ji′rah)
Volvox (vol′voks)

GLOSSARY

Pronunciation Guide

Many of the boldface terms in this glossary are followed by a phonetic spelling in parentheses. These pronunciation aids usually come from *Dorland's Illustrated Medical Dictionary.* The following rules are taken from this dictionary and will help in using its phonetic spelling system.

1. An unmarked vowel ending a syllable (an open syllable) is long; thus *ma* represents the pronunciation of *may; ne,* that of *knee; ri,* of *wry; so,* of *sew; too,* of *two;* and *vu,* of *view.*
2. An unmarked vowel in a syllable ending with a consonant (a closed syllable) is short; thus *kat* represents *cat; bed, bed; hit, hit; not, knot; foot, foot;* and *kusp, cusp.*
3. A long vowel in a closed syllable is indicated by a macron; thus *māt* stands for *mate; sēd,* for *seed; bīl,* for *bile; mōl,* for *mole; fūm,* for *fume;* and *fo͞ol,* for *fool.*
4. A short vowel that ends or itself constitutes a syllable is indicated by a breve; thus *ĕ-fekt´* for *effect, ĭ-mūn´* for *immune,* and *ŏ-klo͞od´* for *occlude.*

Primary (´) and secondary (″) accents are shown in polysyllabic words. Unstressed syllables are followed by hyphens.

Some common vowels are pronounced as indicated here.

ə	sofa	ĕ	met	ŏ	got
ā	mate	ī	bite	ū	fuel
ă	bat	ĭ	bit	ŭ	but
ē	beam	ō	home		

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A

AB toxins The structure and activity of many exotoxins are based on the AB model. In this model, the B portion of the toxin is responsible for toxin binding to a cell but does not directly harm it. The A portion enters the cell and disrupts its function. *797*

accessory pigments Photosynthetic pigments such as carotenoids and phycobiliproteins that aid chlorophyll in trapping light energy. *196*

acetyl coenzyme A (acetyl-CoA) A combination of acetic acid and coenzyme A that is energy rich; it is produced by many catabolic pathways and is the substrate for the tricarboxylic acid cycle, fatty acid biosynthesis, and other pathways. *183*

acid dyes Dyes that are anionic or have negatively charged groups such as carboxyls. *27*

acid fast Refers to bacteria like the mycobacteria that cannot be easily decolorized with acid alcohol after being stained with dyes such as basic fuchsin. *543*

acid-fast staining A staining procedure that differentiates between bacteria based on their ability to retain a dye when washed with an acid alcohol solution. *28*

acidophile (as′id-o-fīl″) A microorganism that has its growth optimum between about pH 0 and 5.5. *123*

acquired enamel pellicle A membranous layer on the tooth enamel surface formed by selectively adsorbing glycoproteins (mucins) from saliva. This pellicle confers a net negative charge to the tooth surface. *934*

acquired immune deficiency syndrome (AIDS) An infectious disease syndrome caused by the human immunodeficiency virus and is characterized by the loss of a normal immune response, followed by increased susceptibility to opportunistic infections and an increased risk of some cancers. *878*

acquired immune tolerance The ability to produce antibodies against nonself antigens while "tolerating" (not producing antibodies against) self-antigens. *758*

acquired immunity Refers to the type of specific immunity that develops after exposure to a suitable antigen or is produced after antibodies are transferred from one individual to another. *729*

actinobacteria (ak″tĭ-no-bak-tēr-e-ah) A group of gram-positive bacteria containing the actinomycetes and their high G + C relatives. *541*

actinomycete (ak″tĭ-no-mi′sēt) An aerobic, gram-positive bacterium that forms branching filaments (hyphae) and asexual spores. *537*

actinorhizae Associations between actinomycetes and plant roots. *682*

activated sludge (sluj) Solid matter or sediment composed of actively growing microorganisms that participate in the aerobic portion of a biological sewage treatment process. The microbes readily use dissolved organic substrates and transform them into additional microbial cells and carbon dioxide. *659*

activation energy The energy required to bring reacting molecules together to reach the transition state in a chemical reaction. *162*

active carrier An individual who has an overt clinical case of a disease and who can transmit the infection to others. *854*

active immunization The induction of active immunity by natural exposure to a pathogen or by vaccination. *764*

active site The part of an enzyme that binds the substrate to form an enzyme-substrate complex and catalyze the reaction. Also called the catalytic site. *162*

active transport The transport of solute molecules across a membrane against an electrochemical gradient; it requires a carrier protein and the input of energy. *101*

acute carrier *See* casual carrier. *854*

acute infections Virus infections with a fairly rapid onset that last for a relatively short time. *410*

acute viral gastroenteritis An inflammation of the stomach and intestines, normally caused by Norwalk and Norwalklike viruses, other caliciviruses, rotaviruses, and astroviruses. *891*

acyclovir (a-si′klo-vir) A synthetic purine nucleoside derivative with antiviral activity against herpes simplex virus. *821*

adenine (ad′e-nēn) A purine derivative, 6-aminopurine, found in nucleosides, nucleotides, coenzymes, and nucleic acids. *217*

adenosine diphosphate (ADP; ah-den′o-sēn) The nucleoside diphosphate usually formed upon the breakdown of ATP when it provides energy for work. *155*

adenosine 5′-triphosphate (ATP) The triphosphate of the nucleoside adenosine, which is a high energy molecule or has high phosphate group transfer potential and serves as the cell's major form of energy currency. *155*

adhesin (ad-he′zin) A molecular component on the surface of a microorganism that is involved in adhesion to a substratum or cell. Adhesion to a specific host tissue usually is a preliminary stage in pathogenesis, and adhesins are important virulence factors. *792*

adjuvant (aj′ə-vənt) Material added to an antigen to increase its immunogenicity. Common examples are alum, killed *Bordetella pertussis,* and an oil emulsion of the antigen, either alone (Freund's incomplete adjuvant) or with killed mycobacteria (Freund's complete adjuvant). *741*

adult T-cell leukemia A type of white blood cell cancer caused by the HTLV-1 virus. *887*

aerobe (a′er-ōb) An organism that grows in the presence of atmospheric oxygen. *127*

aerobic anoxygenic photosynthesis Photosynthetic process in which electron donors such as organic matter or sulfide, which do not result in oxygen evolution, are used under aerobic conditions. *614*

aerobic respiration (res″pĭ-ra′shun) A metabolic process in which molecules, often organic, are oxidized with oxygen as the final electron acceptor. *154, 173*

aerotolerant anaerobes Microorganisms that grow equally well whether or not oxygen is present. *127*

aflatoxin (af″lah-tok′sin) A polyketide secondary fungal metabolite that can cause cancer. *967*

agar (ahg′ar) A complex sulfated polysaccharide, usually extracted from red algae, that is used as a solidifying agent in the preparation of culture media. *105*

agglutinates The visible aggregates or clumps formed by an agglutination reaction. *775*

agglutination reaction (ah-gloo″tĭ-na′shun) The formation of an insoluble immune complex by the cross-linking of cells or particles. *756*

agglutinin (ah-gloo″tĭ-nin) The antibody responsible for an agglutination reaction. *756*

AIDS *See* acquired immune deficiency syndrome. *878*

AIDS-related complex (ARC) A collection of symptoms such as lymphadenopathy (swollen lymph glands), fever, malaise, fatigue, loss of appetite, and weight loss. It results from an HIV infection and may progress to frank AIDS. *879*

airborne transmission The type of infectious organism transmission in which the pathogen is truly suspended in the air and travels over a meter or more from the source to the host. *854*

akinetes Specialized, nonmotile, dormant, thick-walled resting cells formed by some cyanobacteria. *473*

alcoholic fermentation A fermentation process that produces ethanol and CO_2 from sugars. *179*

alga (al′gah) A common term for a series of unrelated groups of photosynthetic eucaryotic microorganisms lacking multicellular sex organs (except for the charophytes) and conducting vessels. *571*

algicide (al′jĭ-sīd) An agent that kills algae. *138*

algology (al-gol′o-je) The scientific study of algae. *571*

alkalophile A microorganism that grows best at pHs from about 8.5 to 11.5. *123*

allergen (al′er-jen) A substance capable of inducing allergy or specific susceptibility. *768*

allergic contact dermatitis An allergic reaction caused by haptens that combine with proteins in the skin to form the allergen that produces the immune response. *771*

allergy (al′er-je) *See* hypersensitivity. *768*

allograft (al′o-graft) A transplant between genetically different individuals of the same species. *773*

allosteric enzyme (al″o-ster′ik) An enzyme whose activity is altered by the binding of a small effector or modulator molecule at a regulatory site separate from the catalytic site; effector binding causes a conformational change in the enzyme and its catalytic site, which leads to enzyme activation or inhibition. *165*

allotype Allelic variants of antigenic determinant(s) found on antibody chains of some, but not all, members of a species, which are inherited as simple Mendelian traits. *734*

alpha hemolysis A greenish zone of partial clearing around a bacterial colony growing on blood agar. *531, 797*

alpha-proteobacteria One of the five subgroups of proteobacteria, each with distinctive 16S rRNA sequences. This group contains most of the oligotrophic proteobacteria; some have unusual metabolic modes such as methylotrophy, chemolithotrophy, and nitrogen fixing ability. Many have distinctive morphological features. *487*

alternative complement pathway An antibody-independent pathway of complement activation that includes the C3–C9 components of the classical pathway and several other serum protein factors (e.g., factor B and properdin). *716*

alveolar macrophage A vigorously phagocytic macrophage located on the epithelial surface of the lung alveoli where it ingests inhaled particulate matter and microorganisms. *711*

amantadine (ah-man′tah-den) An antiviral compound used to prevent type A influenza infections. *821*

amebiasis (amebic dysentery) (am″e-bi′ah-sis) An infection with amoebae, often resulting in dysentery; usually it refers to an infection by *Entamoeba histolytica.* *950*

amensalism (a-men′səl-iz-əm) A relationship in which the product of one organism has a negative effect on another organism. *609*

American trypanosomiasis (Chagas' disease) *See* trypanosomiasis. *957*

Ames test A test that uses a special *Salmonella* strain to test chemicals for mutagenicity and potential carcinogenicity. *253*

amino acid activation The initial stage of protein synthesis in which amino acids are attached to transfer RNA molecules. *266*

aminoacyl or acceptor site (A site) The site on the ribosome that contains an aminoacyl-tRNA at the beginning of the elongation cycle during protein synthesis; the growing peptide chain is transferred to the aminoacyl-tRNA and lengthens by an amino acid. *270*

aminoglycoside antibiotics (am″ĭ-no-gli′ko-sīd) A group of antibiotics synthesized by *Streptomyces* and *Micromonospora,* which contain a cyclohexane ring and amino sugars; all aminoglycoside antibiotics bind to the small ribosomal subunit and inhibit protein synthesis. *816*

amnesic shellfish poisoning (am-ne′sik) The disease arising in humans and animals that eat seafood such as mussels contaminated with domoic acid from diatoms. The disease produces short-term memory loss in its victims. *580*

amoeboid movement Moving by means of cytoplasmic flow and the formation of pseudopodia (temporary cytoplasmic protrusions of the cytoplasm). *590*

amphibolic pathways (am″fe-bol′ik) Metabolic pathways that function both catabolically and anabolically. *176*

amphitrichous (am-fit′rĕ-kus) A cell with a single flagellum at each end. *63*

amphotericin B (am″fo-ter′i-sin) An antibiotic from a strain of *Streptomyces nodosus* that is used to treat systemic fungal infections; it also is used topically to treat candidiasis. *820*

anabolism (ah-nab′o-lizm″) The synthesis of complex molecules from simpler molecules with the input of energy. *173*

anaerobe (an-a′er-ōb) An organism that grows in the absence of free oxygen. *127*

anaerobic digestion (an″a-er-o′bik) The microbiological treatment of sewage wastes under anaerobic conditions to produce methane. *659*

anaerobic respiration (an″a-er-o′bik) An energy-yielding process in which the electron transport chain acceptor is an inorganic molecule other than oxygen. *173*

anammox process The coupled use of nitrite as an oxidant and ammonium ion as a reductant under anaerobic conditions to yield nitrogen gas. *616*

anamnestic response (an″am-nes′tik) The recall, or the remembering, by the immune system of a prior response to a given antigen. *729, 743*

anaphylaxis (an″ah-fĭ-lak′sis) An immediate (type I) hypersensitivity reaction following exposure of a sensitized individual to the appropriate antigen. Mediated by reagin antibodies, chiefly IgE. *768*

anaplerotic reactions (an″ah-plĕ-rot′ik) Reactions that replenish depleted tricarboxylic acid cycle intermediates. *216*

anergy (an′ər-je) A state of unresponsiveness to antigens. Absence of the ability to generate a sensitivity reaction to substances that are expected to be antigenic. *758*

annotation The process of determining the location of specific genes in a genome map after it has been produced by nucleic acid sequencing. *347*

anogenital condylomata (venereal warts) (kon″dĭ-lo″ mah-tah) Warts that are sexually transmitted and caused by types 6, 11, and 42 human papillomavirus. Usually occur around the cervix, vulva, perineum, anus, anal canal, urethra, or glans penis. *894*

anoxic (ə-nok′ sik) Without oxygen present. *635*

anoxygenic photosynthesis Photosynthesis that does not oxidize water to produce oxygen; the form of photosynthesis characteristic of purple and green photosynthetic bacteria. *199, 468*

antheridium (an″ther-id′e-um; pl., **antheridia**) A male gamete-producing organ, which may be unicellular or multicellular. *561, 574*

anthrax (an′thraks) An infectious disease of animals caused by ingesting *Bacillus anthracis* spores. Can also occur in humans and is sometimes called woolsorter's disease. *913*

antibiotic (an″tĭ-bi-ot′ik) A microbial product or its derivative that kills susceptible microorganisms or inhibits their growth. *806*

antibody (immunoglobulin) (an′tĭ-bod″e) A glycoprotein produced in response to the introduction of an antigen; it has the ability to combine with the antigen that stimulated its production. Also known as an immunoglobulin (Ig). *734*

antibody-dependent cell-mediated cytotoxicity (ADCC) The killing of antibody-coated target cells by cells with Fc receptors that recognize the Fc region of the bound antibody. Most ADCC is mediated by NK cells that have the Fc receptor or CD16 on their surface. *723*

antibody-mediated immunity *See* humoral immunity. *729*

anticodon triplet The base triplet on a tRNA that is complementary to the triplet codon on mRNA. *266*

antigen (an′tĭ-jen) A foreign (nonself) substance (such as a protein, nucleoprotein, polysaccharide, or sometimes a glycolipid) to which lymphocytes respond; also known as an immunogen because it induces the immune response. *731*

antigen-binding fragment (Fab) "Fragment antigen binding." A monovalent antigen-binding fragment of an immunoglobulin molecule that consists of one light chain and part of one heavy chain, linked by interchain disulfide bonds. *734*

antigenic determinant site (epitope) *See* epitope. *731*

antigenic drift A small change in the antigenic character of an organism that allows it to avoid attack by the immune system. *852*

antigenic shift A major change in the antigenic character of an organism that alters it to an antigenic strain unrecognized by host immune mechanisms. *852*

antigen-presenting cells Antigen-presenting cells (APCs) are cells that take in protein antigens, process them, and present antigen fragments to B cells and T cells in conjunction with class II MHC molecules so that the cells are activated. Macrophages, B cells, dendritic cells, and Langerhans cells may act as APCs. *745*

antimetabolite (an″tĭ-mĕ-tab′o-līt) A compound that blocks metabolic pathway function by competitively inhibiting a key enzyme's use of a metabolite because it closely resembles the normal enzyme substrate. *812*

antimicrobial agent An agent that kills microorganisms or inhibits their growth. *139*

antisense RNA A single-stranded RNA with a base sequence complementary to a segment of another RNA molecule that can specifically bind to the target RNA and inhibit its activity. *283*

antisepsis (an″tĭ-sep′sis) The prevention of infection or sepsis. *138*

antiseptic (an″tĭ-sep′tik) Chemical agents applied to tissue to prevent infection by killing or inhibiting pathogens. *138*

antiserum (an″tĭ-se′rum) Serum containing induced antibodies. *742*

antitoxin (an″tĭ-tok′sin) An antibody to a microbial toxin, usually a bacterial exotoxin, that combines specifically with the toxin, in vivo and in vitro, neutralizing the toxin. *756, 796*

apical complex (ap′ĭ-kal) A set of organelles characteristic of members of the phylum *Apicomplexa:* polar rings, subpellicular microtubules, conoid, rhoptries, and micronemes. *591*

apicomplexan (a′pĭ-kom-plek′san) A sporozoan protist that lacks special locomotor organelles but has an apical complex and a spore-forming stage. It is either an intra- or extracellular parasite of animals; a member of the phylum *Apicomplexa.* *591*

aplanospore (a′plan-o-spor) A nonflagellated, nonmotile spore that is involved in asexual reproduction. *573*

apoenzyme (ap″o-en′zīm) The protein part of an enzyme that also has a nonprotein component. *161*

apoptosis (ap″o-to′sis) Programmed cell death. The fragmentation of a cell into membrane-bound particles that are eliminated by phagocytosis. Apoptosis is a physiological suicide mechanism that preserves homeostasis and occurs during normal tissue turnover. It is responsible for cell death in pathological circumstances, such as exposure to low concentrations of xenobiotics and infections by HIV and various other viruses. Apoptotic cells display profound structural changes such as plasma membrane blebbing and nuclear collapse. DNA is cleaved into short oligonucleosomal length DNA fragments. Apoptosis usually occurs after the activation of a calcium-dependent endogenous endonuclease. *750, 881*

aporepressor An inactive form of the repressor protein, which becomes the active repressor when the corepressor binds to it. *276*

arbuscular mycorrhizal (AM) fungi The mycorrhizal fungi in a symbiotic fungus-root association that penetrate the outer layer of the root, grow intracellularly, and form characteristic much-branched hyphal structures called arbuscules. *681*

arbuscules Branched, treelike structures formed in cells of plant roots colonized by endotrophic mycorrhizal fungi. *681*

Archaea The domain that contains procaryotes with isoprenoid glycerol diether or diglycerol tetraether lipids in their membranes and archaeal rRNA (among many differences). *424, 451*

arthroconidium (ar′thro-ko-nid′e-um; pl., **arthroconidia**) A thallic conidium released by the fragmentation or lysis of hypha. It is not notably larger than the parental hypha, and separation occurs at a septum. *557*

arthrospore (ar′thro-spōr) A spore resulting from the fragmentation of a hypha. *557*

artificially acquired active immunity The type of immunity that results from immunizing an animal with a vaccine. The immunized animal now produces its own antibodies and activated lymphocytes. *730*

artificially acquired passive immunity The type of immunity that results from introducing into an animal antibodies that have been produced either in another animal or by in vitro methods. Immunity is only temporary. *731*

ascocarp (as′ko-karp) A multicellular structure in ascomycetes lined with specialized cells called asci in which nuclear fusion and meiosis produce ascospores. An ascocarp can be open or closed and may be referred to as a fruiting body. *561*

ascogenous hypha A specialized hypha that gives rise to one or more asci. *561*

ascogonium (as″ko-go′ne-um; pl., **ascogonia**) The receiving (female) organ in ascomycetous fungi which, after fertilization, gives rise to ascogenous hyphae and later to asci and ascospores. *561*

ascomycetes (as″ko-mi-se′tēz) A division of fungi that form ascospores. *560*

ascospore (as′ko-spor) A spore contained or produced in an ascus. *558*

ascus (as′kus) A specialized cell, characteristic of the ascomycetes, in which two haploid nuclei fuse to produce a zygote, which immediately divides by meiosis; at maturity an ascus will contain ascospores. *561*

aseptic meningitis syndrome *See* meningitis. *902*

aspergillosis (as″per-jil-o′sis) A fungal disease caused by species of *Aspergillus.* *948*

assimilatory reduction The reduction of an inorganic molecule to incorporate it into organic material. No energy is made available during this process. *210, 211, 614*

associative nitrogen fixation Nitrogen fixation by bacteria in the plant root zone (rhizosphere). *675*

athlete's foot *See* tinea pedis. *944*

atomic force microscope A type of scanning probe microscope that images a surface by moving a sharp probe over the surface at a constant distance; a very small amount of force is exerted on the tip and probe movement is followed with a laser. *38*

ATP-binding cassette transporters (ABC transporters) Membrane protein complexes that use ATP energy to move substances across membranes without modifying the compound being transported. They require an extracytoplasmic substrate-binding protein for proper function. *101*

attenuation (ah-ten″u-a′shun) 1. A mechanism for the regulation of transcription of some bacterial operons by aminoacyl-tRNAs. 2. A procedure that reduces or abolishes the virulence of a pathogen without altering its immunogenicity. *281, 766*

attenuator A rho-independent termination site in the leader sequence that is involved in attenuation. *279*

autoclave (aw′to-klāv) An apparatus for sterilizing objects by the use of steam under pressure. Its development tremendously stimulated the growth of microbiology. *140*

autogenous infection (aw-toj′e-nus) An infection that results from a patient's own microbiota, regardless of whether the infecting organism became part of the patient's microbiota subsequent to admission to a clinical care facility. *866*

autoimmune disease (aw″to-ĭ-mūn′) A disease produced by the immune system attacking self-antigens. Autoimmune disease results from the activation of self-reactive T and B cells that damage tissues after stimulation by genetic or environmental triggers. *772*

autoimmunity (aw″to-ĭ-mun′ĭ-te) Autoimmunity is a condition characterized by the presence of serum autoantibodies and self-reactive lymphocytes. It may be benign or pathogenic. Autoimmunity is a normal consequence of aging; is readily inducible by infectious agents, organisms, or drugs; and is potentially reversible in that it disappears when the offending "agent" is removed or eradicated. *772*

autolysins (aw-tol′ĭ-sins) Enzymes that partially digest peptidoglycan in growing bacteria so that the peptidoglycan can be enlarged. *223*

autotroph (aw′to-trōf) An organism that uses CO_2 as its sole or principal source of carbon. *96*

auxotroph (awk′so-trōf) A mutated prototroph that lacks the ability to synthesize an essential nutrient and therefore must obtain it or a precursor from its surroundings. *245*

axial filament The organ of motility in spirochetes. It is made of axial fibrils or periplasmic flagella that extend from each end of the protoplasmic cylinder and overlap in the middle of the cell. The outer sheath lies outside the axial filament. *66, 479*

B

bacillus (bah-sil′lus) A rod-shaped bacterium. *43*

bacteremia (bak″ter-e′me-ah) The presence of viable bacteria in the blood. *793*

Bacteria (bak-te′re-ə) The domain that contains procaryotic cells with primarily diacyl glycerol diesters in their membranes and with bacterial rRNA. Bacteria also is a general term for organisms that are composed of procaryotic cells and are not multicellular. *424*

bacterial artificial chromosome (BAC) A cloning vector constructed from the *E. coli* F-factor plasmid that is used to clone foreign DNA fragments in *E. coli.* *335*

bacterial (septic) meningitis *See* meningitis. *902*

bacterial vaginosis (bak-te′re-əl vaj″ĭ-no′sis) Bacterial vaginosis is a sexually transmitted disease caused by *Gardnerella vaginalis, Mobiluncus* spp., *Mycoplasma hominis,* and various anaerobic bacteria. Although a mild disease it is a risk factor for obstetric infections and pelvic inflammatory disease. *914*

bactericide (bak-tēr′ĭ-sid) An agent that kills bacteria. *138*

bacteriochlorophyll (bak-te″re-o-klo′ro-fil) A modified chlorophyll that serves as the primary light-trapping pigment in purple and green photosynthetic bacteria. *199*

bacteriocin (bak-te′re-o-sin) A protein produced by a bacterial strain that kills other closely related strains. *297, 712, 972*

bacteriophage (bak-te′re-o-fāj″) A virus that uses bacteria as its host; often called a phage. *364, 382*

bacteriophage (phage) typing A technique in which strains of bacteria are identified based on their susceptibility to a variety of bacteriophages. *842*

bacteriostatic (bak-te″re-o-stat′ik) Inhibiting the growth and reproduction of bacteria. *138*

bacteroid (bak′tĕ-roid) A modified, often pleomorphic, bacterial cell within the root nodule cells of legumes; after transformation into a symbiosome it carries out nitrogen fixation. *676*

baeocytes Small, spherical, reproductive cells produced by pleurocapsalean cyanobacteria through multiple fission. *475*

balanced growth Microbial growth in which all cellular constituents are synthesized at constant rates relative to each other. *114*

balanitis (bal″ah-ni′tis) Inflammation of the glans penis usually associated with *Candida* fungi; a sexually transmitted disease. *950*

barophilic (bar″o-fil′ik) or **barophile** Organisms that prefer or require high pressures for growth and reproduction. *129, 644*

barotolerant Organisms that can grow and reproduce at high pressures but do not require them. *129, 624*

basal body The cylindrical structure at the base of procaryotic and eucaryotic flagella that attaches them to the cell. *64, 90*

base analogs Molecules that resemble normal DNA nucleotides and can substitute for them during DNA replication, leading to mutations. *246*

basic dyes Dyes that are cationic, or have positively charged groups, and bind to negatively charged cell structures. Usually sold as chloride salts. *27*

basidiocarp (bah-sid′e-o-karp″) The fruiting body of a basidiomycete that contains the basidia. *561*

basidiomycetes (bah-sid″e-o-mi-se′tēz) A division of fungi in which the spores are born on club-shaped organs called basidia. *561*

basidiospore (bah-sid′e-o-spōr) A spore born on the outside of a basidium following karyogamy and meiosis. *558*

basidium (bah-sid′e-um; pl., **basidia**) A structure that bears on its surface a definite number of basidiospores (typically four) that are formed following karyogamy and meiosis. Basidia are found in the basidiomycetes and are usually club-shaped. *561*

basophil (ba′so-fil) A phagocytic leukocyte whose granules stain bluish-black with a basic dye. It has a segmented nucleus. The granules contain histamine and heparin. *707*

batch culture A culture of microorganisms produced by inoculating a closed culture vessel containing a single batch of medium. *113*

B cell, also known as a **B lymphocyte** A type of lymphocyte derived from bone marrow stem cells that matures into an immunologically competent cell under the influence of the bursa of Fabricius in the chicken and bone marrow in nonavian species. Following interaction with antigen, it becomes a plasma cell, which synthesizes and secretes antibody molecules involved in humoral immunity. *705, 751*

B-cell antigen receptor (BCR) A transmembrane immunoglobulin complex on the surface of a B cell that binds an antigen and stimulates the B cell. It is composed of a membrane-bound immunoglobulin, usually IgD or a modified IgM, complexed with another membrane protein (the Ig-α/Ig-β heterodimer). *751*

benthic (ben′thic) Pertaining to the bottom of the sea or another body of water. *571*

beta hemolysis A zone of complete clearing around a bacterial colony growing on blood agar. The zone does not change significantly in color. *532, 797*

β-oxidation pathway The major pathway of fatty acid oxidation to produce NADH, $FADH_2$, and acetyl coenzyme A. *192*

beta-proteobacteria One of the five subgroups of proteobacteria, each with distinctive 16S rRNA sequences. Members of this subgroup are similar to the alpha-proteobacteria metabolically, but tend to use substances that diffuse from organic matter decomposition in anaerobic zones. *495*

binal symmetry The symmetry of some virus capsids (e.g., those of complex phages) that is a combination of icosahedral and helical symmetry. *376*

binary fission Asexual reproduction in which a cell or an organism separates into two cells. *490, 573, 586*

binomial system The nomenclature system in which an organism is given two names; the first is the capitalized generic name, and the second is the uncapitalized specific epithet. *426*

biochemical oxygen demand (BOD) The amount of oxygen used by organisms in water under certain standard conditions; it provides an index of the amount of microbially oxidizable organic matter present. *657*

biodegradation (bi″o-deg″rah-da′shun) The breakdown of a complex chemical through biological processes that can result in minor loss of functional groups, fragmentation into larger constitutents, or complete breakdown to carbon dioxide and minerals. Often the term refers to the undesired microbial-mediated destruction of materials such as paper, paint, and textiles. *1010*

biofilms Organized microbial systems consisting of layers of microbial cells associated with surfaces, often with complex structural and functional characteristics. Biofilms have physical/chemical gradients that influence microbial metabolic processes. They can form on inanimate devices (catheters, medical prosthetic devices) and also cause fouling (e.g., of ships' hulls, water pipes, cooling towers). *620, 920*

biogeochemical cycling The oxidation and reduction of substances carried out by living organisms and/or abiotic processes that results in the cycling of elements within and between different parts of the ecosystem (the soil, aquatic environment, and atmosphere). *611*

bioinsecticide A pathogen that is used to kill or disable unwanted insect pests. Bacteria, fungi, or viruses are used, either directly or after manipulation, to control insect populations. *1018*

biologic transmission A type of vector-borne transmission in which a pathogen goes through some morphological or physiological change within the vector. *858*

bioluminescence (bi″o-loo″mĭ-nes′əns) The production of light by living cells, often through the oxidation of molecules by the enzyme luciferase. *505*

biomagnification The increase in concentration of a substance in higher-level consumer organisms. *618*

biopesticide The use of a microorganism or another biological agent to control a specific pest. *1018*

bioremediation The use of biologically mediated processes to remove or degrade pollutants from specific environments. Bioremediation can be carried out by modification of the environment to accelerate biological processes, either with or without the addition of specific microorganisms. *1012*

biosensor The coupling of a biological process with production of an electrical signal or light to detect the presence of particular substances. *1017*

biosynthesis *See* anabolism. *173*

bioterrorism The intentional or threatened use of viruses, bacteria, fungi, or toxins from living organisms to produce death or disease in humans, animals, and plants. *863*

biotransformation or **microbial transformation** The use of living organisms to modify substances that are not normally used for growth. *1009*

black peidra (pe-a′drah) A fungal infection caused by *Piedraia hortae* that forms hard black nodules on the hairs of the scalp. *943*

blastomycosis (blas″to-mi-ko′sis) A systemic fungal infection caused by *Blastomyces dermatitidis* and marked by suppurating tumors in the skin or by lesions in the lungs. *946*

blastospore (blas′to-spōr) A spore formed by budding from a hypha. *557*

B lymphocyte *See* B cell. *705, 751*

botulism (bochʹoo-lizm) A form of food poisoning caused by a neurotoxin (botulin) produced by *Clostridium botulinum* serotypes A–G; sometimes found in improperly canned or preserved food. *929*

bright-field microscope A microscope that illuminates the specimen directly with bright light and forms a dark image on a brighter background. *19*

Bright's disease *See* glomerulonephritis. *905*

broad-spectrum drugs Chemotherapeutic agents that are effective against many different kinds of pathogens. *808*

bronchial-associated lymphoid tissue (BALT) The type of defensive tissue found in the lungs. Part of the nonspecific immune system. *710*

bronchial asthma An example of an atopic allergy involving the lower respiratory tract. *769*

bubo (buʹbo) A tender, inflamed, enlarged lymph node that results from a variety of infections. *911*

bubonic plague *See* plague. *911*

budding A vegetative outgrowth of yeast and some bacteria as a means of asexual reproduction; the daughter cell is smaller than the parent. *490*

bulking sludge Sludges produced in sewage treatment that do not settle properly, usually due to the development of filamentous microorganisms. *659*

bursa of Fabricius (bərʹsə fə-brisʹe-əs) Found in birds; the blind saclike structure located on the posterior wall of the cloaca; it performs a thymuslike function. A primary lymphoid organ where B-cell maturation occurs. Bone marrow is the equivalent in mammals. *708*

burst *See* rise period. *383*

burst size The number of phages released by a host cell during the lytic life cycle. *383*

butanediol fermentation A type of fermentation most often found in the family *Enterobacteriaceae* in which 2,3-butanediol is a major product; acetoin is an intermediate in the pathway and may be detected by the Voges-Proskauer test. *181, A–18*

C

Calvin cycle The main pathway for the fixation (or reduction and incorporation) of CO_2 into organic material by photoautotrophs during photosynthesis; it also is found in chemolithoautotrophs. *207, A–20*

cancer (kanʹser) A malignant tumor that expands locally by invasion of surrounding tissues, and systemically by metastasis. *411*

candidal vaginitis Vaginitis caused by *Candida* species. *950*

candidiasis (kan″dĭ-diʹah-sis) An infection caused by *Candida* species of dimorphic fungi, commonly involving the skin. *949*

capsid (kapʹsid) The protein coat or shell that surrounds a virion's nucleic acid. *369*

capsomer (kapʹso-mer) The ring-shaped morphological unit of which icosahedral capsids are constructed. *390*

capsule A layer of well-organized material, not easily washed off, lying outside the bacterial cell wall. *61*

carboxysomes Polyhedral inclusion bodies that contain the CO_2 fixation enzyme ribulose 1,5-bisphosphate carboxylase; found in cyanobacteria, nitrifying bacteria, and thiobacilli. *51, 207*

caries (karʹe-ēz) Tooth decay. *936*

carotenoids (kah-rotʹe-noids) Pigment molecules, usually yellowish in color, that are often used to aid chlorophyll in trapping light energy during photosynthesis. *196*

carrier An infected individual who is a potential source of infection for others and plays an important role in the epidemiology of a disease. *854*

caseous lesion (kaʹse-us) A lesion resembling cheese or curd; cheesy. Most caseous lesions are caused by *M. tuberculosis.* *908*

casual carrier An individual who harbors an infectious organism for only a short period. *854*

catabolism (kah-tabʹo-lizm) That part of metabolism in which larger, more complex molecules are broken down into smaller, simpler molecules with the release of energy. *173*

catabolite repression (kah-tabʹo-līt) Inhibition of the synthesis of several catabolic enzymes by a metabolite such as glucose. *281*

catalyst (katʹah-list) A substance that accelerates a reaction without being permanently changed itself. *161*

catalytic site *See* active site. *162*

catheter (kathʹ ĕ-ter) A tubular surgical instrument for withdrawing fluids from a cavity of the body, especially one for introduction into the bladder through the urethra for the withdrawal of urine. *827*

cat-scratch disease (CSD) A loosely defined syndrome caused by either of the following gram-negative bacilli: *Bartonella (Rochalimaea) henselae* or *Afipia felis.* The typical case of CSD is self-limiting, with abatement of symptoms over a period of days to weeks. *914*

CD95 pathway The CD95 receptor is found on many nucleated eucaryotic cells. When the receptor is bound to a specific ligand (CD95L), the CD95-CD95L complex activates several cytoplasmic proteins that initiate a cellular suicide cascade leading to apoptosis. *750*

cell cycle The sequence of events in a cell's growth-division cycle between the end of one division and the end of the next. In eucaryotic cells, it is composed of the G_1 period, the S period in which DNA and histones are synthesized, the G_2 period, and the M period (mitosis). *87, 285*

cell-mediated immunity The type of immunity that results from T cells coming into close contact with foreign cells or infected cells to destroy them; it can be transferred to a nonimmune individual by the transfer of cells. *729*

cellular slime molds Slime molds with a vegetative phase consisting of amoeboid cells that aggregate to form a multicellular pseudoplasmodium; they belong to the division *Acrasiomycota.* *565*

cellulitis (sel″u-liʹtis) A diffuse spreading infection of subcutaneous skin tissue caused by streptococci, staphylococci, or other organisms. The tissue is inflamed with edema, redness, pain, and interference with function. *903*

cell wall The strong layer or structure that lies outside the plasma membrane; it supports and protects the membrane and gives the cell shape. *88*

cephalosporin (sef″ah-lo-spōrʹin) A group of β-lactam antibiotics derived from the fungus *Cephalosporium,* which share the 7-aminocephalosporanic acid nucleus. *814*

chancre (shangʹker) The primary lesion of syphilis, occurring at the site of entry of the infection. *923*

chancroid (shangʹkroid) A sexually transmitted disease caused by the gram-negative bacterium *Haemophilus ducreyi.* Worldwide, chancroid is an important cofactor in the transmission of the AIDS virus. Also known as genital ulcer disease due to the painful circumscribed ulcers that form on the penis or entrance to the vagina. *914*

chemical oxygen demand (COD) The amount of chemical oxidation required to convert organic matter in water and wastewater to CO_2. *657*

chemiosmotic hypothesis (kem″e-o-os-motʹik) The hypothesis that a proton gradient and an electrochemical gradient are generated by electron transport and then used to drive ATP synthesis by oxidative phosphorylation. *187*

chemoheterotroph (ke″mo-hetʹer-o-trōf″) *See* chemoorganotrophic heterotrophs. *98*

chemolithotroph (ke″mo-lithʹo-trōf) *See* chemolithotrophic autotrophs. *98, 193*

chemolithotrophic autotrophs Microorganisms that oxidize reduced inorganic compounds to derive both energy and electrons; CO_2 is their carbon source. Also called chemolithoautotrophs. *98*

chemoorganotrophic heterotrophs Organisms that use organic compounds as sources of energy, hydrogen, electrons, and carbon for biosynthesis. *98*

chemoreceptors Special protein receptors in the plasma membrane or periplasmic space that bind chemicals and trigger the appropriate chemotaxic response. *67*

chemostat (keʹmo-stat) A continuous culture apparatus that feeds medium into the culture vessel at the same rate as medium containing microorganisms is removed; the medium in a chemostat contains one essential nutrient in a limiting quantity. *120*

chemotaxis (ke″mo-takʹsis) The pattern of microbial behavior in which the microorganism moves toward chemical attractants and/or away from repellents. *67*

chemotherapeutic agents (ke″mo-ther-ah-puʹtik) Compounds used in the treatment of disease that destroy pathogens or inhibit their growth at concentrations low enough to avoid doing undesirable damage to the host. *806*

chemotrophs (keʹmo-trōfs) Organisms that obtain energy from the oxidation of chemical compounds. *97*

chickenpox (**varicella;** chikʹen-poks) A highly contagious skin disease, usually affecting 2- to 7-year-old children; it is caused by the varicella-zoster virus, which is acquired by droplet inhalation into the respiratory system. *871*

chimera (ki-meʹrah) A recombinant plasmid containing foreign DNA, which is used as a cloning vector in genetic engineering. *334*

chiral (kiʹ rəl) Having handedness: consisting of one or another stereochemical form. *1010*

chitin (ki′tin) A tough, resistant, nitrogen-containing polysaccharide forming the walls of certain fungi, the exoskeleton of arthropods, and the epidermal cuticle of other surface structures of certain protists and animals. *554*

chlamydiae (klə-mid′e-e) Members of the genus *Chlamydia:* gram-negative, coccoid cells that reproduce only within the cytoplasmic vesicles of host cells using a life cycle that alternates between elementary bodies and reticulate bodies. *477*

chlamydial pneumonia (klə-mid′e-əl noo-mo′ ne-ə) A pneumonia caused by *Chlamydia pneumoniae.* Clinically, infections are mild and 50% of adults have antibodies to the chlamydiae. *914*

chlamydospore (klam′ĭ-do-spōr″) An asexually produced, thick-walled resting spore formed by some fungi. *557*

chloramphenicol (klo″ram-fen′ĭ-kol) A broad-spectrum antibiotic that is produced by *Streptomyces venezuelae* or synthetically; it binds to the large ribosomal subunit and inhibits the peptidyl transferase reaction. *817*

chlorophyll (klor′o-fil) The green photosynthetic pigment that consists of a large tetrapyrrole ring with a magnesium atom in the center. *196*

chloroplast (klo′ra-plast) A eucaryotic plastid that contains chlorophyll and is the site of photosynthesis. *85*

cholera (kol′er-ah) An acute infectious enteritis, endemic and epidemic in Asia, which periodically spreading to the Middle East, Africa, Southern Europe, and South America; caused by *Vibrio cholerae.* *930*

choleragen (kol′er-ah-gen) The cholera toxin; an extremely potent protein molecule elaborated by strains of *Vibrio cholerae* in the small intestine after ingestion of feces-contaminated water or food. It acts on epithelial cells to cause hypersecretion of chloride and bicarbonate and an outpouring of large quantities of fluid from the mucosal surface. *930*

chromatin (kro′mah-tin) The DNA-containing portion of the eucaryotic nucleus; the DNA is almost always complexed with histones. It can be very condensed (heterochromatin) or more loosely organized and genetically active (euchromatin). *86*

chromoblastomycosis (kro″mo-blas″to-mi-ko′sis) A chronic fungal infection of the skin, producing wartlike nodules that may ulcerate. It is caused by the black molds *Phialophora verrucosa* or *Fonsecaea pedrosoi.* *945*

chromogen (kro′me-jen) A colorless substrate that is acted on by an enzyme to produce a colored end product. *779*

chromophore group (kro′mo-fōr) A chemical group with double bonds that absorbs visible light and gives a dye its color. *27*

chromosomes (kro′mo-somz) The bodies that have most or all of the cell's DNA and contain most of its genetic information (mitochondria and chloroplasts also contain DNA and genes). *86*

chronic carrier An individual who harbors a pathogen for a long time. *854*

chrysolaminarin The polysaccharide storage product of the chrysophytes and diatoms. *577*

chytrids A group of chytridiomycetes, which are simple terrestrial and aquatic fungi that produce motile zoospores with single, posterior, whiplash flagella. Also considered protists. *564, 641*

cilia (sil′e-ah) Threadlike appendages extending from the surface of some protozoa that beat rhythmically to propel them; cilia are membrane-bound cylinders with a complex internal array of microtubules, usually in a 9 + 2 pattern. *89*

citric acid cycle *See* tricarboxylic acid (TCA) cycle. *183, A–16*

classical complement pathway The antibody-dependent pathway of complement activation; it leads to the lysis of pathogens and stimulates phagocytosis and other host defenses. *758*

classification The arrangement of organisms into groups based on mutual similarity or evolutionary relatedness. *422*

clone (klōn) A group of genetically identical cells or organisms derived by asexual reproduction from a single parent. *228, 741*

clostridial myonecrosis (klo-strid′e-al mi″o-ne-kro′sis) Death of individual muscle cells caused by clostridia. Also called gas gangrene. *915*

cluster of differentiation molecules (CDs) Functional cell surface proteins or receptors that can be measured in situ from peripheral blood, biopsy samples, or other body fluids. They can be used to identify leukocyte subpopulations. Some examples include interleukin-2 receptor (IL-2R), CD4, CD8, CD25, and intercellular adhesion molecule-1 (ICAM-1). *733*

coaggregation The collection of a variety of bacteria on a surface such as a tooth surface because of cell-to-cell recognition of genetically distinct bacterial types. Many of these interactions appear to be mediated by a lectin on one bacterium that interacts with a complementary carbohydrate receptor on another bacterium. *934*

coagulase (ko-ag′u-las) An enzyme that induces blood clotting; it is characteristically produced by pathogenic staphylococci. *529*

coccidioidomycosis (kok-sid″e-oi″do-mi-ko′sis) A fungal disease caused by *Coccidioides immitis* that exists in dry, highly alkaline soils. Also known as valley fever, San Joaquin fever, or desert rheumatism. *946*

coccus (kok′us, pl. **cocci,** kok′si) A roughly spherical bacterial cell. *42*

code degeneracy The genetic code is organized in such a way that often there is more than one codon for each amino acid. *240*

codon (ko′don) A sequence of three nucleotides in mRNA that directs the incorporation of an amino acid during protein synthesis or signals the start or stop of translation. *240*

coenocytic (se″no-sit′ik) Refers to a multinucleate cell or hypha formed by repeated nuclear divisions not accompanied by cell divisions. *113, 556*

coenzyme (ko-en′zīm) A loosely bound cofactor that often dissociates from the enzyme active site after product has been formed. *161*

cofactor The nonprotein component of an enzyme; it is required for catalytic activity. *161*

cold sore A lesion caused by the herpes simplex virus; usually occurs on the border of the lips or nares. Also known as a fever blister or herpes labialis. *884*

colicin (kol′ĭ-sin) A plasmid-encoded protein that is produced by enteric bacteria and binds to specific receptors on the cell envelope of sensitive target bacteria, where it may cause lysis or attack specific intracellular sites such as ribosomes. *712*

coliform (ko′lĭ-form) A gram-negative, nonsporing, facultative rod that ferments lactose with gas formation within 48 hours at 35°C. *654*

colonization (kol″ə-nĭ-za′shən) The establishment of a site of microbial reproduction on an inanimate surface or organism without necessarily resulting in tissue invasion or damage. *792*

colony A cluster or assemblage of microorganisms growing on a solid surface such as the surface of an agar culture medium; the assemblage often is directly visible, but also may be seen only microscopically. *106*

colony forming units (CFU) The number of microorganisms that can form colonies when cultured using spread plates or pour plates, an indication of the number of viable microorganisms in a sample. *118*

Colorado tick fever A disease that occurs in the mountainous regions of the western United States. It is caused by an RNA virus of the genus *Coltivirus* that is spread from ground squirrels, rabbits, and deer to humans by the tick, *Dermacentor andersoni.* Complications are rare. *878*

colorless sulfur bacteria A diverse group of nonphotosynthetic proteobacteria that can oxidize reduced sulfur compounds such as hydrogen sulfide. Many are lithotrophs and derive energy from sulfur oxidation. Some are unicellular, whereas others are filamentous gliding bacteria. *496*

combinatorial biology Introduction of genes from one microorganism into another microorganism to synthesize a new product or a modified product, especially in relation to antibiotic synthesis. *995*

comedo (kom′ĕ-do; pl., **comedones**) A plug of dried sebum in an excretory duct of the skin. *701*

cometabolism The modification of a compound not used for growth by a microorganism, which occurs in the presence of another organic material that serves as a carbon and energy source. *1013*

commensal (kŏ-men′sal) Living on or within another organism without injuring or benefiting the other organism. *606*

commensalism (kŏ-men′sal-izm″) A type of symbiosis in which one individual gains from the association and the other is neither harmed nor benefited. *606*

common cold An acute, self-limiting, and highly contagious virus infection of the upper respiratory tract that produces inflammation, profuse discharge, and other symptoms. *884*

common-source epidemic An epidemic that is characterized by a sharp rise to a peak and then a rapid, but not as pronounced, decline in the number of individuals infected; it usually involves a single contaminated source from which individuals are infected. *851*

common vehicle transmission The transmission of a pathogen to a host by means of an inanimate medium or vehicle. *857*

communicable disease A disease associated with a pathogen that can be transmitted from one host to another. *854*

community An assemblage of different types of organisms or a mixture of different microbial populations. *596*

competent A bacterial cell that can take up free DNA fragments and incorporate them into its genome during transformation. *305*

competition An interaction between two organisms attempting to use the same resource (nutrients, space, etc.). *609*

competitive exclusion principle Two competing organisms overlap in resource use, which leads to the exclusion of one of the organisms. *609, 987*

complementary DNA (cDNA) A DNA copy of an RNA molecule (e.g., a DNA copy of an mRNA). *321*

complement system A group of plasma proteins that plays a major role in an animal's defensive immune response. *714, 758*

complex medium Culture medium that contains some ingredients of unknown chemical composition. *105*

complex viruses Viruses with capsids having a complex symmetry that is neither icosahedral nor helical. *369*

composting The microbial processing of fresh organic matter under moist, aerobic conditions, resulting in the accumulation of a stable humified product, which is suitable for soil improvement and stimulation of plant growth. *686*

compromised host A host with lowered resistance to infection and disease for any of several reasons. The host may be seriously debilitated (due to malnutrition, cancer, diabetes, leukemia, or another infectious disease), traumatized (from surgery or injury), immunosuppressed, or have an altered microbiota due to prolonged use of antibiotics. *704, 948*

concatemer A long DNA molecule consisting of several genomes linked together in a row. *387*

conditional mutations Mutations that are expressed only under certain environmental conditions. *245*

confocal scanning laser microscope (CSLM) A light microscope in which monochromatic laser-derived light scans across the specimen at a specific level and illuminates one spot at a time to form an image. Stray light from other parts of the specimen is blocked out to give an image with excellent contrast and resolution. *36*

congenital (neonatal) herpes An infection of a newborn caused by transmission of the herpesvirus during vaginal delivery. *886*

congenital rubella syndrome A wide array of congenital defects affecting the heart, eyes, and ears of a fetus during the first trimester of pregnancy, and caused by the rubella virus. *876*

congenital syphilis Syphilis that is acquired in utero from the mother. *923*

conidiospore (ko-nid′e-o-spōr) An asexual, thin-walled spore borne on hyphae and not contained within a sporangium; it may be produced singly or in chains. *537, 557*

conidium (ko-nid′e-um; pl., **conidia**) *See* conidiospore. *537*

conjugants (kon′joo-gants) Complementary mating types that participate in a form of protozoan sexual reproduction called conjugation. *586*

conjugation (kon″ju-ga′shun) 1. The form of gene transfer and recombination in bacteria that requires direct cell-to-cell contact. 2. A complex form of sexual reproduction commonly employed by protozoa. *302, 586*

conjugative plasmid A plasmid that carries the genes for sex pili and can transfer copies of itself to other bacteria during conjugation. *294*

conjunctivitis of the newborn *See* ophthalmia neonatorum. *916*

conoid (ko′noid) A hollow cone of spirally coiled filaments in the anterior tip of certain apicomplexan protozoa. *591*

consortium A physical association of two different organisms, usually beneficial to both organisms. *596*

constant region (C_L and C_H) The part of an antibody molecule that does not vary greatly in amino acid sequence among molecules of the same class, subclass, or type. *734*

constitutive mutant A strain that produces an inducible enzyme continually, regardless of need, because of a mutation in either the operator or regulator gene. *276*

constructed wetlands Intentional creation of marshland plant communities and their associated microorganisms for environmental restoration or to purify water by the removal of bacteria, organic matter, and chemicals as the water passes through the aquatic plant communities. *662*

consumer An organism that feeds directly on living or dead animals, by ingestion or by phagocytosis. *622*

contact transmission Transmission of the pathogen by contact of the source or reservoir of the pathogen with the host. *856*

continuous culture system A culture system with constant environmental conditions maintained through continual provision of nutrients and removal of wastes. *120*

contractile vacuole (vak′u-ōl) In protists and some animals, a clear fluid-filled cell vacuole that takes up water from within the cell and then contracts, releasing it to the outside through a pore in a cyclical manner. Contractile vacuoles function primarily in osmoregulation and excretion. *585*

convalescent carrier (kon″vah-les′ent) An individual who has recovered from an infectious disease but continues to harbor large numbers of the pathogen. *854*

copiotrophic Having a high nutrient level. *638*

corepressor (ko″re-pre′sor) A small molecule that inhibits the synthesis of a repressible enzyme. *276*

cortex (kor′teks) The layer of a bacterial endospore that is thought to be particularly important in conferring heat resistance on the endospore. *69*

coryza (kŏ-ri′zah) *See* common cold. *884*

cosmid (koz′mid) A plasmid vector with lambda phage cos sites that can be packaged in a phage capsid; it is useful for cloning large DNA fragments. *335*

cristae (kris′te) Infoldings of the inner mitochondrial membrane. *83*

cross-feeding *See* syntropism. *604*

crossing-over A process in which segments of two adjacent DNA strands are exchanged; breaks occur in both strands, and the exposed ends of each strand join to those of the opposite segment on the other strand. *292*

cryptins Peptides produced by Paneth cells in the intestines. Cryptins are toxic for some bacteria, although their mode of action is not known. *711*

cryptococcosis (krip″to-kok-o′sis) An infection caused by the basidiomycete, *Cryptococcus neoformans,* which may involve the skin, lungs, brain, or meninges. *561, 947*

cryptosporidiosis (krip″to-spo-rid″e-o′sis) Infection with protozoa of the genus *Cryptosporidium.* The most common symptoms are prolonged diarrhea, weight loss, fever, and abdominal pain. *952*

crystallizable fragment (Fc) The stem of the Y portion of an antibody molecule. Cells such as macrophages bind to the Fc region, and it also is involved in complement activation. *734*

cutaneous anthrax (ku-ta′ne-us an′thraks) A form of anthrax involving the skin. *913*

cutaneous diphtheria (ku-ta′ne-us dif-the′re-ah) A skin disease caused by *Corynebacterium diphtheriae* that infects wound or skin lesions, causing a slow-healing ulceration. *901*

cyanobacteria (si″ah-no-bak-te′re-ah) A large group of bacteria that carry out oxygenic photosynthesis using a system like that present in photosynthetic eucaryotes. *471*

cyclic photophosphorylation (fo″to-fos″for-ĭ-la′shun) The formation of ATP when light energy is used to move electrons cyclically through an electron transport chain during photosynthesis; only photosystem I participates. *198*

cyst (sist) A general term used for a specialized microbial cell enclosed in a wall. Cysts are formed by protozoa and a few bacteria. They may be dormant, resistant structures formed in response to adverse conditions or reproductive cysts that are a normal stage in the life cycle. *586*

cytochromes (si′to-krōms) Heme proteins that carry electrons, usually as members of electron transport chains. *159*

cytokine (si′to-kīn) A general term for nonantibody proteins, released by a cell in response to inducing stimuli, which are mediators that influence other cells. Are produced by lymphocytes, monocytes, macrophages, and other cells. *720*

cytomegalovirus inclusion disease (si″to-meg″ah-lo-vi′rus) An infection caused by the cytomegalovirus and marked by nuclear inclusion bodies in enlarged infected cells. *885*

cytopathic effect (si″to-path′ik) The observable change that occurs in cells as a result of viral replication. Examples include ballooning, binding together, clustering, or even death of the cultured cells. *364, 832*

cytoplasmic matrix (si″to-plaz′mik) The protoplasm of a cell that lies within the plasma membrane and outside any other organelles. In bacteria it is the substance between the cell membrane and the nucleoid. *49, 76*

cytoproct (si′to-prokt) Site on a protozoan where undigestible matter is expelled. *592*

cytosine (si′to-sēn) A pyrimidine 2-oxy-4-aminopyrimidine found in nucleosides, nucleotides, and nucleic acids. *217*

cytoskeleton (si″to-skel′ĕ-ton) A network of microfilaments, microtubules, intermediate filaments, and other components in the cytoplasm of eucaryotic cells that helps give them shape. *79*

cytostome (si′to-stōm) A permanent site in the protozoan ciliate body through which food is ingested. *586*

cytotoxic T (T_C) cell (si″to-tok′sik) A cell that is capable of recognizing virus-infected cells through the major histocompatability molecules and developing into an activated cell that destroys the infected cells. *748*

cytotoxic T lymphocyte (CTL) The activated T cell that can attack and destroy virus-infected cells, tumor cells, and foreign cells. *748*

cytotoxin (si′to-tok′sin) A toxin or antibody that has a specific toxic action upon cells; cytotoxins are named according to the cell for which they are specific (e.g., nephrotoxin). *797*

D

Dane particle A 42 nm spherical particle that is one of three that are seen in hepatitis B virus infections. The Dane particle is the complete virion. *889*

dark-field microscopy Microscopy in which the specimen is brightly illuminated while the background is dark. *22*

dark reactivation The excision and replacement of thymine dimers in DNA that occurs in the absence of light. *130*

deamination (de-am″i-na′shun) The removal of amino groups from amino acids. *192*

death phase The decrease in viable microorganisms that occurs after the completion of growth in a batch culture. *115*

decimal reduction time (*D* or *D* value) The time required to kill 90% of the microorganisms or spores in a sample at a specified temperature. *140*

decomposer An organism that breaks down complex materials into simpler ones, including the release of simple inorganic products. Often a decomposer such as an insect or earthworm physically reduces the size of substrate particles. *622*

defensin (de-fens′sin) Specific peptides produced by neutrophils that permeabilize the outer and inner membranes of certain microorganisms, thus killing them. *720*

defined medium Culture medium made with components of known composition. *105*

Delta agent A defective RNA virus that is transmitted as an infectious agent, but cannot cause disease unless the individual is also infected with the hepatitis B virus. *See* hepatitis D. *891*

delta-proteobacteria One of the five subgroups of proteobacteria, each with distinctive 16S rRNA sequences. Chemoorganotrophic bacteria that usually are either predators on other bacteria or anaerobes that generate sulfide from sulfate and sulfite. *507*

denaturation (de-na″chur-a′shun) A change in the shape of an enzyme that destroys its activity; the term is also applied to changes in nucleic-acid shape. *163*

dendritic cell (den-drit′ ik) An antigen-presenting cell that has long membrane extensions resembling the dendrites of neurons. These cells are found in the lymph nodes, spleen, and thymus (interdigitating dendritic cells); skin (Langerhans cells); and other tissues (interstitial dendritic cells). Express MHC class II and B7 costimulatory molecules and thus are efficient presenters of antigens to T-helper cells. *708*

dendrogram A treelike diagram that is used to graphically summarize mutual similarities and relationships between organisms. *427*

denitrification (de-ni″trĭ-fĭ-ka′shən) The reduction of nitrate to gas products, primarily nitrogen gas, during anaerobic respiration. *190, 616*

dental plaque (plak) A thin film on the surface of teeth consisting of bacteria embedded in a matrix of bacterial polysaccharides, salivary glycoproteins, and other substances. *934*

deoxyribonucleic acid (DNA; de-ok″se-ri″bo-nu-kle′ik) The nucleic acid that constitutes the genetic material of all cellular organisms. It is a polynucleotide composed of deoxyribonucleotides connected by phosphodiester bonds. *54, 230*

dermatophyte (der′mah-to-fīt″) A fungus parasitic on the skin. *943*

dermatomycosis (der′ma-to-mi-ko′sis) A fungal infection of the skin; the term is a general term that comprises the various forms of tinea, and it is sometimes used to specifically refer to athlete's foot (tinea pedis). *943*

desensitization (de-sen″si-ti-za′shun) To make a sensitized or hypersensitive individual insensitive or nonreactive to a sensitizing agent. *769*

desert crust A crust formed by microbial binding of sand grains in the surface zone of desert soil; crust formation primarily involves cyanobacteria. *673*

detergent (de-ter′jent) An organic molecule, other than a soap, that serves as a wetting agent and emulsifier; it is normally used as cleanser, but some may be used as antimicrobial agents. *148*

deuteromycetes (doo″tər-o-mi-se′tēz) In some classification systems, the deuteromycetes or Fungi Imperfecti are a class of fungi. These organisms either lack a sexual stage or it has not yet been discovered. *564*

diatoms (di′ah-toms) Algal protists with siliceous cell walls called frustules. They constitute a substantial subfraction of the phytoplankton. *577*

diauxic growth (di-awk′sik) A biphasic growth pattern or response in which a microorganism, when exposed to two nutrients, initially uses one of them for growth and then alters its metabolism to make use of the second. *281*

differential interference contrast (DIC) microscope A light microscope that employs two beams of plane polarized light. The beams are combined after passing through the specimen and their intereference is used to create the image. *25*

differential media (dif″er-en′shal) Culture media that distinguish between groups of microorganisms based on differences in their growth and metabolic products. *106*

differential staining procedures Staining procedures that divide bacteria into separate groups based on staining properties. *28*

diffusely adhering *E. coli* (DAEC) DAEC strains of *E. coli* adhere over the entire surface of epithelial cells and usually cause diarrheal disease in immunologically naive and malnourished children. *932*

dikaryotic stage (di-kar-e-ot′ik) In fungi, having pairs of nuclei within cells or compartments. Each cell contains two separate haploid nuclei, one from each parent. *557*

dinoflagellate (di″no-flaj′e-lāt) An algal protist characterized by two flagella used in swimming in a spinning pattern. Many are bioluminescent and an important part of marine phytoplankton, some also are important marine pathogens. *579*

diphtheria (dif-the′re-ah) An acute, highly contagious childhood disease that generally affects the membranes of the throat and less frequently the nose. It is caused by *Corynebacterium diphtheriae.* *900*

dipicolinic acid A substance present at high concentrations in the bacterial endospore. It is thought to contribute to the endospore's heat resistance. *69*

diplococcus (dip″lo-kok′us) A pair of cocci. *43*

directed- or adaptive mutation A mutation that seems to be chosen so the organism can better adapt to its surroundings. *246*

disease (di-zez) A deviation or interruption of the normal structure or function of any part of the body that is manifested by a characteristic set of symptoms and signs. *848*

disease syndrome (sin′drŏm) A set of signs and symptoms that are characteristic of the disease. *850*

disinfectant (dis″in-fek′tant) An agent, usually chemical, that disinfects; normally, it is employed only with inanimate objects. *138*

disinfection (dis″in-fek′shun) The killing, inhibition, or removal of microorganisms that may cause disease. It usually refers to the treatment of inanimate objects with chemicals. *138*

disinfection by-products (DBPs) Chlorinated organic compounds formed during chlorine use for water disinfection. Many are carcinogens. *653*

dissimilatory nitrate reduction The process in which some bacteria use nitrate as the electron acceptor at the end of their electron transport chain to produce ATP. The nitrate is reduced to nitrite or nitrogen gas. *190*

dissimilatory reduction The use of a substance as an electron acceptor in energy generation. The acceptor (e.g., sulfate or nitrate) is reduced but not incorporated into organic matter during biosynthetic processes. *614*

diurnal oxygen shifts (di-er′nal) The changes in oxygen levels that occur in waters when algae produce and use oxygen on a cyclic basis during day and night. *650*

DNA ligase An enzyme that joins two DNA fragments together through the formation of a new phosphodiester bond. *239*

DNA microarrays (DNA chips) Solid supports that have DNA attached in highly organized arrays and are normally used to evaluate gene expression. *354, 1018*

DNA polymerase (pol-im′er-ās) An enzyme that synthesizes new DNA using a parental DNA strand as a template. *236*

DNA vaccine A vaccine that contains DNA which encodes antigenic proteins. It is injected directly into the muscle; the DNA is taken up by the muscle cells and encoded protein antigens are synthesized. This produces both humoral and cell-mediated responses. *767*

domains (do-mān′) 1. Compact, self-folding, structurally independent regions of proteins (usually around 100–300 amino acids in length); large proteins may have two or more domains connected by less structured stretches of polypeptide. In the antibody molecule, they are the loops, along with about 25 amino acids on each side, that form compact, globular sections. 2. The primary taxonomic groups above the kingdom level; all living organisms may be placed in one of three domains. *274, 424, 734*

double diffusion agar assay (Ouchterlony technique) An immunodiffusion reaction in which both antibody and antigen diffuse through agar to form stable immune complexes, which can be observed visually. *780*

doubling time *See* generation time. *115*

DPT (diphtheria-pertussis-tetanus) vaccine A vaccine containing three antigens that is used to immunize people against diphtheria, pertussis or whooping cough, and tetanus. *901*

droplet nuclei Small particles (1 to 4 µm in diameter) that represent what is left from the evaporation of larger particles (10 µm or more in diameter) called droplets. *856*

***D* value** *See* decimal reduction time. *140*

E

early mRNA Messenger RNA produced early in a virus infection that codes for proteins needed to take over the host cell and manufacture viral nucleic acids. *385*

Ebola virus hemorrhagic fever (a′bo-lə) An acute infection caused by a virus that produces varying degrees of hemorrhage, shock, and sometimes death. *877*

eclipse period (e-klips′) The initial part of the latent period in which infected host bacteria do not contain any complete virions. *383*

ecosystem (ek″o-sis′tem) A self-regulating biological community and its associated physical and chemical environment. *596*

ectomycorrhizal Referring to a mutualistic association between fungi and plant roots in which the fungus surrounds the root tip with a sheath. *681*

ectoparasite (ek″to-par′ah-sīt) A parasite that lives on the surface of its host. *788*

ectoplasm (ek′to-plazm) The outer stiffer portion or region of the cytoplasm in a protozoan, which may be differentiated in texture from the inner portion or endoplasm. *585*

ectosymbiosis A type of symbiosis in which one organism remains outside of the other organism. *701*

effacing lesion (le′zhən) The type of lesion caused by enteropathogenic strains of *E. coli* (EPEC) when the bacteria attach to and destroy the brush border of intestinal epithelial cells. The term AE (attaching-effacing) *E. coli* is now used to designate true EPEC strains that are an important cause of diarrhea in children from developing countries and in traveler's diarrhea. *932*

ehrlichiosis (ar-lik″e-o′sis) A tick-borne (*Dermacentor andersoni, Amblyomma americanum*) rickettsial disease caused by *Ehrlichia chaffeensis.* Once inside leukocytes, a nonspecific illness develops that resembles Rocky Mountain spotted fever. *909*

electron transport chain A series of electron carriers that operate together to transfer electrons from donors such as NADH and $FADH_2$ to acceptors such as oxygen. *184*

electrophoresis (e-lek″tro-fo-re′sis) A technique that separates substances through differences in their migration rate in an electrical field due to variations in the number and kinds of charged groups they have. *327*

electroporation (e-lek″tro-pə-ra′shən) The application of an electric field to create temporary pores in the plasma membrane in order to insert DNA into the cell and transform it. *335*

elementary body A small, dormant body that serves as the agent of transmission between host cells in the chlamydial life cycle. *477*

elongation cycle The cycle in protein synthesis that results in the addition of an amino acid to the growing end of a peptide chain. *270*

Embden-Meyerhof pathway (em′den mi′er-hof) A pathway that degrades glucose to pyruvate; the six-carbon stage converts glucose to fructose 1,6-bisphosphate, and the three-carbon stage produces ATP while changing glyceraldehyde 3-phosphate to pyruvate. *176, A–13*

encystation (en-sis-′ta′shen) The formation of a cyst. *586*

endemic disease (en-dem′ik) A disease that is commonly or constantly present in a population, usually at a relatively steady low frequency. *849*

endemic (murine) typhus (mu′rin ti′fus) A form of typhus fever caused by the rickettsia *Rickettsia typhi* that occurs sporadically in individuals who come into contact with rats and their fleas. *909*

endergonic reaction (end″er-gon′ik) A reaction that does not spontaneously go to completion as written; the standard free energy change is positive, and the equilibrium constant is less than one. *156*

endocytosis (en″do-si-to′sis) The process in which a cell takes up solutes or particles by enclosing them in vesicles pinched off from its plasma membrane. *80*

endogenote (en″do-je′nōt) The recipient bacterial cell's own genetic material into which the donor DNA can integrate. *294*

endogenous infection (en-doj′ĕ-nus in-fek′shun) An infection by a member of an individual's own normal body microbiota. *905*

endogenous pyrogen (en-doj′e-nus pi′ro-jen) A substance such as the lymphokine interleukin-1, which is produced by host cells and induces a fever response in the host. It also is called simply a pyrogen. *801*

endomycorrhizal Referring to a mutualistic association of fungi and plant roots in which the fungus penetrates into the root cells and arbuscules and vesicles are formed. *681*

endoparasite (en″do-par′ah-sīt) A parasite that lives inside the body of its host. *789*

endophyte (en′do-fīt) A microorganism living within a plant, but not necessarily parasitic on it. *679*

endoplasm (en′do-plazm) The central portion of the cytoplasm in a protozoan. *585*

endoplasmic reticulum (en″do-plas′mik rĕ-tik′u-lum) A system of membranous tubules and flattened sacs (cisternae) in the cytoplasmic matrix of eucaryotic cells. Rough or granular endoplasmic reticulum (RER or GER) bears ribosomes on its surface; smooth or agranular endoplasmic reticulum (SER or AER) lacks them. *79*

endosome (en′do-sōm) A membranous vesicle formed by endocytosis. *80*

endospore (en′do-spōr) An extremely heat- and chemical-resistant, dormant, thick-walled spore that develops within bacteria. *68*

endosymbiont (en″do-sim′be-ont) An organism that lives within the body of another organism in a symbiotic association. *596*

endosymbiosis (en″do-sim″bi-o′sis) A type of symbiosis in which one organism is found within another organism. *701*

endosymbiotic theory or **hypothesis** The theory that eucaryotic organelles such as mitochondria and chloroplasts arose when bacteria established an endosymbiotic relationship with the eucaryotic ancestor and then evolved into eucaryotic organelles. *85*, *424*

endotoxin (en″do-tox′sin) The heat-stable lipopolysaccharide in the outer membrane of the cell wall of gram-negative bacteria that is released when the bacterium lyses, or sometimes during growth, and is toxic to the host. *799*

end product inhibition *See* feedback inhibition. *169*

energy The capacity to do work or cause particular changes. *154*

enology The science of wine making. *982*

enteric bacteria (**enterobacteria;** en-ter′ik) Members of the family *Enterobacteriaceae* (gram-negative, peritrichous or nonmotile, facultatively anaerobic, straight rods with simple nutritional requirements); also used for bacteria that live in the intestinal tract. *505*

enterohemorrhagic *E. coli* (EHEC) (en′tər-o-hem″ə-raj′ik) EHEC strains of *E. coli* (0157:H7) produce several cytotoxins that provoke fluid secretion in traveler's diarrhea; however, their mode of action is unknown. *932*

enteroinvasive *E. coli* (EIEC) (en′tər-o-in-va′siv) EIEC strains of *E. coli* cause traveler's diarrhea by penetrating and binding to the intestinal epithelial cells. EIEC may also produce a cytotoxin and enterotoxin. *932*

enteropathogenic *E. coli* (EPEC) (en′tər-o-path-o-jen′ik) EPEC strains of *E. coli* attach to the brush border of intestinal epithelial cells and cause a specific type of cell damage called effacing lesions that lead to traveler's diarrhea. *932*

enterotoxigenic *E. coli* (ETEC) (en′tər-o-tok″sĭ-jen′ik) ETEC strains of *E. coli* produce two plasmid-encoded enterotoxins (which are responsible for traveler's diarrhea) and are distinguished by their heat stability: heat-stable enterotoxin (ST) and heat-labile enterotoxin (LT). *932*

enterotoxin (en″ter-o-tok′sin) A toxin specifically affecting the cells of the intestinal mucosa, causing vomiting and diarrhea. *797*, *927*

Entner-Doudoroff pathway A pathway that converts glucose to pyruvate and glyceraldehyde 3-phosphate by producing 6-phosphogluconate and then dehydrating it. *179, A–15*

entropy (en′tro-pe) A measure of the randomness or disorder of a system; a measure of that part of the total energy in a system that is unavailable for useful work. *156*

envelope (en′vĕ-lōp) 1. All the structures outside the plasma membrane in bacterial cells. 2. In virology it is an outer membranous layer that surrounds the nucleocapsid in some viruses. *55*, *369*

enzootic (en″zo-ot′ik) The moderate prevalence of a disease in a given animal population. *849*

enzyme (en′zīm) A protein catalyst with specificity for both the reaction catalyzed and its substrates. *161*

enzyme-linked immunosorbent assay (ELISA) A technique used for detecting and quantifying specific antibodies and antigens. *778*

eosinophil (e″ o-sin,′o-fil) A polymorphonuclear leukocyte that has a two-lobed nucleus and cytoplasmic granules that stain yellow-red. A mobile phagocyte that is highly antiparasitic. *707*

epidemic (ep″ĭ-dem′ik) A disease that suddenly increases in occurrence above the normal level in a given population. *849*

epidemic (louse-borne) typhus (ep″i-dem′ik ti′fus) A disease caused by *Rickettsia prowazekii* that is transmitted from person to person by the body louse. *909*

epidemiologist (ep″ĭ-de″me-ol′o-jist) A person who specializes in epidemiology. *849*

epidemiology (epi″-de″me-ol′o-je) The study of the factors determining and influencing the frequency and distribution of disease, injury, and other health-related events and their causes in defined human populations. *848*

episome (ep′ĭ-sōm) A plasmid that can exist either independently of the host cell's chromosome or be integrated into it. *294*

epitheca (ep″ĭ-the′kah) The larger of two halves of a diatom frustule (shell). *577*

epitope (ep′i-tōp) An area of the antigen molecule that stimulates the production of, and combines with, specific antibodies; also known as the antigenic determinant site. *731*

epizootic (ep″ĭ-zo-ot′ik) A sudden outbreak of a disease in an animal population. *849*

epizootiology (ep″i-zo-ot″e-ol′o-je) The field of science that deals with factors determining the frequency and distribution of a disease within an animal population. *849*

epsilon-proteobacteria One of the five subgroups of proteobacteria, each with distinctive 16S rRNA sequences. Slender gram-negative rods, some of which are medically important (*Campylobacter* and *Helicobacter*). *514*

equilibrium (e″kwĭ-lib′re-um) The state of a system in which no net change is occurring and free energy is at a minimum; in a chemical reaction at equilibrium, the rates in the forward and reverse directions exactly balance each other out. *156*

ergot (er′got) The dried sclerotium of *Claviceps purpurea.* Also, an ascomycete that parasitizes rye and other higher plants causing the disease called ergotism. *561*

ergotism (er′got-izm) The disease or toxic condition caused by eating grain infected with ergot; it is often accompanied by gangrene, psychotic delusions, nervous spasms, abortion, and convulsions in humans and in animals. *561, 967*

erysipelas (er″ĭ-sip′ĕ-las) An acute inflammation of the dermal layer of the skin, occurring primarily in infants and persons over 30 years of age with a history of streptococcal sore throat. *903*

erythema infectiosum (er″ə-the′-mə) A disease in children caused by the parvovirus B19. This disease is common in children between 4 and 11 years of age and is sometimes called fifth disease, since it was the fifth of six erythematous rash diseases in children in an older classification. *887*

erythromycin (ĕ-rith″ro-mi′sin) An intermediate spectrum macrolide antibiotic produced by *Streptomyces erythreus.* *817*

eschar (es′kar) A slough produced on the skin by a thermal burn, gangrene, or the anthrax bacillus. *914*

Eucarya The domain that contains organisms composed of eucaryotic cells with primarily glycerol fatty acyl diesters in their membranes and eucaryotic rRNA. *424*

eucaryotic cells (u″kar-e-ot′ik) Cells that have a membrane-delimited nucleus and differ in many other ways from procaryotic cells; protists, algae, fungi, plants, and animals are all eucaryotic. *11, 91*

euglenoids (u-gle′noids) A group of algae (the division *Euglenophyta*) or protozoa (order *Euglenida*) that normally have chloroplasts containing chlorophyll *a* and *b*. They usually have a stigma and one or two flagella emerging from an anterior gullet. *576*

Eumycota (u″mi-ko′tə) A division of fungi in some classification systems. These are the true fungi consisting of the Zygomycetes, Ascomycetes, Basidiomycetes, and Chytridiomycetes. *553*

eumycotic mycetoma (mi″se-to′mah) *See* maduromycosis. *945*

eutrophic (u-trof′ik) A nutrient-enriched environment. *648*

eutrophication (u″tro-fĭ-ka′shun) The enrichment of an aquatic environment with nutrients. *648*

evolutionary distance A quantitative indication of the number of positions that differ between two aligned macromolecules, and presumably a measure of evolutionary similarity between molecules and organisms. *433*

excystation (ek″sis-ta′shun) The escape of one or more cells or organisms from a cyst. *586*

exergonic reaction (ek″ser-gon′ik) A reaction that spontaneously goes to completion as written; the standard free energy change is negative, and the equilibrium constant is greater than one. *156*

exfoliative toxin (eks-fo′le-a″tiv) or **exfoliatin** (eks-fō″le-a′tin) An exotoxin produced by *Staphylococcus aureus* that causes the separation of epidermal layers and the loss of skin surface layers. It produces the symptoms of the scaled skin syndrome. *922*

exit site (E site) The location on a ribosome to which an empty tRNA moves from the P site before it finally leaves the ribosome during protein synthesis. *270*

exoenzymes (ek″so-en′zīms) Enzymes that are secreted by cells. *55*

exogenote (eks″o-je′nōt) The piece of donor DNA that enters a bacterial cell during gene exchange and recombination. *294*

exon (eks′on) The region in a split or interrupted gene that codes for RNA which ends up in the final product (e.g., mRNA). *263*

exotoxin (ek″so-tok′sin) A heat-labile, toxic protein produced by a bacterium as a result of its normal metabolism or because of the acquisition of a plasmid or prophage that redirects its metabolism. It is usually released into the bacterium's surroundings. *794*

exponential phase (eks″po-nen′shul) The phase of the growth curve during which the microbial population is growing at a constant and maximum rate, dividing and doubling at regular intervals. *114*

expressed sequence tag (EST) A partial gene sequence unique to a gene that can be used to identify and position the gene during genomic analysis. *354*

expression vector A special cloning vector used to express recombinant genes in host cells; the recombinant gene is transcribed and its protein synthesized. *336*

exteins Polypeptide sequences of precursor self-splicing proteins that are joined together during formation of the final, functional protein. They are separated from one another by intein sequences, which they flank. *275*

extracutaneous sporotrichosis (spo″ro-tri-ko′sis) An infection by the fungus *Sporothrix schenckii* that spreads throughout the body. *945*

extreme barophilic bacteria Bacteria that require a high-pressure environment to function. *624*

extreme environment An environment in which physical factors such as temperature, pH, salinity, and pressure are outside of the normal range for growth of most microorganisms; these conditions allow unique organisms to survive and function. *624*

extremophiles Microorganisms that grow under harsh or extreme environmental conditions such as very high temperatures or low pHs. *121, 624*

extrinsic factor An environmental factor such as temperature that influences microbial growth in food. *964*

F

facilitated diffusion Diffusion across the plasma membrane that is aided by a carrier. *100*

facultative anaerobes (fak′ul-ta″tiv an-a′er-ōbs) Microorganisms that do not require oxygen for growth, but do grow better in its presence. *127*

facultative psychrophile (fak′ul-ta″tiv si′kro-fīl) *See* psychrotroph. *126*

fas gene The gene that is active in target cells which are susceptible to killing by cells expressing the Fas ligand, a member of the TNF family of cytokines and cell surface molecules. *750*

fatty acid synthetase (sin′thĕ-tās) The multienzyme complex that makes fatty acids; the product usually is palmitic acid. *218*

fecal coliform (fe′kal ko′lĭ-form) Coliforms whose normal habitat is the intestinal tract and that can grow at 44.5°C. *654*

fecal enterococci (fe′kal en″ter-o-kok′si) Enterococci found in the intestine of humans and other warm-blooded animals. They are used as indicators of the fecal pollution of water. *656*

feedback inhibition A negative feedback mechanism in which a pathway end product inhibits the activity of an enzyme in the sequence leading to its formation; when the end product accumulates in excess, it inhibits its own synthesis. *169*

fermentation (fer″men-ta′shun) An energy-yielding process in which an energy substrate is oxidized without an exogenous electron acceptor. Usually organic molecules serve as both electron donors and acceptors. *173, 1000*

fever A complex physiological response to disease mediated by pyrogenic cytokines and characterized by a rise in core body temperature and activation of the immune system. *722*

fever blister *See* cold sore. *884*

F factor The fertility factor, a plasmid that carries the genes for bacterial conjugation and makes its *E. coli* host cell the gene donor during conjugation. *295*

fimbria (fim′bre-ah; pl., **fimbriae**) A fine, hairlike protein appendage on some gram-negative bacteria that helps attach them to surfaces. *62*

final host The host on/in which a parasitic organism either attains sexual maturity or reproduces. *789*

first law of thermodynamics Energy can be neither created nor destroyed (even though it can be changed in form or redistributed). *155*

fixation (fik-sa′shun) The process in which the internal and external structures of cells and organisms are preserved and fixed in position. *27*

flagellin (flaj′ĕ-lin) The protein used to construct the filament of a bacterial flagellum. *64*

flagellum (flah-jel′um; pl., **flagella**) A thin, threadlike appendage on many procaryotic and eucaryotic cells that is responsible for their motility. *63, 89*

flat or plane warts Small, smooth, slightly raised warts. *894*

flavin adenine dinucleotide (FAD; fla′vin ad′ĕ-nēn) An electron carrying cofactor often involved in energy production (for example, in the tricarboxylic acid cycle and the β-oxidation pathway). *159*

fluid mosaic model The currently accepted model of cell membranes in which the membrane is a lipid bilayer with integral proteins buried in the lipid, and peripheral proteins more loosely attached to the membrane surface. *47*

fluorescence microscope A microscope that exposes a specimen to light of a specific wavelength and then forms an image from the fluorescent light produced. Usually the specimen is stained with a fluorescent dye or fluorochrome. *25*

fluorescent light (floo″o-res′ent) The light emitted by a substance when it is irradiated with light of a shorter wavelength. *25*

fomite (fo′mīt; pl., **fomites**) An object that is not in itself harmful but is able to harbor and transmit pathogenic organisms. Also called fomes. *792, 857*

food-borne infection Gastrointestinal illness caused by ingestion of microorganisms, followed by their growth within the host. Symptoms arise from tissue invasion and/or toxin production. *926, 973*

food chain The flow of energy and matter in living organisms through a producer-consumer sequence (*See also* food web). *584*

food intoxication Food poisoning caused by microbial toxins produced in a food prior to consumption. The presence of living bacteria is not required. *927, 975*

food poisoning A general term usually referring to a gastrointestinal disease caused by the ingestion of food contaminated by pathogens or their toxins. *926*

food web A network of many interlinked food chains, encompassing primary producers, consumers, decomposers, and detritivores. *584*

F_1 particle Particle on the inner mitochondrial membrane, which is the site of ATP synthesis by oxidative phosphorylation. *83, 187*

F′ plasmid An F plasmid that carries some bacterial genes and transmits them to recipient cells when the F′ cell carries out conjugation; the transfer of bacterial genes in this way is often called sexduction. *305*

fragmentation (frag″men-ta′shun) A type of asexual reproduction in which a thallus breaks into two or more parts, each of which forms a new thallus. *573*

frameshift mutations Mutations arising from the loss or gain of a base or DNA segment, leading to a change in the codon reading frame and thus a change in the amino acids incorporated into protein. *251*

free energy change The total energy change in a system that is available to do useful work as the system goes from its initial state to its final state at constant temperature and pressure. *156*

French polio *See* Guillain-Barré syndrome. *874*

fruiting body A specialized structure that holds sexually or asexually produced spores; found in fungi and in some bacteria. *512, 565*

frustule (frus′tūl) A silicified cell wall in the diatoms. *577*

fungicide (fun′jĭ-sīd) An agent that kills fungi. *138*

fungistatic (fun″jĭ-stat′ik) Inhibiting the growth and reproduction of fungi. *138*

fungus (fung′gus; pl., **fungi**) Achlorophyllous, heterotrophic, spore-bearing eucaryotes with absorptive nutrition; usually, they have a walled thallus. *553*

***F* value** The time in minutes at a specific temperature (usually 250°F) needed to kill a population of cells or spores. *140*

G

gametangium (gam-ĕ-tan′je-um; pl., **gametangia**) A structure that contains gametes or in which gametes are formed. *557*

gamma-proteobacteria One of the five subgroups of proteobacteria, each with distinctive 16S rRNA sequences. This is the largest subgroup and is very diverse physiologically; many important genera are facultatively anaerobic chemoorganotrophs. *498*

gas gangrene (gang′grēn) A type of gangrene that arises from dirty, lacerated wounds infected by anaerobic bacteria, especially species of *Clostridium.* As the bacteria grow, they release toxins and ferment carbohydrates to produce carbon dioxide and hydrogen gas. *915*

gastritis (gas-tri′tis) Inflammation of the stomach. *918*

gastroenteritis (gas″tro-en-ter-i′tis) An acute inflammation of the lining of the stomach and intestines, characterized by anorexia, nausea, diarrhea, abdominal pain, and weakness. It has various causes including food poisoning due to such organisms as *E. coli, S. aureus, Campylobacter* (campylobacteriosis), and *Salmonella* species; consumption of irritating food or drink; or psychological factors such as anger, stress, and fear. Also called enterogastritis. *929*

gas vacuole A gas-filled vacuole found in cyanobacteria and some other aquatic bacteria that provides flotation. It is composed of gas vesicles, which are made of protein. *51*

gene (jēn) A DNA segment or sequence that codes for a polypeptide, rRNA, or tRNA. *241*

gene gun A device that uses high-pressure gas or another propellant to shoot a spray of DNA-coated microprojectiles into cells and transform them. Sometimes it is called a biolistic device. *335*

generalized transduction The transfer of any part of a bacterial genome when the DNA fragment is packaged within a phage capsid by mistake. *308*

general recombination Recombination involving a reciprocal exchange of a pair of homologous DNA sequences; it can occur any place on the chromosome. *292*

generation time The time required for a microbial population to double in number. *115*

genetic engineering The deliberate modification of an organism's genetic information by directly changing its nucleic acid genome. *320*

genital herpes (her′pēz) A sexually transmitted disease caused by the herpes simplex virus type 2. *885*

genital ulcer disease *See* chancroid. *914*

genome (je′nōm) The full set of genes present in a cell or virus; all the genetic material in an organism; a haploid set of genes in a cell. *228*

genomics The study of the molecular organization of genomes, their information content, and the gene products they encode. *345*

genus (je′nəs) A well-defined group of one or more species that is clearly separate from other genera. *426*

geographic information system (GIS) A data management system that organizes and displays digital map data from remote sensing and aids in the analysis of relationships between mapped features. *850*

German measles *See* rubella. *875*

germicide (jer′mĭ-sīd) An agent that kills pathogens and many nonpathogens but not necessarily bacterial endospores. *138*

germination (jer″mĭ-na′shun) The stage following bacterial endospore activation in which the endospore breaks its dormant state. Germination is followed by outgrowth. *71*

Ghon complex (gon) The initial focus of parenchymal infection in primary pulmonary tuberculosis. *908*

giardiasis (je″ar-di′ah-sis) A common intestinal disease caused by the parasitic protozoan *Giardia lamblia.* *953*

gingivitis (jin-jĭ-vi′tis) Inflammation of the gingival tissue. *936*

gingivostomatitis (jin″jĭ-vo-sto″mə-ti′tis) Inflammation of the gingiva and other oral mucous membranes. *884*

gliding motility A type of motility in which a microbial cell glides along when in contact with a solid surface. *66, 482*

global regulatory systems Regulatory systems that simultaneously affect many genes and pathways. *281*

glomerulonephritis (glo-mer″u-lo-nĕ-fri′tis) An inflammatory disease of the renal glomeruli. *905*

glucans Polysaccharides composed of glucose units held together by glycosidic linkages. Some types of glucans have $\alpha(1 \rightarrow 3)$ and $\alpha(1 \rightarrow 6)$ linkages and bind bacterial cells together on teeth forming a plaque ecosystem. *936*

gluconeogenesis (gloo″ko-ne″o-jen′e-sis) The synthesis of glucose from noncarbohydrate precursors such as lactate and amino acids. *209*

glycocalyx (gli″ko-kal′iks) A network of polysaccharides extending from the surface of bacteria and other cells. *61*

glycogen (gli′ko-jen) A highly branched polysaccharide containing glucose, which is used to store carbon and energy. *49, A–6*

glycolysis (gli-kol′ĭ-sis) The anaerobic conversion of glucose to lactic acid by use of the Embden-Meyerhof pathway. *176*

glycolytic pathway (gli″ko-lit′ik) *See* Embden-Meyerhof pathway. *176, A–13*

glyoxylate cycle (gli-ok′sĭ-lat) A modified tricarboxylic acid cycle in which the decarboxylation reactions are bypassed by the enzymes isocitrate lyase and malate synthase; it is used to convert acetyl-CoA to succinate and other metabolites. *216*

gnotobiotic (no″to-bi-ot′ik) Animals that are germfree (microorganism free) or live in association with one or more known microorganisms. *698*

Golgi apparatus (gol′je) A membranous eucaryotic organelle composed of stacks of flattened sacs (cisternae), which is involved in packaging and modifying materials for secretion and many other processes. *80*

gonococci (gon′o-kok′si) Bacteria of the species *Neisseria gonorrhoeae*—the organism causing gonorrhea. *915*

gonorrhea (gon″o-re′ah) An acute infectious sexually transmitted disease of the mucous membranes of the genitourinary tract, eye, rectum, and throat. It is caused by *Neisseria gonorrhoeae.* *915*

graft-versus-host disease A disease that results when mature post-thymic T cells in donor grafts (e.g., bone marrow) recognize the host as foreign and attack it. *773*

Gram stain A differential staining procedure that divides bacteria into gram-positive and gram-negative groups based on their ability to retain crystal violet when decolorized with an organic solvent such as ethanol. *28*

grana (gra′nah) A stack of thylakoids in the chloroplast stroma. *85*

granuloma (gran″u-lo′mə) Term applied to nodular inflammatory lesions containing phagocytic cells. *714*

greenhouse gases Gases released from the earth's surface through chemical and biological processes that interact with the chemicals in the stratosphere to decrease the release of radiation from the earth. It is believed that this leads to global warming. *689*

griseofulvin (gris″e-o-ful′vin) An antibiotic from *Penicillium griseofulvum* given orally to treat chronic dermatophytic infections of skin and nails. *820*

group translocation A transport process in which a molecule is moved across a membrane by carrier proteins while being chemically altered at the same time. *103*

growth An increase in cellular constituents. *113*

growth factors Organic compounds that must be supplied in the diet for growth because they are essential cell components or precursors of such components and cannot be synthesized by the organism. *99*

guanine (gwan′in) A purine derivative, 2-amino-6-oxypurine, found in nucleosides, nucleotides, and nucleic acids. *217*

Guillain-Barré syndrome (ge-yan′bar-ra′) A relatively rare disease affecting the peripheral nervous system, especially the spinal nerves, but also the cranial nerves. The cause is unknown, but it most often occurs after an influenza infection or flu vaccination. Also called French Polio. *874*

gumma (gum′ah) A soft, gummy tumor occurring in tertiary syphilis. *924*

gut-associated lymphoid tissue (GALT) The defensive lymphoid tissue present in the intestines. *See* Peyer's patches. *710*

H

H-2 complex Term for the MHC in the mouse. *745*

halobacteria or **extreme halophiles** A group of archaea that have an absolute dependence on high NaCl concentrations for growth and will not survive at a concentration below about 1.5 M NaCl. *461*

halophile (hal′o-fīl) A microorganism that requires high levels of sodium chloride for growth. *123*

Hansen's disease *See* leprosy. *916*

hantavirus pulmonary syndrome The disease in humans caused by the pulmonary syndrome hantavirus. Deer mice shed the virus in their feces, humans inhale the virus and first develop ordinary flulike aches and pains. Within a few days the hantavirus causes lung damage and capillary leakage. After about a week the infected person enters a crisis phase and may die. *877*

hapten (hap′ten) A molecule not immunogenic by itself but that, when coupled to a macromolecular carrier, can elicit antibodies directed against itself. *731*

harborage transmission The mode of transmission in which an infectious organism does not undergo morphological or physiological changes within the vector. *858*

hay fever Allergic rhinitis; a type of atopic allergy involving the upper respiratory tract. *768*

health (helth) A state of optimal physical, mental, and social well-being, and not merely the absence of disease and infirmity. *848*

healthy carrier An individual who harbors a pathogen, but is not ill. *854*

heat-shock proteins Proteins produced when cells are exposed to high temperatures or other stressful conditions. They protect the cells from damage and often aid in the proper folding of proteins. *273*

helical (hel′ĭ-kal) In virology this refers to a virus with a helical capsid surrounding its nucleic acid. *369*

helicases Enzymes that use ATP energy to unwind DNA ahead of the replication fork. *236*

hemadsorption (hem″ad-sorp′shun) The adherence of red blood cells to the surface of something, such as another cell or a virus. *832*

hemagglutination (hem″ah-gloo″tĭ-na′shun) The agglutination of red blood cells by antibodies. *756*

hemagglutinin (hem″ah-gloo′tĭ-nin) The antibody responsible for a hemagglutination reaction. *756*

hemoflagellate (he″mo-flaj′ĕ-lāt) A flagellated protozoan parasite that is found in the bloodstream. *956*

hemolysin (he-mol′ĭ-sin) A substance that causes hemolysis (the lysis of red blood cells). At least some hemolysins are enzymes that destroy the phospholipids in erythrocyte plasma membranes. *797*

hemolysis (he-mol′ĭ-sis) The disruption of red blood cells and release of their hemoglobin. There are several types of hemolysis when bacteria such as streptococci and staphylococci grow on blood agar. In α-hemolysis, a narrow greenish zone of incomplete hemolysis forms around the colony. A clear zone of complete hemolysis without any obvious color change is formed during β-hemolysis. *531, 797*

hemolytic uremic syndrome A kidney disease characterized by blood in the urine and often by kidney failure. It is caused by enterohemorrhagic strains of *Escherichia coli* O157:H7 that produce a Shiga-like toxin, which attacks the kidneys. *932*

hemorrhagic fever A fever usually caused by a specific virus that may lead to hemorrhage, shock, and sometimes death. *877*

hepatitis (hep″ah-ti′tis) Any infection that results in inflammation of the liver. Also refers to liver inflammation as such. *889*

hepatitis A (formerly infectious hepatitis; hep″ah-ti′tis) A type of hepatitis that is transmitted by fecal-oral contamination; it primarily affects children and young adults, especially in environments where there is poor sanitation and overcrowding. It is caused by the hepatitis A virus, a single-stranded RNA virus. *892*

hepatitis B (formerly serum hepatitis; hep″ah-ti′tis) This form of hepatitis is caused by a double-stranded DNA virus (HBV) formerly called the "Dane particle." The virus is transmitted by body fluids. *889*

hepatitis C About 90% of all cases of viral hepatitis can be traced to either HAV or HBV. The remaining 10% is believed to be caused by one and possibly several other types of viruses. At least one of these is hepatitis C (formerly non-A, non-B). *890*

hepatitis D (formerly delta hepatitis) The liver diseases caused by the hepatitis D virus in those individuals already infected with the hepatitis B virus. *891*

hepatitis E (formerly enteric-transmitted NANB hepatitis) The liver disease caused by the hepatitis E virus. Usually, a subclinical, acute infection results; however, there is a high mortality in women in their last trimester of pregnancy. *892*

herd immunity The resistance of a population to infection and spread of an infectious organism due to the immunity of a high percentage of the population. *851*

herpes labialis *See* cold sore. *884*

herpetic keratitis (her-pet′ik ker″ah-ti′tis) An inflammation of the cornea and conjunctiva of the eye resulting from a herpes simplex virus infection. *884*

heterocysts Specialized cells produced by cyanobacteria that are the sites of nitrogen fixation. *473*

heteroduplex DNA A double-stranded stretch of DNA formed by two slightly different strands that are not completely complementary. *292*

heterogeneous nuclear RNA (hnRNA) The RNA transcript of DNA made by RNA polymerase II; it is then processed to form mRNA. *263*

heterolactic fermenters (het″er-o-lak′tik) Microorganisms that ferment sugars to form lactate, and also other products such as ethanol and CO_2. *181*

heterotroph (het′er-o-trōf″) An organism that uses reduced, preformed organic molecules as its principal carbon source. *96*

heterotrophic nitrification Nitrification carried out by chemoheterotrophic microorganisms. *615*

hexon or **hexamer** A capsomer composed of six protomers. *370*

hexose monophosphate pathway (hek′sōs mon″o-fos′fāt) *See* pentose phosphate pathway. *177*

Hfr strain A bacterial strain that donates its genes with high frequency to a recipient cell during conjugation because the F factor is integrated into the bacterial chromosome. *303*

high-energy molecule A molecule whose hydrolysis under standard conditions makes available a large amount of free energy (the standard free energy change is more negative than about −7 kcal/mole); a high-energy molecule readily decomposes and transfers groups such as phosphate to acceptors. *157*

high oxygen diffusion environment A microbial environment in close contact with air and through which oxygen can move at a rapid rate (in comparison with the slow diffusion rate of oxygen through water). *635*

histone (his′tōn) A small basic protein with large amounts of lysine and arginine that is associated with eucaryotic DNA in chromatin. *234*

histoplasmosis (his″to-plaz-mo′sis) A systemic fungal infection caused by *Histoplasma capsulatum* var *capsulatum.* *947*

hives (hĭvz) An eruption of the skin. *769*

holdfast A structure produced by some bacteria and algae that attaches the cell to a solid object. *491*

holoenzyme A complete enzyme consisting of the apoenzyme plus a cofactor. *161*

holozoic nutrition (hol″o-zo′ik) In this type of nutrition, nutrients (such as bacteria) are acquired by phagocytosis and the subsequent formation of a food vacuole or phagosome. *586*

homolactic fermenters (ho″mo-lak′tik) Organisms that ferment sugars almost completely to lactic acid. *181*

horizontal gene transfer The process in which genes are transferred from one mature, independent organism to another. *292*

hormogonia Small motile fragments produced by fragmentation of filamentous cyanobacteria; used for asexual reproduction and dispersal. *473*

host (hōst) The body of an organism that harbors another organism. It can be viewed as a microenvironment that shelters and supports the growth and multiplication of a parasitic organism. *788*

host restriction The degradation of foreign genetic material by nucleases after the genetic material enters a host cell. *294*

human herpesvirus 6 (HHV-6, type A and B) (hər′pēz) HHV-6 was discovered in 1986 and was initially called the human B-lymphotropic virus. The virus was later shown to have a marked tropism for $CD4^+$ T cells and was renamed HHV-6. HHV-6 is genetically similar to cytomegalovirus. HHV-6 causes exanthem subitum (roseola infantum or sixth disease) in infants and has been suspected of involvement in many conditions, including opportunistic infections in immunocompromised patients, hepatitis, lymphoproliferative diseases, synergistic interactions with HIV, lymphadenitis, and chronic fatigue syndrome. *887*

human immunodeficiency virus (HIV) A lentivirus of the family *Retroviridae* that is associated with the onset of AIDS. *878*

human leukocyte antigen complex (HLA) An antigen on the surface of cells of human tissues and organs that is recognized by the immune system cells and therefore is important in graft rejection and regulation of the immune response. This is the same as MHC class II. *745*

humoral (antibody-mediated) immunity (hu′mor-al) The type of immunity that results from the presence of soluble antibodies in blood and lymph; also known as antibody-mediated immunity. *729*

hybridoma (hi″brĭ-do′mah) A fast-growing cell line produced by fusing a cancer cell (myeloma) to another cell, such as an antibody-producing cell. *743*

hydrogenosome (hi-dro-jen′osom) A microbodylike organelle that contains a unique electron transfer pathway in which hydrogenase transfers electrons to protons (which act as the terminal electron acceptors) and molecular hydrogen is formed. *585*

hydrophilic (hi″dro-fil′ik) A polar substance that has a strong affinity for water (or is readily soluble in water). *46*

hydrophobic (hi″dro-fo′bik) A nonpolar substance lacking affinity for water (or which is not readily soluble in water). *46*

hyperendemic disease (hi″per-en-dem′ik) A disease that has a gradual increase in occurrence beyond the endemic level, but not at the epidemic level, in a given population; also may refer to a disease that is equally endemic in all age groups. *849*

hypermutation A rapid production of multiple mutations in a gene or genes through the activation of special mutator genes. The process may be deliberately used to maximize the possibility of creating desirable mutants. *246*

hypersensitivity (hi″per-sen′si-tiv″i-te) A condition of increased immune sensitivity in which the body reacts to an antigen with an exaggerated immune response that usually harms the individual. Also termed an allergy. *768*

hyperthermophile (hi″per-ther′mo-fīl) A bacterium that has its growth optimum between 80°C and about 113°C. Hyperthermophiles usually do not grow well below 55°C. *126, 626*

hypha (hi′fah; pl., **hyphae**) The unit of structure of most fungi and some bacteria; a tubular filament. *556*

hypoferremia (hi″po-fĕ-re′me-ah) Deficiency of iron in the blood. *723*

hypotheca (hi-po-theca) The smaller half of a diatom frustule. *577*

hypothesis A tentative assumption or educated guess developed to explain a set of observations. *8*

hypoxic (hi pok′sik) Having a low oxygen level. *635*

I

icosahedral In virology this term refers to a virus with an icosahedral capsid, which has the shape of a regular polyhedron having 20 equilateral triangular faces and 12 corners. *369*

identification (i-den″tĭ-fĭ-ka′shun) The process of determining that a particular isolate or organism belongs to a recognized taxon. *422*

idiotype (id′e-o-tīp′) A set of one or more unique epitopes in the variable region of an immunoglobulin that distinguishes it from immunoglobulins produced by different plasma cells. *734*

IgA Immunoglobulin A; the class of immunoglobulins that is present in dimeric form in many body secretions (e.g., saliva, tears, and bronchial and intestinal secretions) and protects body surfaces. IgA also is present in serum. *736*

IgD Immunoglobulin D; the class of immunoglobulins found on the surface of many B lymphocytes; thought to serve as an antigen receptor in the stimulation of antibody synthesis. *738*

IgE Immunoglobulin E; the immunoglobulin class that binds to mast cells and basophils, and is responsible for type I or anaphylactic hypersensitivity reactions such as hay fever and asthma. IgE is also involved in resistance to helminth parasites. *738*

IgG Immunoglobulin G; the predominant immunoglobulin class in serum. Has functions such as neutralizing toxins, opsonizing bacteria, activating complement, and crossing the placenta to protect the fetus and neonate. *736*

IgM Immunoglobulin M; the class of serum antibody first produced during an infection. It is a large, pentameric molecule that is active in agglutinating pathogens and activating complement. The monomeric form is present on the surface of some B lymphocytes. *736*

immobilization (im-mo″bil-i-za′shun) The incorporation of a simple, soluble substance into the body of an organism, making it unavailable for use by other organisms. *613*

immune complex (ĭ-mūn′kom′pleks) The product of an antigen-antibody reaction, which may also contain components of the complement system. *756*

immune surveillance (ĭ-mūn′sur-vāl′ans) The still somewhat hypothetical process in which lymphocytes such as natural killer (NK) cells recognize and destroy tumor cells; other cells with abnormal surface antigens (e.g., virus-infected cells) also may be destroyed. *760*

immune system The defensive system in a host consisting of the nonspecific and specific immune responses. It is composed of widely distributed cells, tissues, and organs that recognize foreign substances and microorganisms and acts to neutralize or destroy them. *705*

immunity (ĭ-mu′nĭ-te) Refers to the overall general ability of a host to resist a particular disease; the condition of being immune. *705*

immunoblotting The electrophoretic transfer of proteins from polyacrylamide gels to nitrocellulose sheets to demonstrate the presence of specific proteins through reaction with labeled antibodies. *779*

immunodeficiency (im″u-no-dĕ-fish′en-se; pl., **immunodeficiencies**) The inability to produce a normal complement of antibodies or immunologically sensitized T cells in response to specific antigens. *774*

immunodiffusion A technique involving the diffusion of antigen and/or antibody within a semisolid gel to produce a precipitin reaction where they meet in proper proportions. Often both the antibody and antigen diffuse through the gel; sometimes an antigen diffuses through a gel containing antibody. *779*

immunoelectrophoresis (ĭ-mu″no-e-lek″tro-fo-re′sis; pl., **immunoelectrophoreses**) The electrophoretic separation of protein antigens followed by diffusion and precipitation in gels using antibodies against the separated proteins. *781*

immunofluorescence (im″u-no-floo″o-res′ens) A technique used to identify particular antigens microscopically in cells or tissues by the binding of a fluorescent antibody conjugate. *781*

immunoglobulin (Ig; im″u-no-glob′u-lin) *See* antibody. *734*

immunology (im″u-nol′o-je) The branch of science that deals with the immune system and attempts to understand the many phenomena that are responsible for both acquired and innate immunity. It also includes the use of antibody-antigen reactions in other laboratory work (serology and immunochemistry). *705*

immunopathology (im″u-no-pə-thol′o-je) The study of diseases or conditions resulting from immune reactions. *790*

immunoprecipitation (im″u-no-pre-sip″i-ta′shun) A reaction involving soluble antigens reacting with antibodies to form a large aggregate that precipitates out of solution. *781*

immunotoxin (im′u-no-tok″sin) A monoclonal antibody that has been attached to a specific toxin or toxic agent (antibody + toxin = immunotoxin) and can kill specific target cells. *744*

impetigo (im″pə-ti′go) This superficial cutaneous disease, most commonly seen in children, is characterized by crusty lesions, usually located on the face; the lesions typically have vesicles surrounded by a red border. It is the most frequently diagnosed skin infection caused by *S. pyogenes* (impetigo can also be caused by *S. aureus*). *903*

inclusion bodies Granules of organic or inorganic material lying in the cytoplasmic matrix of bacteria. *49, 410*

inclusion conjunctivitis (in-klu′zhun kon-junk″tĭ-vi′tis) An acute infectious disease that occurs throughout the world. It is caused by *Chlamydia trachomatis* that infects the eye and causes inflammation and the occurrence of large inclusion bodies. *916*

incubation period The period after pathogen entry into a host and before signs and symptoms appear. *850*

incubatory carrier An individual who is incubating a pathogen in large numbers but is not yet ill. *854*

index case The first disease case in an epidemic within a given population. *849*

indicator organism An organism whose presence indicates the condition of a substance or environment, for example, the potential presence of pathogens. Coliforms are used as indicators of fecal pollution. *654*

inducer (in-dūs′er) A small molecule that stimulates the synthesis of an inducible enzyme. *275*

inducible enzyme An enzyme whose level rises in the presence of a small molecule that stimulates its synthesis. *275*

industrial ecology The study of the ecology of industrial societies with a major focus on material cycling, energy flow, and the ecological impacts of such societies. *1022*

infantile paralysis (in′fan-til pah-ral′i-sis) *See* poliomyelitis. *892*

infection (in-fek′shun) The invasion of a host by a microorganism with subsequent establishment and multiplication of the agent. An infection may or may not lead to overt disease. *789*

infection thread A tubular structure formed during the infection of a root by nitrogen-fixing bacteria. The bacteria enter the root by way of the infection thread and stimulate the formation of the root nodule. *676*

infectious disease Any change from a state of health in which part or all of the host's body cannot carry on its normal functions because of the presence of an infectious agent or its products. *789*

infectious disease cycle (chain of infection) The chain or cycle of events that describes how an infectious organism grows, reproduces, and is disseminated. *852*

infectious dose 50 (ID_{50}) Refers to the dose or number of organisms that will infect 50% of an experimental group of hosts within a specified time period. *368, 790*

infectious mononucleosis (mono; mon″o-nu″kle-o′sis) An acute, self-limited infectious disease of the lymphatic system caused by the Epstein-Barr virus and characterized by fever, sore throat, lymph node and spleen swelling, and the proliferation of monocytes and abnormal lymphocytes. *888*

infectivity (in″fek-tiv′i-te) Infectiousness; the state or quality of being infectious or communicable. *790*

inflammation (in″flah-ma′shun) A localized protective response to tissue injury or destruction. Acute inflammation is characterized by pain, heat, swelling, and redness in the injured area. *712*

influenza or **flu** (in″flu-en′zah) An acute viral infection of the respiratory tract, occurring in isolated cases, epidemics, and pandemics. Influenza is caused by three strains of influenza virus, labeled types A, B, and C, based on the antigens of their protein coats. *872*

ingoldian fungi Aquatic hyphomycetes that often have a characteristic tetraradiate hyphal development form and which sporulate under water. Discovered by the British mycologist, C. T. Ingold, in the 1940s. *641*

initial body *See* reticulate body (RB). *477*

innate or **natural immunity** *See* nonspecific resistance. *705*

insertion sequence (in-ser′shun se′kwens) A simple transposon that contains genes only for those enzymes, such as the transposase, that are required for transposition. *298*

integration The incorporation of one DNA segment into a second DNA molecule to form a new hybrid DNA. Integration occurs during such processes as genetic recombination, episome incorporation into host DNA, and prophage insertion into the bacterial chromosome. *394*

integrins (in′tə-grin) A large and broadly distributed family of α/β heterodimers. Integrins are cellular adhesion receptors that mediate cell-cell and cell-substratum interactions. Integrins usually recognize linear amino acid sequences on protein ligands. *712*

inteins Internal intervening sequences of precursor self-splicing proteins that separate exteins and are removed during formation of the final protein. *275*

intercalating agents Molecules that can be inserted between the stacked bases of a DNA double helix, thereby distorting the DNA and inducing insertion and deletion mutations. *248*

interdigitating dendritic cell Special dendritic cells in the lymph nodes that function as potent antigen-presenting cells and develop from Langerhans cells. *710*

interferon (IFN; in″tər-fēr′on) A glycoprotein that has nonspecific antiviral activity by stimulating cells to produce antiviral proteins, which inhibit the synthesis of viral RNA and proteins. Interferons also regulate the growth, differentiation, and/or function of a variety of immune system cells. Their production may be stimulated by virus infections, intracellular pathogens (chlamydiae and rickettsias), protozoan parasites, endotoxins, and other agents. *721, 822*

interleukin (in″tər-loo′kin) A glycoprotein produced by macrophages and T cells that regulates growth and differentiation, particularly of lymphocytes. Interleukins promote cellular and humoral immune responses. *720*

intermediate filaments (in″ter-me′de-it fil′ah-ments) Small protein filaments, about 8 to 10 nm in diameter, in the cytoplasmic matrix of eucaryotic cells that are important in cell structure. *79*

intermediate host (in″ter-me′de-it hōst) The host that serves as a temporary but essential environment for development of a parasite and completion of its life cycle. *789*

interspecies hydrogen transfer The linkage of hydrogen production from organic matter by anaerobic heterotrophic microorganisms to the use of hydrogen by other anaerobes in the reduction of carbon dioxide to methane. This avoids possible hydrogen toxicity. *604*

intertriginous candidiasis A skin infection caused by *Candida* species. Involves those areas of the body, usually opposed skin surfaces, that are warm and moist (axillae, groin, skin folds). *950*

intoxication (in-tok″si-ka′shun) A disease that results from the entrance of a specific toxin into the body of a host. The toxin can induce the disease in the absence of the toxin-producing organism. *794*

intraepidermal lymphocytes T cells found in the epidermis of the skin that express the $\gamma\delta$ T-cell receptor. *710*

intranuclear inclusion body (in″trah-nu′kel-ar) A structure found within cells infected with the cytomegalovirus. *885*

intrinsic factors Food-related factors such as moisture, pH, and available nutrients that influence microbial growth. *964*

intron (in′tron) A noncoding intervening sequence in a split or interrupted gene, which codes for RNA that is missing from the final RNA product. *263*

invasiveness (in-va′siv-nes) The ability of a microorganism to enter a host, grow and reproduce within the host, and spread throughout its body. *790*

ionizing radiation Radiation of very short wavelength or high energy that causes atoms to lose electrons or ionize. *130, 144*

isotype (i′so-tīp) A variant form of an immunoglobulin (e.g., an immunoglobulin class, subclass, or type) that occurs in every normal individual of a particular species. Usually the characteristic antigenic determinant is in the constant region of H and L chains. *734*

J

Jaccard coefficient (S_J) An association coefficient used in numerical taxonomy; it is the proportion of characters that match, excluding those that both organisms lack. *427*

J chain A polypeptide present in polymeric IgM and IgA that links the subunits together. *736*

jock itch *See* tinea cruris. *944*

K

kelp (kelp) A common name for any of the larger members of the order *Laminariales* of the brown algae. *578*

keratinocyte Cell found in skin-associated lymphoid tissue; secretes cytokines that may induce an inflammatory response. *709*

keratitis (ker″ah-ti′tis) Inflammation of the cornea of the eye. *953*

kinetoplast (ki-ne′to-plast) A special structure in the mitochondrion of kinetoplastid protozoa. It contains the mitochondrial DNA. *588*

Kirby-Bauer method A disk diffusion test to determine the susceptibility of a microorganism to chemotherapeutic agents. *809*

Koch's postulates (koks pos′tu-lāts) A set of rules for proving that a microorganism causes a particular disease. *7*

Koplik's spots (kop′liks) Lesions of the oral cavity caused by the measles (rubeola) virus that are characterized by a bluish white speck in the center of each. *874*

Korean hemorrhagic fever An acute infection caused by a virus that produces varying degrees of hemorrhage, shock, and sometimes death. *877*

Krebs cycle *See* tricarboxylic acid (TCA) cycle. *183*

L

lactic acid fermentation (lak′tik) A fermentation that produces lactic acid as the sole or primary product. *179, A–19*

lager Pertaining to the process of aging beers to allow flavor development. *983*

lag phase A period following the introduction of microorganisms into fresh culture medium when there is no increase in cell numbers or mass during batch culture. *113*

laminarin (lam″i-na′rin) One of the principal storage products of the golden-brown algae; a polymer of glucose. *578*

Lancefield system (group; lans′feld) One of the serologically distinguishable groups (as group A, group B) into which streptococci can be divided. *532, 784*

land farming The addition of waste material, such as a hydrocarbon waste, to the soil surface so that it will be degraded. The soil may be moistened or mixed to stimulate the desired degradation process. *1011*

Langerhans cell Cell found in the skin that internalizes antigen and moves in the lymph to lymph nodes where it differentiates into a dendritic cell. *709*

late mRNA Messenger RNA produced later in a virus infection, which codes for proteins needed in capsid construction and virus release. *387*

latent period (la′tent) The initial phase in the one-step growth experiment in which no phages are released. *383*

latent virus infections Virus infections in which the virus quits reproducing and remains dormant for a period before becoming active again. *410*

leader sequence A nontranslated sequence at the 5′ end of mRNA that lies between the operator and the initiation codon; it aids in the initiation and regulation of transcription. *244, 261*

lectin complement pathway (lek′tin) The lectin pathway for complement activation is triggered by the binding of a serum lectin (mannan-binding lectin; MBL) to mannose-containing proteins or to carbohydrates on viruses or bacteria. *716*

legionellosis (le″jə-nel-o′sis) *See* Legionnaires' disease. *901*

Legionnaires' disease (legionellosis) A pulmonary form of legionellosis, resulting from infection with *Legionella pneumophila.* *901*

leishmanias (lēsh″ma′ne-ăs) Zooflagellates, members of the genus *Leishmania,* that cause the disease leishmaniasis. *956*

leishmaniasis (lēsh″mah-ni′ah-sis) The disease caused by the protozoa called leishmanias. *956*

lepromatous (progressive) leprosy (lep-ro′mah-tus lep′ro-se) A relentless, progressive form of leprosy in which large numbers of *Mycobacterium leprae* develop in skin cells, killing the skin cells and resulting in the loss of features. Disfiguring nodules form all over the body. *916*

leprosy (lep′ro-se) or **Hansen's disease** A severe disfiguring skin disease caused by *Mycobacterium leprae.* *916*

lethal dose 50 (LD_{50}) Refers to the dose or number of organisms that will kill 50% of an experimental group of hosts within a specified time period. *368, 790*

leukemia (loo-ke′me-ah) A progressive, malignant disease of blood-forming organs, marked by distorted proliferation and development of leukocytes and their precursors in the blood and bone marrow. Certain leukemias are caused by viruses (HTLV-1, HTLV-2). *887*

leukocidin (loo″ko-si′din) A microbial toxin that can damage or kill leukocytes. *797*

leukocyte (loo′ko-sīt) Any colorless white blood cell. Can be classified into granular and agranular lymphocytes. *705*

lichen (li′ken) An organism composed of a fungus and either green algae or cyanobacteria in a symbiotic association. *598*

Liebig's law of the minimum (le′bigz) Living organisms and populations will grow until lack of a resource begins to limit further growth. *131*

lipopolysaccharide (lip″o-pol″e-sak′ah-rīd) A molecule containing both lipid and polysaccharide, which is important in the outer membrane of the gram-negative cell wall. *58*

liposome (lip′o-som) A spherical particle formed by a lipid bilayer enclosing an aqueous solution. It may be used to administer chemotherapeutic agents or in diagnostic testing. *782*

listeriosis (lis-ter″e-o′sis) A sporadic disease of animals and humans, particularly those who are immunocompromised or pregnant, caused by the bacterium *Listeria monocytogenes.* *931*

lithotroph (lith′o-trōf) An organism that uses reduced inorganic compounds as its electron source. *97*

log phase *See* exponential phase. *114*

lophotrichous (lo-fot′rĭ-kus) A cell with a cluster of flagella at one or both ends. *63*

low oxygen diffusion environment An aquatic environment in which microorganisms are surrounded by deep water layers that limit oxygen diffusion to the cell surface. In contrast, microorganisms in thin water films have good oxygen transfer from air to the cell surface. *635*

LPS-binding protein A special plasma protein that binds bacterial lipopolysaccharides and then attaches to receptors on monocytes, macrophages, and other cells. This triggers the release of IL-1 and other cytokines that stimulate the development of fever and additional endotoxin effects. *801*

Lyme disease (LD, Lyme borreliosis; līm) A tick-borne disease caused by the spirochete *Borrella burgdorferi.* *910*

lymph node A small secondary lymphoid organ that contains lymphocytes, macrophages, and dendritic cells. It serves as a site for (1) filtration and removal of foreign antigens and (2) the activation and proliferation of lymphocytes. *709*

lymphocyte (lim′fo-sīt) A nonphagocytic, mononuclear leukocyte (white blood cell) that is an immunologically competent cell, or its precursor. Lymphocytes are present in the blood, lymph, and lymphoid tissues. *See* B cell and T cell. *705*

lymphogranuloma venereum (LGV; lim″fo-gran″u-lo′mah) A sexually transmitted disease caused by *Chlamydia trachomatis* serotypes L_1–L_3, which affect the lymph organs in the genital area. *917*

lymphokine (lim′fo-kin) A biologically active glycoprotein (e.g., IL-1) secreted by activated lymphocytes, especially sensitized T cells. It acts as an intercellular mediator of the immune response and transmits growth, differentiation, and behavioral signals. *720*

lysis (li′sis) The rupture or physical disintegration of a cell. *61*

lysogenic (li-so-jen′ik) *See* lysogens. *308, 390*

lysogens (li′so-jens) Bacteria that are carrying a viral prophage and have the potential of producing bacteriophages under the proper conditions. *308, 390*

lysogeny (li-soj′e-ne) The state in which a phage genome remains within the bacterial host cell after infection and reproduces along with it rather than taking control of the host and destroying it. *307, 390*

lysosome (li′so-sōm) A spherical membranous eucaryotic organelle that contains hydrolytic enzymes and is responsible for the intracellular digestion of substances. *80*

lysozyme (li′sō-zīm) An enzyme that degrades peptidoglycan by hydrolyzing the $\beta(1 \rightarrow 4)$ bond that joins *N*-acetylmuramic acid and *N*-acetylglucosamine. *61, 710*

lytic cycle (lit′ik) A virus life cycle that results in the lysis of the host cell. *383*

M

M cell Specialized cell of the intestinal mucosa and other sites, such as the urogenital tract, that delivers the antigen from the apical face of the cell to lymphocytes clustered within the pocket in its basolateral face. *710*

macrolide antibiotic (mak′ro-lĭd) An antibiotic containing a macrolide ring, a large lactone ring with multiple keto and hydroxyl groups, linked to one or more sugars. *817*

macromolecule (mak″ro-mol′ĕ-kūl) A large molecule that is a polymer of smaller units joined together. *205*

macromolecule vaccine A vaccine made of specific, purified macromolecules derived from pathogenic microorganisms. *767*

macronucleus (mak″ro-nu′kle-us) The larger of the two nuclei in ciliate protozoa. It is normally polyploid and directs the routine activities of the cell. *585*

macrophage (mak′ro-fāj) The name for a large mononuclear phagocytic cell, present in blood, lymph, and other tissues. Macrophages are derived from monocytes. They phagocytose and destroy pathogens; some macrophages also activate B cells and T cells. *705*

maduromycosis (mah-du′ro-mi-ko′sis) A subcutaneous fungal infection caused by *Madurella mycetoma;* also termed an eumycotic mycetoma. *945*

madurose The sugar derivative 3-O-methyl-D-galactose, which is characteristic of several actinomycete genera that are collectively called maduromycetes. *548*

magnetosomes Magnetite particles in magnetotactic bacteria that are tiny magnets and allow the bacteria to orient themselves in magnetic fields. *52*

maintenance energy The energy a cell requires simply to maintain itself or remain alive and functioning properly. It does not include the energy needed for either growth or reproduction. *121*

major histocompatibility complex (MHC) A large set of cell surface molecules in each individual, encoded by a family of genes, that serves as a unique biochemical marker of individual identity. It can trigger T-cell responses that may lead to rejection of transplanted tissues and organs. MHC molecules are also involved in the regulation of the immune response and the interactions between immune cells. *745*

malaria (mah-la′re-ah) A serious infectious illness caused by the parasitic protozoan *Plasmodium.* Malaria is characterized by bouts of high chills and fever that occur at regular intervals. *954*

malt (mawlt) Grain soaked in water to soften it, induce germination, and activate its enzymes. The malt is then used in brewing and distilling. *983*

Marburg viral hemorrhagic fever An acute infection caused by a virus that produces varying degrees of hemorrhage, shock, and sometimes death. *877*

mash The soluble materials released from germinated grains and prepared as a microbial growth medium. *983*

mashing The process in which cereals are mixed with water and incubated in order to degrade their complex carbohydrates (e.g., starch) to more readily usable forms such as simple sugars. *982*

mast cell A bone marrow–derived cell present in a variety of tissues that resembles peripheral blood-borne basophils and contains an Fc receptor for IgE. It undergoes IgE-mediated degranulation. *707*

mean growth rate constant (*k*) The rate of microbial population growth expressed in terms of the number of generations per unit time. *116*

measles (rubeola; me′zelz) A highly contagious skin disease that is endemic throughout the world. It is caused by a morbilli virus in the family *Paramyxoviridae,* which enters the body through the respiratory tract or through the conjunctiva. *873*

medical mycology (mi-kol′o-je) The discipline that deals with the fungi that cause human disease. *942*

meiosis (mi-o′sis) The sexual process in which a diploid cell divides and forms two haploid cells. *88*

melting temperature (T_m) The temperature at which double-stranded DNA separates into individual strands; it is dependent on the G + C content of the DNA and is used to compare genetic material in microbial taxonomy. *430*

membrane attack complex (MAC) The complex complement components (C5b–C9) that create a pore in the plasma membrane of a target cell and leads to cell lysis. C9 probably forms most of the actual pore. *716, 758*

membrane-disrupting exotoxin A type of exotoxin that lyses host cells by disrupting the integrity of the plasma membrane. *797*

membrane filter technique The use of a thin porous filter made from cellulose acetate or some other polymer to collect microorganisms from water, air, and food. *118, 654*

memory B cell A lymphocyte capable of initiating the antibody-mediated immune response upon detection of a specific antigen molecule for which it is genetically programmed. It circulates freely in the blood and lymph and may live for years. *741*

meningitis (men″in-ji′tis) A condition that refers to inflammation of the brain or spinal cord meninges (membranes). The disease can be divided into bacterial (septic) meningitis (caused by bacteria) and aseptic meningitis syndrome (caused by nonbacterial sources). *902*

mesophile (mes′o-fīl) A microorganism with a growth optimum around 20 to 45°C, a minimum of 15 to 20°C, and a maximum about 45°C or lower. *126*

messenger RNA (mRNA) Single-stranded RNA synthesized from a DNA template during transcription that binds to ribosomes and directs the synthesis of protein. *230*

metabolic channeling (mĕt″ah-bol′ik) The localization of metabolites and enzymes in different parts of a cell. *165*

metabolic control engineering Modification of the controls for biosynthetic pathways without altering the pathways themselves in order to improve process efficiency. *997*

metabolic pathway engineering (MPE) The use of molecular techniques to improve the efficiency of pathways that synthesize specific products. *997*

metabolism (me-tab′o-lizm) The total of all chemical reactions in the cell; almost all are enzyme catalyzed. *173*

metachromatic granules (met″ah-kro-mat′ik) Granules of polyphosphate in the cytoplasm of some bacteria that appear a different color when stained with a blue basic dye. They are storage reservoirs for phosphate. Sometimes called volutin granules. *52*

metastasis (mĕ-tas′tah-sis) The transfer of a disease like cancer from one organ to another not directly connected with it. *411*

methanogens (meth′ə-no-jens″) Strictly anaerobic archaeons that derive energy by converting CO_2, H_2, formate, acetate, and other compounds to either methane or methane and CO_2. *458*

methylotroph A bacterium that uses reduced one-carbon compounds such as methane and methanol as its sole source of carbon and energy. *491, 502*

Michaelis constant (K_m; mĭ-ka′lis) A kinetic constant for an enzyme reaction that equals the substrate concentration required for the enzyme to operate at half maximal velocity. *163*

microaerophile (mi″kro-a′er-o-fīl) A microorganism that requires low levels of oxygen for growth, around 2 to 10%, but is damaged by normal atmospheric oxygen levels. *127*

microarray technology Profiling of gene expression by measuring binding of RNA from growing cells to an array of function-specific oligonucleotides attached to an inert surface. *354, 1018*

microbial dietary adjuvant A substance added to the diet to stimulate specific microbial processes and populations. *986*

microbial ecology The study of microorganisms in their natural environments, with a major emphasis on physical conditions, processes, and interactions that occur on the scale of individual microbial cells. *596*

microbial transformation (mi-kro′be-al) *See* bioconversion. *1009*

microbial loop The mineralization of organic matter synthesized by photosynthetic phytoplankton through the activity of microorganisms such as bacteria and protozoa. This process “loops” minerals and carbon dioxide back for reuse by the primary producers and makes the organic matter unavailable to higher consumers. *608, 638*

microbial mat A firm structure of layered microorganisms with complementary physiological activities that can develop on surfaces in aquatic environments. *621*

microbiology (mi″kro-bi-ol′o-je) The study of organisms that are usually too small to be seen with the naked eye. Special techniques are required to isolate and grow them. *2*

microbivory The use of microorganisms as a food source by organisms that can ingest or phagocytose them. *672*

microenvironment (mi″kro-en-vi′ron-ment) The immediate environment surrounding a microbial cell or other structure, such as a root. *619*

microfilaments (mi″kro-fil′ah-ments) Protein filaments, about 4 to 7 nm in diameter, that are present in the cytoplasmic matrix of eucaryotic cells and play a role in cell structure and motion. *77*

micronucleus (mi″kro-nu′kle-us) The smaller of the two nuclei in ciliate protozoa. Micronuclei are diploid and involved only in genetic recombination and the regeneration of macronuclei. *585*

micronutrients Nutrients such as zinc, manganese, and copper that are required in very small quantities for growth and reproduction. Also called trace elements. *96*

microorganism (mi″kro-or′gan-izm) An organism that is too small to be seen clearly with the naked eye. *2*

microtubules (mi″kro-tu′buls) Small cylinders, about 25 nm in diameter, made of tubulin proteins and present in the cytoplasmic matrix and flagella of eucaryotic cells; they are involved in cell structure and movement. *78*

miliary tuberculosis (mil′e-a-re) An acute form of tuberculosis in which small tubercles are formed in a number of organs of the body because of dissemination of *M. tuberculosis* throughout the body by the bloodstream. Also known as reactivation tuberculosis. *908*

mineralization The release of inorganic nutrients from organic matter during microbial growth and metabolism. *504, 613*

minimal inhibitory concentration (MIC) The lowest concentration of a drug that will prevent the growth of a particular microorganism. *809*

minimal lethal concentration (MLC) The lowest concentration of a drug that will kill a particular microorganism. *809*

minus, or negative, strand The virus nucleic acid strand that is complementary in base sequence to the viral mRNA. *374*

missense mutation A single base substitution in DNA that changes a codon for one amino acid into a codon for another. *250*

mitochondrion (mi″to-kon′dre-on) The eucaryotic organelle that is the site of electron transport, oxidative phosphorylation, and pathways such as the Krebs cycle; it provides most of a nonphotosynthetic cell's energy under aerobic conditions. It is constructed of an outer membrane and an inner membrane, which contains the electron transport chain. *83*

mitosis (mi-to′sis) A process that takes place in the nucleus of a eucaryotic cell and results in the formation of two new nuclei, each with the same number of chromosomes as the parent. *87*

mixed acid fermentation A type of fermentation carried out by members of the family *Enterobacteriaceae* in which ethanol and a complex mixture of organic acids are produced. *181, A–17*

mixotrophic (mik″so-trof′ik) Refers to microorganisms that combine autotrophic and heterotrophic metabolic processes (they use inorganic electron sources and organic carbon sources). *98*

modified atmosphere packaging (MAP) Addition of gases such as nitrogen and carbon dioxide to packaged foods in order to inhibit the growth of spoilage organisms. *966*

mold Any of a large group of fungi that cause mold or moldiness and that exist as multicellular filamentous colonies; also the deposit or growth caused by such fungi. Molds typically do not produce macroscopic fruiting bodies. *556*

molecular chaperones Proteins that aid in the proper folding of unfolded polypeptides or partly denatured proteins and often also help transport proteins across membranes. *272*

molecular chronometers Nucleic acid and protein sequences that gradually change over time in a random fashion and at a steady rate, and which therefore can be used to determine phylogenetic relationships. *432*

monoclonal antibody (MAb; mon″o-klōn′al) An antibody of a single type that is produced by a population of genetically identical plasma cells (a clone); a monoclonal antibody is typically produced from a cell culture derived from the fusion product of a cancer cell and an antibody-producing cell (a hybridoma). *743*

monocyte (mon′ o-sīt) A mononuclear phagocytic leukocyte that circulates briefly in the bloodstream before migrating to the tissues where it becomes a macrophage. *705*

monocyte-macrophage system The collection of fixed phagocytic cells (including macrophages, monocytes, and specialized endothelial cells) located in the liver, spleen, lymph nodes, and bone marrow. This system is an important component of the host's general nonspecific defense against pathogens. *705*

monokine (mon′o-kīn) A generic term for a cytokine produced by mononuclear phagocytes (macrophages or monocytes). *720*

monotrichous (mon-ot′rĭ-kus) Having a single flagellum. *63*

morbidity rate (mor-bid′i-te) Measures the number of individuals who become ill as a result of a particular disease within a susceptible population during a specific time period. *849*

mordant (mor′dant) A substance that helps fix dye on or in a cell. *28*

mortality rate (mor-tal′i-te) The ratio of the number of deaths from a given disease to the total number of cases of the disease. *849*

most probable number (MPN) The statistical estimation of the probable population in a liquid by diluting and determining end points for microbial growth. *654*

mucociliary blanket The layer of cilia and mucus that lines certain portions of the respiratory system; it traps microorganisms up to 10 μm in diameter and then transports them by ciliary action away from the lungs. *711*

mucosal-associated lymphoid tissue (MALT) The defensive immune lymphoid tissue located in the intestinal mucosa. *710*

multi-drug-resistant strains of tuberculosis (MDR-TB) A multi-drug-resistant strain is defined as *Mycobacterium tuberculosis* resistant to isoniazid and rifampin, with or without resistance to other drugs. *908*

mumps An acute generalized disease that occurs primarily in school-age children and is caused by a paramyxovirus that is transmitted in saliva and respiratory droplets. The principal manifestation is swelling of the parotid salivary glands. *875*

murein *See* peptidoglycan. *55*

must The juices of fruits, including grapes, that can be fermented for the production of alcohol. *982*

mutagen (mu′tah-jen) A chemical or physical agent that causes mutations. *246*

mutation (mu-ta′shun) A permanent, heritable change in the genetic material. *244*

mutualism (mu′tu-al-izm″) A type of symbiosis in which both partners gain from the association and are unable to survive without it. The mutualist and the host are metabolically dependent on each other. *598*

mutualist (mu′tu-al-ist) An organism associated with another in a relationship that is beneficial to both (and often obligatory). *598*

mycelium (mi-se′le-um) A mass of branching hyphae found in fungi and some bacteria. *43, 556*

mycobiont The fungal partner in a lichen. *598*

mycolic acids Complex 60 to 90 carbon fatty acids with a hydroxyl on the β-carbon and an aliphatic chain on the α-carbon; found in the cell walls of mycobacteria. *543*

mycologist (mi-kol′o-jist) A person specializing in mycology; a student of mycology. *553*

mycology (mi-kol′o-je) The science and study of fungi. *553*

mycoplasma (mi″ko-plaz′mah) Bacteria that are members of the class *Mollicutes* and order *Mycoplasmatales;* they lack cell walls and cannot synthesize peptidoglycan precursors; most require sterols for growth; they are the smallest organisms capable of independent reproduction. *520*

mycoplasmal pneumonia (mi″ko-plaz′mal nu-mo′ne-ah) A type of pneumonia caused by *Mycoplasma pneumoniae.* Spread involves airborne droplets and close contact. *917*

mycorrhizosphere The region around a mycorrhizal fungus in which nutrients released from the fungus increase the microbial population and its activities. *681*

mycosis (mi-ko′sis; pl., **mycoses**) Any disease caused by a fungus. *553, 942*

mycotoxicology (mi-ko′tok″si-kol′o-je) The study of fungal toxins and their effects on various organisms. *553*

myeloma cell (mi″e-lo′mah) A tumor cell that is similar to the cell type found in bone marrow. Also, a malignant, neoplastic plasma cell that produces large quantities of antibodies and can be readily cultivated. *743*

myositis (mi″o-si′tis) Inflammation of a striated or voluntary muscle. *904*

myxamoeba (mik-sah-me′bah; pl., **myxamoebae**) A free-living amoeboid cell that can aggregate with other myxamoeba to form a plasmodium or pseudoplasmodium. Found in cellular slime molds and the myxomycetes. *565*

myxobacteria A group of gram-negative, aerobic soil bacteria characterized by gliding motility, a complex life cycle with the production of fruiting bodies, and the formation of myxospores. *512*

myxospores (mik′so-spōrs) Special dormant spores formed by the myxobacteria. *512*

N

napkin (diaper) candidiasis Typically found in infants whose diapers are not changed frequently and are therefore not kept dry. Caused by *Candida* species of fungi. *950*

narrow-spectrum drugs Chemotherapeutic agents that are effective only against a limited variety of microorganisms. *808*

natural attenuation The decrease in the level of an enviromental contaminant that results from natural chemical, physical, and biological processes. *1016*

natural classification A classification system that arranges organisms into groups whose members share many characteristics and reflect as much as possible the biological nature of organisms. *426*

natural killer (NK) cell A non-T, non-B lymphocyte present in nonimmunized individuals that exhibits MHC-independent cytolytic activity against tumor cells. *723, 760*

naturally acquired active immunity The type of active immunity that develops when an individual's immunologic system comes into contact with an appropriate antigenic stimulus during the course of normal activities; it usually arises as the result of recovering from an infection and lasts a long time. *729*

naturally acquired passive immunity The type of temporary immunity that involves the transfer of antibodies from one individual to another. *729*

necrotizing fasciitis (nek′ro-tīz″ing fas″e-i′tis) A disease that results from a severe invasive group A streptococcus infection. Necrotizing fasciitis is an infection of the subcutaneous soft tissues, particularly of fibrous tissue, and is most common on the extremities. It begins with skin reddening, swelling, pain, and cellulitis, and proceeds to skin breakdown and gangrene after 3 to 5 days. *904*

negative staining A staining procedure in which a dye is used to make the background dark while the specimen is unstained. *28*

Negri bodies (na′gre) Masses of viruses or unassembled viral subunits found within the brain neurons of rabies-infected animals. *888*

neurotoxin (nu″ro-tok′sin) A toxin that is poisonous to or destroys nerve tissue; especially the toxins secreted by *C. tetani, Corynebacterium diphtheriae,* and *Shigella dysenteriae.* *797*

neustonic (nu′ston″ik) The microorganisms that live at the atmospheric interface of a water body. *571*

neutrophil (noo′tro-fil) A mature white blood cell in the granulocyte lineage formed in bone marrow. It has a nucleus with three to five lobes and is very phagocytic. *123, 707*

neutrophile (nu′tro-fīl″) Microorganisms that grow best at a neutral pH range between pH 5.5 and 8.0. *123*

niche (nich) The function of an organism in a complex system, including place of the organism, the resources used in a given location, and the time of use. *619*

nicotinamide adenine dinucleotide (NAD$^+$; nik″o-tin′ah-mīd) An electron-carrying coenzyme; it is particularly important in catabolic processes and usually donates its electrons to the electron transport chain under aerobic conditions. *157*

nicotinamide adenine dinucleotide phosphate (NADP$^+$; nik″o-tin′ah-mīd) An electron-carrying coenzyme that most often participates as an electron carrier in biosynthetic metabolism. *158*

nitrification (ni″trĭ-fĭ-ka′shun) The oxidation of ammonia to nitrate. *193, 495, 615*

nitrifying bacteria (ni′trĭ-fi″ing) Chemolithotrophic, gram-negative bacteria that are members of the family *Nitrobacteriaceae* and convert ammonia to nitrate and nitrite to nitrate. *193, 493*

nitrogenase (ni′tro-jen-ās) The enzyme that catalyzes biological nitrogen fixation. *213*

nitrogen fixation The metabolic process in which atmospheric molecular nitrogen is reduced to ammonia; carried out by cyanobacteria, *Rhizobium,* and other nitrogen-fixing bacteria. *212, 616, 676*

nitrogen oxygen demand (*NOD*) The demand for oxygen in sewage treatment, caused by nitrifying microorganisms. *657*

nitrogen saturation point The point at which mineral nitrogen, when added to an ecosystem, can no longer be incorporated into organic matter through biological processes. *686*

nocardioforms Bacteria that resemble members of the genus *Nocardia;* they develop a substrate mycelium that readily breaks up into rods and coccoid elements (a quality sometimes called fugacity). *544*

nomenclature (no′men-kla″tūr) The branch of taxonomy concerned with the assignment of names to taxonomic groups in agreement with published rules. *422*

noncyclic photophosphorylation (fo″to-fos″for-i-la′shun) The process in which light energy is used to make ATP when electrons are moved from water to NADP$^+$ during photosynthesis; both photosystem I and photosystem II are involved. *198*

nongonococcal urethritis (NGU) (u″rə-thri′tis) Any inflammation of the urethra not caused by *Neisseria gonorrhoeae.* *918*

nonsense codon A codon that does not code for an amino acid but is a signal to terminate protein synthesis. *241, 270*

nonsense mutation A mutation that converts a sense codon to a nonsense or stop codon. *251*

nonspecific immune response (innate or natural immunity) *See* nonspecific resistance. *705*

nonspecific resistance Refers to those general defense mechanisms that are inherited as part of the innate structure and function of each animal; also known as nonspecific, innate or natural immunity. *705*

normal microbiota (also **indigenous microbial population, microflora, microbial flora;** mi″kro-bi-o′tah) The microorganisms normally associated with a particular tissue or structure. *699*

nosocomial infection (nos″o-ko′me-al) An infection that develops within a hospital (or other type of clinical care facility) and is produced by an infectious organism acquired during the stay of the patient. *866*

nuclear envelope (nu′kle-ar) The complex double-membrane structure forming the outer boundary of the eucaryotic nucleus. It is covered by pores through which substances enter and leave the nucleus. *86*

nucleic acid hybridization (nu-kle′ik) The process of forming a hybrid double-stranded DNA molecule using a heated mixture of single-stranded DNAs from two different sources; if the sequences are fairly complementary, stable hybrids will form. *431*

nucleocapsid (nu″kle-o-kap′sid) The nucleic acid and its surrounding protein coat or capsid; the basic unit of virion structure. *369*

nucleoid (nu′kle-oid) An irregularly shaped region in the procaryotic cell that contains its genetic material. *54*

nucleolus (nu-kle′o-lus) The organelle, located within the eucaryotic nucleus and not bounded by a membrane, that is the location of ribosomal RNA synthesis and the assembly of ribosomal subunits. *87*

nucleoside (nu′kle-o-sīd″) A combination of ribose or deoxyribose with a purine or pyrimidine base. *217*

nucleosome (nu′kle-o-sōm″) A complex of histones and DNA found in eucaryotic chromatin; the DNA is wrapped around the surface of the beadlike histone complex. *235*

nucleotide (nu′kle-o-tīd) A combination of ribose or deoxyribose with phosphate and a purine or pyrimidine base; a nucleoside plus one or more phosphates. *217*

nucleus (nu′kle-us) The eucaryotic organelle enclosed by a double-membrane envelope that contains the cell's chromosomes. *86*

numerical aperture The property of a microscope lens that determines how much light can enter and how great a resolution the lens can provide. *20*

numerical taxonomy The grouping by numerical methods of taxonomic units into taxa based on their character states. *426*

nutrient (nu′tre-ent) A substance that supports growth and reproduction. *96*

nystatin (nis′tah-tin) A polyene antibiotic from *Streptomyces noursei* that is used in the treatment of *Candida* infections of the skin, vagina, and alimentary tract. *820*

O

O antigen A polysaccharide antigen extending from the outer membrane of some gram-negative bacterial cell walls; it is part of the lipopolysaccharide. *58*

obligate aerobes Organisms that grow only in the presence of oxygen. *127*

obligate anaerobes Microorganisms that cannot tolerate the presence of oxygen and die when exposed to it. *127*

odontopathogens Dental pathogens. *933*

Okazaki fragments Short stretches of polynucleotides produced during discontinuous DNA replication. *239*

oligotrophic environment (ol″ĭ-go-trof′ik) An environment containing low levels of nutrients, particularly nutrients that support microbial growth. *131, 648*

oncogene (ong′ko-jēn) A gene whose activity is associated with the conversion of normal cells to cancer cells. *411*

one-step growth experiment An experiment used to study the reproduction of lytic phages in which one round of phage reproduction occurs and ends with the lysis of the host bacterial population. *383*

onychomycosis (on″i-ko-mi-ko′sis) A fungal infection of the nail plate producing nails that are opaque, white, thickened, friable, and brittle. Also called ringworm of the nails and tinea unguium. Caused by *Trichophyton* and other fungi such as *C. albicans.* *950*

oocyst (o′o-sist) Cyst formed around a zygote of malaria and related protozoa. *591*

oogonia (o″o-go′ne-a) Mitotically dividing female structures that produce primary oocytes and gametes. *574*

oomycetes (o″o-mi-se′tēz) A collective name for members of the division *Oomycota;* also known as the water molds. *565*

open reading frame (ORF) A reading frame sequence not interrupted by a stop codon; it is usually determined by nucleic acid sequencing studies. *347*

operator The segment of DNA to which the repressor protein binds; it controls the expression of the genes adjacent to it. *276*

operon (op′er-on) The sequence of bases in DNA that contains one or more structural genes together with the operator controlling their expression. *277*

ophthalmia neonatorum (of-thal′me-ah ne″o-nat-or-um) A gonorrheal eye infection in a newborn, which may lead to blindness. Also called conjunctivitis of the newborn. *916*

opportunistic microorganism or **pathogen** A microorganism that is usually free-living or a part of the host's normal microbiota, but which may become pathogenic under certain circumstances, such as when the immune system is compromised. *704, 789, 948*

opsonization (op″so-ni-za′shun) The action of opsonins in making bacteria and other cells more readily phagocytosed. Antibodies, complement (especially C3b), and fibronectin are potent opsonins. *718, 756*

optical tweezer The use of a focused laser beam to drag and isolate a specific microorganism from a complex microbial mixture. *627*

oral candidiasis *See* thrush. *949*

orchitis (or-ki′tis) Inflammation of the testes. *875*

organelle (or″gah-nel′) A structure within or on a cell that performs specific functions and is related to the cell in a way similar to that of an organ to the body. *76*

organotrophs Organisms that use reduced organic compounds as their electron source. *97*

ornithosis *See* psittacosis. *919*

osmophilic microorganisms (oz″mo-fil′ik) Microorganisms that grow best in or on media of high solute concentration. *965*

osmosis (oz-mo′sis) The movement of water across a selectively permeable membrane from a dilute solution (higher water concentration) to a more concentrated solution. *61*

osmotolerant Organisms that grow over a fairly wide range of water activity or solute concentration. *122*

Ouchterlony technique *See* double diffusion agar assay. *780*

outbreak The sudden, unexpected occurrence of a disease in a given population. *849*

outer membrane A special membrane located outside the peptidoglycan layer in the cell walls of gram-negative bacteria. *55*

oxidation-reduction (redox) reactions Reactions involving electron transfers; the reductant donates electrons to an oxidant. *157*

oxidative phosphorylation (fos″for-ĭ-la′shun) The synthesis of ATP from ADP using energy made available during electron transport. *184*

oxidizing agent or oxidant (ok′sĭ-dant) The electron acceptor in an oxidation-reduction reaction. *157*

oxygenic photosynthesis Photosynthesis that oxidizes water to form oxygen; the form of photosynthesis characteristic of eucaryotic algae and cyanobacteria. *199, 468*

P

pacemaker enzyme The enzyme in a metabolic pathway that catalyzes the slowest or rate-limiting reaction; if its rate changes, the pathway's activity changes. *169*

pandemic (pan-dem′ik) An increase in the occurrence of a disease within a large and geographically widespread population (often refers to a worldwide epidemic). *849*

Paneth cell (pah′ net) The granular cell located at the base of glands in the small intestine; it produces the enzyme lysozyme. *711*

pannus (pan′us) A superficial vascularization of the cornea with infiltration of granulation tissue. *926*

panzootic (pan″zo-ot′ik) The wide dissemination of a disease in an animal population. *849*

paralytic shellfish poisoning (par″ə-lit′ik) Dinoflagellates (*Gonyaulax* spp.) produce a powerful neurotoxin called saxitoxin. Shellfish accumulate saxitoxin and are poisonous when consumed by animals and humans. Saxitoxin paralyzes the striated respiratory muscles by inhibiting sodium transport. Paralytic shellfish poisoning is characterized by numbness of the mouth, lips, face, and extremities. *580*

parasite (par′ah-sīt) An organism that lives on or within another organism (the host) and benefits from the association while harming its host. Often the parasite obtains nutrients from the host. *788*

parasitism (par′ah-si″tizm) A type of symbiosis in which one organism adversely affects the other (the host), but cannot live without it. *609, 788*

parenteral route (pah-ren′ter-al) A route of drug administration that is nonoral (e.g., by injection). *812*

parfocal (par-fo′kal) A microscope that retains proper focus when the objectives are changed. *20*

paronychia (par″o-nik′e-ah) Inflammation involving the folds of tissue surrounding the nail; usually caused by *Candida albicans.* *950*

passive diffusion The process in which molecules move from a region of higher concentration to one of lower concentration as a result of random thermal agitation. *100*

passive immunization The induction of temporary immunity by the transfer of immune products, such as antibodies or sensitized T cells, from an immune vertebrate to a nonimmune one. *765*

Pasteur effect (pas-tur′) The decrease in the rate of sugar catabolism and change to aerobic respiration that occurs when microorganisms are switched from anaerobic to aerobic conditions. *189*

pasteurization (pas″ter-ĭ-za′shun) The process of heating milk and other liquids to destroy microorganisms that can cause spoilage or disease. *142, 970*

pathogen (path′o-jən) Any virus, bacterium, or other agent that causes disease. *698, 789*

pathogenicity (path″o-je-nis′ĭ-te) The condition or quality of being pathogenic, or the ability to cause disease. *698, 789*

pathogenicity island A large segment of DNA in some pathogens that contains the genes responsible for virulence; often it codes for the type III secretion system that allows the pathogen to secrete virulence proteins and damage host cells. A pathogen may have more than one pathogenicity island. *794*

pathogenic potential The degree that a pathogen causes morbid signs and symptoms. *790*

pathway architecture The analysis, design, and modification of biochemical pathways to increase process efficiency. *997*

pébrine (pa-brēn′) An infectious disease of silkworms caused by the protozoan *Nosema bombycis.* *591*

ped A natural soil aggregate, formed partly through bacterial and fungal growth in the soil. *670*

pellicle (pel′ĭ-k′l) A relatively rigid layer of proteinaceous elements just beneath the plasma membrane in many protozoa and algae. The plasma membrane is sometimes considered part of the pellicle. *89, 576, 585*

pelvic inflammatory disease (PID) A severe infection of the female reproductive organs. The disease that results when gonococci and chlamydiae infect the uterine tubes and surrounding tissue. *915, 918*

penicillins (pen″ĭ-sil-ins) A group of antibiotics containing a β-lactam ring, which are active against gram-positive bacteria. *61, 814*

penton or pentamer A capsomer composed of five protomers. *370*

pentose phosphate pathway (pen′tōs) The pathway that oxidizes glucose 6-phosphate to ribulose 5-phosphate and then converts it to a variety of three to seven carbon sugars; it forms several important products (NADPH for biosynthesis, pentoses, and other sugars) and also can be used to degrade glucose to CO_2. *177, A–14*

peplomer or spike (pep′lo-mer) A protein or protein complex that extends from the virus envelope and often is important in virion attachment to the host cell surface. *374*

peptic ulcer disease A gastritis caused by *Helicobacter pylori.* *918*

peptide interbridge (pep′tīd) A short peptide chain that connects the tetrapeptide chains in some peptidoglycans. *56*

peptidoglycan (pep″tĭ-do-gli′kan) A large polymer composed of long chains of alternating *N*-acetylglucosamine and *N*-acetylmuramic acid residues. The polysaccharide chains are linked to each other through connections between tetrapeptide chains attached to the *N*-acetylmuramic acids. It provides much of the strength and rigidity possessed by bacterial cell walls. *55, 521*

peptidyl or **donor site (P site)** The site on the ribosome that contains the peptidyl-tRNA at the beginning of the elongation cycle during protein synthesis. *270*

peptidyl transferase The enzyme that catalyzes the transpeptidation reaction in protein synthesis; in this reaction, an amino acid is added to the growing peptide chain. *270*

peptones (pep′tōns) Water-soluble digests or hydrolysates of proteins that are used in the preparation of culture media. *105*

perforin pathway The cytotoxic pathway that uses perforin protein, which polymerizes to form membrane pores that help destroy cells during cell-mediated cytotoxicity. Perforin is produced by cytotoxic T cells and NK cells and stored in granules that are released when a target cell is contacted. *750*

period of infectivity Refers to the time during which the source of an infectious disease is infectious or is disseminating the pathogen. *854*

periodontal disease (per″e-o-don′tal) A disease located around the teeth or in the periodontium—the tissue investing and supporting the teeth, including the cementum, periodontal ligament, alveolar bone, and gingiva. *936*

periodontitis (per″e-o-don-ti′tis) An inflammation of the periodontium. *936*

periodontium (per″e-o-don′she-um) *See* periodontal disease. *936*

periodontosis (per″e-o-don-to′sis) A degenerative, noninflammatory condition of the periodontium, which is characterized by destruction of tissue. *936*

periplasm (per′ĭ-plaz-əm) The substance that fills the periplasmic space. *55*

periplasmic flagella The flagella that lie under the outer sheath and extend from both ends of the spirochete cell to overlap in the middle and form the axial filament. Also called axial fibrils and endoflagella. *479*

periplasmic space (per″i-plas′mik) or **periplasm** (per′ĭ-plazm) The space between the plasma membrane and the outer membrane in gram-negative bacteria, and between the plasma membrane and the cell wall in gram-positive bacteria. *55*

peritrichous (pĕ-rit′rĭ-kus) A cell with flagella evenly distributed over its surface. *63*

permease (per′me-ās) A membrane-bound carrier protein or a system of two or more proteins that transports a substance across the membrane. *100*

pertussis (pər-tus′is) An acute, highly contagious infection of the respiratory tract, most frequently affecting young children, usually caused by *Bordetella pertussis* or *B. parapertussis.* Consists of peculiar paroxysms of coughing, ending in a prolonged crowing or whooping respiration; hence the name whooping cough. *903*

petri dish (pe′tre) A shallow dish consisting of two round, overlapping halves that is used to grow microorganisms on solid culture medium; the top is larger than the bottom of the dish to prevent contamination of the culture. *108*

phage (fāj) *See* bacteriophage. *364*

phagocytic vacuole (fag″o-sit′ik vak′u-ol) A membrane-delimited vacuole produced by cells carrying out phagocytosis. It is formed by the invagination of the plasma membrane and contains solid material. *585*

phagocytosis (fag″o-si-to′sis) The endocytotic process in which a cell encloses large particles in a membrane-delimited phagocytic vacuole or phagosome and engulfs them. *80, 718*

phagolysosome (fag″o-li′so-sōm) The vacuole that results from the fusion of a phagosome with a lysosome. *718*

phagovar (fag′o-var) A specific phage type. *842*

pharyngitis (far″in-ji′tis) Inflammation of the pharynx, often due to a *S. pyogenes* infection. *905*

phase-contrast microscope A microscope that converts slight differences in refractive index and cell density into easily observed differences in light intensity. *22*

phenetic system A classification system that groups organisms together based on the similarity of their observable charactistics. *426*

phenol coefficient test A test to measure the effectiveness of disinfectants by comparing their activity against test bacteria with that of phenol. *149*

phosphatase (fos′fah-tās″) An enzyme that catalyzes the hydrolytic removal of phosphate from molecules. *210*

phosphate group transfer potential A measure of the ability of a phosphorylated molecule such as ATP to transfer its phosphate to water and other acceptors. It is the negative of the $\Delta G^{o\prime}$ for the hydrolytic removal of phosphate. *157*

photoautotroph (fo′to-aw′to-trōf) *See* photolithotrophic autotrophs. *97*

photolithotrophic autotrophs Organisms that use light energy, an inorganic electron source (e.g., H_2O, H_2, H_2S), and CO_2 as a carbon source. *97*

photoorganotrophic heterotrophs Microorganisms that use light energy and organic electron donors, and also employ simple organic molecules rather than CO_2 as their carbon source. *98*

photoreactivation (fo″to-re-ak″tĭ-va′shun) The process in which blue light is used by a photoreactivating enzyme to repair thymine dimers in DNA by splitting them apart. *130, 254*

photosynthesis (fo″to-sin′thĕ-sis) The trapping of light energy and its conversion to chemical energy, which is then used to reduce CO_2 and incorporate it into organic form. *154, 195, 207*

photosystem I The photosystem in eucaryotic cells that absorbs longer wavelength light, usually greater than about 680 nm, and transfers the energy to chlorophyll P700 during photosynthesis; it is involved in both cyclic photophosphorylation and noncyclic photophosphorylation. *196*

photosystem II The photosystem in eucaryotic cells that absorbs shorter wavelength light, usually less than 680 nm, and transfers the energy to chlorophyll P680 during photosynthesis; it participates in noncyclic photophosphorylation. *196*

phototrophs Organisms that use light as their energy source. *97*

phycobiliproteins Photosynthetic pigments that are composed of proteins with attached tetrapyrroles; they are often found in cyanobacteria and red algae. *196*

phycobilisomes Special particles on the membranes of cyanobacteria that contain photosynthetic pigments and electron transport chains. *471*

phycobiont (fi″ko-bi′ont) The algal or cyanobacterial partner in a lichen. *599*

phycocyanin (fi″ko-si′an-in) A blue phycobiliprotein pigment used to trap light energy during photosynthesis. *196*

phycoerythrin (fi″ko-er′i-thrin) A red photosynthetic phycobiliprotein pigment used to trap light energy. *196*

phycology (fi-kol′o-je) The study of algae; algology. *571*

phyllosphere The surface of plant leaves. *674*

phylogenetic or **phyletic classification system** (fi″lo-jĕ-net′ik, fi-let′ik) A classification system based on evolutionary relationships rather than the general similarity of contemporary characteristics. *428*

phylogenetic tree A graph made of nodes and branches, much like a tree in shape, that shows phylogenetic relationships between groups of organisms and sometimes also indicates the evolutionary development of groups. *433*

phytoplankton (fi″to-plank′ton) A community of floating photosynthetic organisms, largely composed of algae and cyanobacteria. *571, 638*

phytoremediation The use of plants and their associated microorganisms to remove, contain, or degrade environmental contaminants. *1014*

piedra (pe-a′drah) A fungal disease of the hair in which white or black nodules of fungi form on the shafts. *943*

pinocytosis (pi″no-si-to′sis) The endocytotic process in which a cell encloses a small amount of the surrounding liquid and its solutes in tiny pinocytotic vesicles or pinosomes. *80*

pitched Pertaining to inoculation of a nutrient medium with yeast, for example, in beer brewing. *983*

plague (plāg) An acute febrile, infectious disease, caused by the bacillus *Yersinia pestis,* which has a high mortality rate; the two major types are bubonic plague and pneumonic plague. *911*

plankton (plank′ton) Free-floating, mostly microscopic microorganisms that can be found in almost all waters; a collective name. *571, 584*

planktonic (adj.) *See* plankton. *571*

plaque (plak) 1. A clear area in a lawn of bacteria or a localized area of cell destruction in a layer of animal cells that results from the lysis of the bacteria by bacteriophages or the destruction of the animal cells by animal viruses. 2. The term also refers to dental plaque, a film of food debris, polysaccharides, and dead cells that cover the teeth. It provides a medium for the growth of bacteria (which may be considered a part of the plaque), leading to a microbial plaque ecosystem that can produce dental decay. *364, 934*

plasma cell A mature, differentiated B lymphocyte chiefly occupied with antibody synthesis and secretion; a plasma cell lives for only 5 to 7 days. *709*

plasma membrane The selectively permeable membrane surrounding the cell's cytoplasm; also called the cell membrane, plasmalemma, or cytoplasmic membrane. *46*

plasmid (plaz′mid) A double-stranded DNA molecule that can exist and replicate independently of the chromosome or may be integrated with it. A plasmid is stably inherited, but is not required for the host cell's growth and reproduction. *54, 294, 819*

plasmid fingerprinting A technique used to identify microbial isolates as belonging to the same strain because they contain the same number of plasmids with the identical molecular weights and similar phenotypes. *843*

plasmodial (acellular) slime mold (plaz-mo′de-al) A member of the division *Myxomycota* that exists as a thin, streaming, multinucleate mass of protoplasm, which creeps along in an amoeboid fashion. *564*

plasmodium (plaz-mo′de-um; pl., **plasmodia**) A stage in the life cycle of myxomycetes (plasmodial slime molds); a multinucleate mass of protoplasm surrounded by a membrane. Also, a parasite of the genus *Plasmodium.* *565*

plasmolysis (plaz-mol′ĭ-sis) The process in which water osmotically leaves a cell, which causes the cytoplasm to shrivel up and pull the plasma membrane away from the cell wall. *61*

plastid (plas′tid) A cytoplasmic organelle of algae and higher plants that contains pigments such as chlorophyll, stores food reserves, and often carries out processes such as photosynthesis. *85*

pleomorphic (ple″o-mor′fik) Refers to bacteria that are variable in shape and lack a single, characteristic form. *44*

plus strand or **positive strand** The virus nucleic-acid strand that is equivalent in base sequence to the viral mRNA. *374*

pneumocystis pneumonia, ***Pneumocystis carinii*** **pneumonia (PCP)**; (noo″mo-sis-tis) A type of pneumonia caused by the protist *Pneumocystis carinii.* *950*

pneumonic plague *See* plague. *911*

point mutation A mutation that affects only a single base pair in a specific location. *249*

polar flagellum A flagellum located at one end of an elongated cell. *63*

poliomyelitis (po″le-o-mi″e-li′tis) An acute, contagious viral disease that attacks the central nervous system, injuring or destroying the nerve cells that control the muscles and sometimes causing paralysis; also called polio or infantile paralysis. *892*

poly-β-hydroxybutyrate (hi-drok″se-bu′tĭ-rāt) A linear polymer of β-hydroxybutyrate used as a reserve of carbon and energy by many bacteria. *49*

polymerase chain reaction (PCR) An in vitro technique used to synthesize large quantities of specific nucleotide sequences from small amounts of DNA. It employs oligonucleotide primers complementary to specific sequences in the target gene and special heat-stable DNA polymerases. *326*

polymorphonuclear leukocyte (PMN) (pol″e-mor″fo-noo′kle-ər) A leukocyte that has a variety of nuclear forms. *707*

polyphasic taxonomy An approach in which taxonomic schemes are developed using a wide range of phenotypic and genotypic information. *435*

polyribosome (pol″e-ri′bo-sōm) A complex of several ribosomes with a messenger RNA; each ribosome is translating the same message. *83, 266*

Pontiac fever A bacterial disease caused by *Legionella pneumophila* that resembles an allergic disease more than an infection. First described from Pontiac, Michigan. *See* Legionnaires' disease. *902*

population An assemblage of organisms of the same type. *596*

porin proteins Proteins that form channels across the outer membrane of gram-negative bacterial cell walls. Small molecules are transported through these channels. *60*

postherpetic neuralgia The severe pain after a herpes infection. *872*

posttranscriptional modification The processing of the initial RNA transcript, heterogeneous nuclear RNA, to form mRNA. *263*

potable (po′tah-b′l) Refers to water suitable for drinking. *654*

pour plate A petri dish of solid culture medium with isolated microbial colonies growing both on its surface and within the medium, which has been prepared by mixing microorganisms with cooled, still liquid medium and then allowing the medium to harden. *107*

precipitation (or **precipitin) reaction** (pre-sip″ĭ-ta′shun) The reaction of an antibody with a soluble antigen to form an insoluble precipitate. *756*

precipitin (pre-sip′ĭ-tin) The antibody responsible for a precipitation reaction. *756*

prevalence rate Refers to the total number of individuals infected at any one time in a given population regardless of when the disease began. *849*

Pribnow box A special base sequence in the promoter that is recognized by the RNA polymerase and is the site of initial polymerase binding. *244, 262*

primary amebic meningoencephalitis An infection of the meninges of the brain by the free-living amoebae *Naegleria* or *Acanthamoeba.* *953*

primary (frank) pathogen Any organism that causes a disease in the host by direct interaction with or infection of the host. *789*

primary metabolites Microbial metabolites produced during the growth phase of an organism. *1002*

primary producer Photoautotrophic and chemoautotrophic organisms that incorporate carbon dioxide into organic carbon and thus provide new biomass for the ecosystem. *622*

primary production The incorporation of carbon dioxide into organic matter by photosynthetic organisms and chemoautotrophic bacteria. *622*

primary treatment The first step of sewage treatment, in which physical settling and screening are used to remove particulate materials. *658*

prion (pri′on) An infectious particle that is the cause of slow diseases like scrapie in sheep and goats; it has a protein component, but no nucleic acid has yet been detected. *416*

probe (prōb) A short, labeled nucleic acid segment complementary in base sequence to part of another nucleic acid, which is used to identify or isolate the particular nucleic acid from a mixture through its ability to bind specifically with the target nucleic acid. *322, 976*

probiotic (1) The oral administration of either living microorganisms or substances to promote the health and growth of an animal or human. (2) A living organism that may provide health benefits beyond its nutritional value when it is ingested. *703, 986*

procaryotic cells (pro″kar-e-ot′ik) Cells that lack a true, membrane-enclosed nucleus; bacteria are procaryotic and have their genetic material located in a nucleoid. *11, 91*

procaryotic species A collection of strains that share many stable properties and differ significantly from other groups of strains. *425*

prodromal stage (pro-dro′məl) The period during the course of a disease in which there is the appearance of signs and symptoms, but they are not yet distinctive and characteristic enough to make an accurate diagnosis. *850*

progametangium (pro-gam-ĕ″tan′je-um; pl., **progametangia**) The cell that gives rise to a gametangium and a proximal suspensor during the early stages of sexual reproduction in zygomycetous fungi. *560*

proliferative kidney disease (pro-lif′er-a-tiv) A protozoan disease caused by an unclassified myxozoan in salmonids throughout the world. *591*

promoter The region on DNA at the start of a gene that the RNA polymerase binds to before beginning transcription. *242, 262*

propagated epidemic An epidemic that is characterized by a relatively slow and prolonged rise and then a gradual decline in the number of individuals infected. It usually results from the introduction of an infected individual into a susceptible population, and the pathogen is transmitted from person to person. *851*

prophage (pro′fāj) The latent form of a temperate phage that remains within the lysogen, usually integrated into the host chromosome. *308, 390*

prostheca (pros-the′kah) An extension of a bacterial cell, including the plasma membrane and cell wall, that is narrower than the mature cell. *490*

prosthetic group (pros-thet′ik) A tightly bound cofactor that remains at the active site of an enzyme during its catalytic activity. *161*

protease (pro′te-ās) An enzyme that hydrolyzes proteins to their constituent amino acids. Also called a proteinase. *192*

proteasome A large, cylindrical protein complex that degrades ubiquitin-labeled proteins to peptides in an ATP-dependent process. *82*

protein engineering (pro′tēn) The rational design of proteins by constructing specific amino acid sequences through molecular techniques, with the objective of modifying protein characteristics. *994*

protein splicing The post-translational process in which part of a precursor polypeptide is removed before the mature polypeptide folds into its final shape; it is carried out by self-splicing proteins that remove inteins and join the remaining exteins. *275*

proteobacteria (pro″te-o-bak-tēr′-e-ah) A large group of bacteria, primarily gram-negative, that 16S rRNA sequence comparisons show to be phylogenetically related; proteobacteria contain the purple photosynthetic bacteria and their relatives and are composed of the α, β, γ, δ, and ϵ subgroups. *487*

proteome The complete collection of proteins that an organism produces. *356*

protists (pro′tist) Eucaryotes with unicellular organization, either in the form of solitary cells or colonies of cells lacking true tissues. *438*

protocooperation A positive, but not obligatory, interaction between two different organisms in which both parties benefit. *604*

protomer An individual subunit of a viral capsid; a capsomer is made of protomers. *369*

proton motive force (PMF) The force arising from a gradient of protons and a membrane potential that is thought to power ATP synthesis and other processes. *187*

protoplast (pro′to-plast) A bacterial or fungal cell with its cell wall completely removed. It is spherical in shape and osmotically sensitive. *49, 61*

protoplast fusion The joining of cells that have had their walls weakened or completely removed. *994*

protothecosis (pro″to-the-ko′sis; pl., **protothecoses**) A disease of humans and animals produced by the green alga *Prototheca moriformis.* *575*

prototroph (pro′to-trōf) A microorganism that requires the same nutrients as the majority of naturally occurring members of its species. *245*

protozoan or **protozoon** (pro″to-zo′an, pl. **protozoa**) A microorganism belonging to the *Protozoa* subkingdom. A unicellular or acellular eucaryotic protist whose organelles have the functional role of organs and tissues in more complex forms. Protozoa vary greatly in size, morphology, nutrition, and life cycle. *584*

protozoology (pro″to-zo-ol′o-je) The study of protozoa. *584*

proviral DNA Viral DNA that has been integrated into host cell DNA. In retroviruses it is the double-stranded DNA copy of the RNA genome. *407*

pseudomurein A modified peptidoglycan lacking D-amino acids and containing *N*-acetyltalosaminuronic acid instead of *N*-acetylmuramic acid; found in methanogenic archaea. *452*

pseudoplasmodium (soo″do-plaz-mo′de-um; pl., **pseudoplasmodia**) A sausage-shaped amoeboid structure consisting of many myxamoebae and behaving as a unit; the result of myxamoebal aggregation in the cellular slime molds; also called a slug. *565*

pseudopodium or **pseudopod** (soo″do-po′de-um) A nonpermanent cytoplasmic extension of the cell body by which amoebae and amoeboid organisms move and feed. *586*

psittacosis (ornithosis; sit″ah-ko′sis) A disease due to a strain of *Chlamydia psittaci,* first seen in parrots and later found in other birds and domestic fowl (in which it is called ornithosis). It is transmissible to humans. *919*

psychrophile (si′kro-fīl) A microorganism that grows well at 0°C and has an optimum growth temperature of 15°C or lower and a temperature maximum around 20°C. *126*

psychrotroph A microorganism that grows at 0°C, but has a growth optimum between 20 and 30°C, and a maximum of about 35°C. *126*

puerperal fever (pu-er′per-al) An acute, febrile condition following childbirth; it is characterized by infection of the uterus and/or adjacent regions and is caused by streptococci. *857*

pulmonary anthrax (pul′mo-ner″e) A form of anthrax involving the lungs. Also known as woolsorter's disease. *913*

pulmonary syndrome hantavirus *See* hantavirus pulmonary syndrome. *877*

pure culture A population of cells that are identical because they arise from a single cell. *106*

purine (pu′rin) A basic, heterocyclic, nitrogen-containing molecule with two joined rings that occurs in nucleic acids and other cell constituents; most purines are oxy or amino derivatives of the purine skeleton. The most important purines are adenine and guanine. *216*

purple membrane An area of the plasma membrane of *Halobacterium* that contains bacteriorhodopsin and is active in photosynthetic light energy trapping. *461*

putrefaction (pu″trĕ-fak′shun) The microbial decomposition of organic matter, especially the anaerobic breakdown of proteins, with the production of foul-smelling compounds such as hydrogen sulfide and amines. *965*

pyrenoid (pi′rĕ-noid) The differentiated region of the chloroplast that is a center of starch formation in green algae and stoneworts. *85, 573*

pyrimidine (pi-rim′i-dēn) A basic, heterocyclic, nitrogen-containing molecule with one ring that occurs in nucleic acids and other cell constituents; pyrimidines are oxy or amino derivatives of the pyrimidine skeleton. The most important pyrimidines are cytosine, thymine, and uracil. *216*

Q

Q fever An acute zoonotic disease caused by the rickettsia *Coxiella burnetii.* *912*

Quellung reaction The increase in visibility or the swelling of the capsule of a microorganism in the presence of antibodies against capsular antigens. *784*

quorum sensing The process in which bacteria monitor their own population density by sensing the levels of signal molecules that are released by the microorganisms. When these signal molecules reach a threshold concentration, the population density has attained a critical level or quorum, and quorum-dependent genes are expressed. *132*

R

rabies (ra′bēz) An acute infectious disease of the central nervous system, which affects all warm-blooded animals (including humans). It is caused by an ssRNA virus belonging to the genus *Lyssavirus* in the family *Rhabdoviridae.* *888*

racking The removal of sediments from wine bottles. *982*

radappertization The use of gamma rays from a cobalt source for control of microorganisms in foods. *972*

radioimmunoassay (RIA; ra″de-o-im″u-no-as′a) A very sensitive assay technique that uses a purified radioisotope-labeled antigen or antibody to compete for antibody or antigen with unlabeled standard and samples to determine the concentration of a substance in the samples. *783*

reactivation tuberculosis *See* miliary tuberculosis. *908*

reading frame The way in which nucleotides in DNA and mRNA are grouped into codons or groups of three for reading the message contained in the nucleotide sequence. *241*

reagin (re′ah-jin) Antibody that mediates immediate hypersensitivity reactions. IgE is the major reagin in humans. *768*

recombinant DNA technology The techniques used in carrying out genetic engineering; they involve the identification and isolation of a specific gene, the insertion of the gene into a vector such as a plasmid to form a recombinant molecule, and the production of large quantities of the gene and its products. *320*

recombinant-vector vaccine The type of vaccine that is produced by the introduction of one or more of a pathogen's genes into attenuated viruses or bacteria. The attenuated virus or bacterium serves as a vector, replicating within the vertebrate host and expressing the gene(s) of the pathogen. The pathogen's antigens induce an immune response. *767*

recombination (re″kom-bĭ-na′shun) The process in which a new recombinant chromosome is formed by combining genetic material from two organisms. *292*

recombination repair A DNA repair process that repairs damaged DNA when there is no remaining template; a piece of DNA from a sister molecule is used. *255*

Redfield ratio The carbon-nitrogen-phosphorus ratio of aquatic microorganisms. This ratio is important for predicting limiting factors for microbial growth. *638*

red tides Red tides occur frequently in coastal areas and often are associated with population blooms of dinoflagellates. Dinoflagellate pigments are responsible for the red color of the water. Under these conditions, the dinoflagellates often produce saxitoxin, which can lead to paralytic shellfish poisoning. *580*

reducing agent or **reductant** (re-duk′tant) The electron donor in an oxidation-reduction reaction. *157*

reductive dehalogenation The cleavage of carbon-halogen bonds by anaerobic bacteria that creates a strong electron-donating environment. *1010*

refraction (re-frak′shun) The deflection of a light ray from a straight path as it passes from one medium (e.g., glass) to another (e.g., air). *18*

refractive index (re-frak′tiv) The ratio of the velocity of light in the first of two media to that in the second as it passes from the first to the second. *18*

regulator T cell Regulator T cells control the development of effector T cells. Two types exist: T-helper cells ($CD4^+$ cells) and T-suppressor cells. There are three subsets of T-helper cells: T_H1, T_H2, and T_H0. T_H1 cells produce IL-2, IFN-γ, and TNF-β. They effect cell-mediated immunity and are responsible for delayed-type hypersensitivity reactions and macrophage activation. T_H2 cells produce IL-4, IL-5, IL-6, IL-10, IL-13. They are helpers for B-cell antibody responses and humoral immunity; they also support IgE responses and eosinophilia. T_H0 cells exhibit an unrestricted cytokine profile. *751*

regulatory mutants Mutant organisms that have lost the ability to limit synthesis of a product, which normally occurs by regulation of activity of an earlier step in the biosynthetic pathway. *1005*

regulon A collection of genes or operons that is controlled by a common regulatory protein. *281*

replica plating A technique for isolating mutants from a population by plating cells from each colony growing on a nonselective agar medium onto plates with selective media or environmental conditions, such as the lack of a nutrient or the presence of an antibiotic or a phage; the location of mutants on the original plate can be determined from growth patterns on the replica plates. *252*

replication (rep″lĭ-ka′shun) The process in which an exact copy of parental DNA or RNA is made with the parental molecule serving as a template. *230*

replication fork The Y-shaped structure where DNA is replicated. The arms of the Y contain template strand and a newly synthesized DNA copy. *235*

replicative form A double-stranded form of nucleic acid that is formed from a single-stranded virus genome and used to synthesize new copies of the genome. *388, 406*

replicon (rep′lĭ-kon) A unit of the genome that contains an origin for the initiation of replication and in which DNA is replicated. *235, 294*

repressible enzyme An enzyme whose level drops in the presence of a small molecule, usually an end product of its metabolic pathway. *276*

repressor protein (re-pres′or) A protein coded for by a regulator gene that can bind to the operator and inhibit transcription; it may be active by itself or only when the corepressor is bound to it. *276*

reservoir (rez′er-vwar) A site, alternate host, or carrier that normally harbors pathogenic organisms and serves as a source from which other individuals can be infected. *791, 854*

reservoir host An organism other than a human that is infected with a pathogen that can also infect humans. *789*

residuesphere The region surrounding organic matter such as a seed or plant part in which microbial growth is stimulated by increased organic matter availability. *690*

resolution (rez″o-lu′shun) The ability of a microscope to separate or distinguish between small objects that are close together. *20*

respiration (res″ pĭ-ra′ shən) An energy-yielding process in which the energy substrate is oxidized using an exogenous or externally derived electron acceptor. *173*

respiratory burst The respiratory burst occurs when an activated phagocytic cell increases its oxygen consumption to support the increased metabolic activity of phagocytosis. The burst generates highly toxic oxygen products such as singlet oxygen, superoxide radical, hydrogen peroxide, hydroxyl radical, and hypochlorite. *720*

respiratory syncytial virus (RSV; sin-sish′al) A member of the family *Paramyxoviridae* and genus *Pneumovirus;* it is a negative-sense ssRNA virus that causes respiratory infections in children. *875*

restricted transduction A transduction process in which only a specific set of bacterial genes are carried to another bacterium by a temperate phage; the bacterial genes are acquired because of a mistake in the excision of a prophage during the lysogenic life cycle. *309*

restriction enzymes Enzymes produced by host cells that cleave virus DNA at specific points and thus protect the cell from virus infection; they are used in carrying out genetic engineering. *320, 386*

reticulate body (RB) The form in the chlamydial life cycle whose role is growth and reproduction within the host cell. *477*

retroviruses (re″tro-vi′rus-es) A group of viruses with RNA genomes that carry the enzyme reverse transcriptase and form a DNA copy of their genome during their reproductive cycle. *407*

reverse transcriptase (RT) An RNA-dependent DNA polymerase that uses a viral RNA genome as a template to form a DNA copy; this is a reverse of the normal flow of genetic information, which proceeds from DNA to RNA. *407, 879*

reversible covalent modification A mechanism of enzyme regulation in which the enzyme's activity is either increased or decreased by the reversible covalent addition of a group such as phosphate or AMP to the protein. *167*

Reye's syndrome An acute, potentially fatal disease of childhood that is characterized by severe edema of the brain and increased intracranial pressure, vomiting, hypoglycemia, and liver dysfunction. The cause is unknown but is almost always associated with a previous viral infection (e.g., influenza or varicella-zoster virus infections). *874*

R factors or **R plasmids** Plasmids bearing one or more drug resistant genes. *297, 819*

rheumatic fever (roo-mat′ik) An autoimmune disease characterized by inflammatory lesions involving the heart valves, joints, subcutaneous tissues, and central nervous system. The disease is associated with hemolytic streptococci in the body. It is called rheumatic fever because two common symptoms are fever and pain in the joints similar to that of rheumatism. *905*

rhizosphere A region around the plant root where materials released from the root increase the microbial population and its activities. *675*

rho factor (ro) The protein that helps RNA polymerase dissociate from the terminator after it has stopped transcription. *263*

rhoptry Saclike, electron dense structure in the anterior portion of a zoite of a member of the phylum *Apicomplexa;* perhaps involved in the penetration of host cells. *591*

ribonucleic acid (RNA; ri″bo-nu-kle′ik) A polynucleotide composed of ribonucleotides joined by phosphodiester bridges. *230*

ribosomal RNA (rRNA) The RNA present in ribosomes; ribosomes contain several sizes of single-stranded rRNA that contribute to ribosome structure and are also directly involved in the mechanism of protein synthesis. *261*

ribosome (ri′bo-sōm) The organelle where protein synthesis occurs; the message encoded in mRNA is translated here. *52, 267*

ribotyping Ribotyping is the use of *E. coli* rRNA to probe chromosomal DNA in Southern blots for typing bacterial strains. This method is based on the fact that rRNA genes are scattered throughout the chromosome of most bacteria and therefore polymorphic restriction endonuclease patterns result when chromosomes are digested and probed with rRNA. *843*

ribulose-1,5-bisphosphate carboxylase (ri′bu-lōs) The enzyme that catalyzes the incorporation of CO_2 in the Calvin cycle. *208*

ringworm (ring′werm) The common name for a fungal infection of the skin, even though it is not caused by a worm and is not always ring-shaped in appearance. *943*

rise period or **burst** The period during the one-step growth experiment when host cells lyse and release phage particles. *383*

RNA polymerase The enzyme that catalyzes the synthesis of mRNA under the direction of a DNA template. *261*

Rocky Mountain spotted fever A disease caused by *Rickettsia rickettsii. 913*

rolling-circle mechanism A mode of DNA replication in which the replication fork moves around a circular DNA molecule, displacing a strand to give a tail that is also copied to produce a new double-stranded DNA. *236*

root nodule Gall-like structures on roots that contain endosymbiotic nitrogen-fixing bacteria (e.g., *Rhizobium* or *Bradyrhizobium* is present in legume nodules). *676*

roseola infantum (ro-ze′o-lə) A skin eruption that produces a rose-colored rash in infants. Caused by the human herpesvirus 6. The disease is short-lived and characterized by a high fever of 3 to 4 days' duration. *887*

rubella A moderately contagious skin disease that occurs primarily in children 5 to 9 years of age that is caused by the rubella virus, which is acquired by droplet inhalation into the respiratory system; German measles. *875*

rubeola *See* measles. *873*

rumen (roo-men) The expanded upper portion or first compartment of the stomach of ruminants. *602*

ruminant (roo′mĭ-nant) An herbivorous animal that has a stomach divided into four compartments and chews a cud consisting of regurgitated, partially digested food. *602*

run The straight line movement of a bacterium. *67*

S

salmonellosis (sal″mo-nel-o′sis) An infection with certain species of the genus *Salmonella,* usually caused by ingestion of food containing salmonellae or their products. Also known as *Salmonella* gastroenteritis or *Salmonella* food poisoning. *931*

sanitization (san″ĭ-ti-za′shun) Reduction of the microbial population on an inanimate object to levels judged safe by public health standards; usually, the object is cleaned. *138*

saprophyte (sap′ro-fit) An organism that takes up nonliving organic nutrients in dissolved form and usually grows on decomposing organic matter. *557*

saprozoic nutrition (sap″ro-zo′ik) Having the type of nutrition in which organic nutrients are taken up in dissolved form; normally refers to animals or animal-like organisms. *586*

satellite phenomenon *See* syntrophism. *604*

scaffolding proteins Special proteins that are used to aid procapsid construction during the assembly of a bacteriophage capsid and are removed after the completion of the procapsid. *388*

scale (skāl) A platelike organic structure found on the surface of some cells (chrysophytes). *577*

scanning electron microscope (SEM) An electron microscope that scans a beam of electrons over the surface of a specimen and forms an image of the surface from the electrons that are emitted by it. *34*

scanning probe microscope A microscope used to study surface features by moving a sharp probe over the object's surface (e.g., the scanning tunneling microscope). *38*

scanning tunneling microscope A type of scanning probe microscope used to image a surface by moving a fine probe over it at a constant height, which is maintained by keeping a constant electron flow (tunneling current) between the tip and surface. *38*

scarlatina (skahr″la-te′nah) *See* scarlet fever. *905*

scarlet fever (scarlatina; skar′let) A disease that results from infection with a strain of *Streptococcus pyogenes* that carries a lysogenic phage with the gene for erythrogenic (rash-inducing) toxin. The toxin causes shedding of the skin. This is a communicable disease spread by respiratory droplets. *905*

schizogony (skĭ-zog′o-ne) Multiple asexual fission. *591*

secondary metabolites Products of metabolism that are synthesized after growth has been completed. *1002*

secondary treatment The biological degradation of dissolved organic matter in the process of sewage treatment; the organic material is either mineralized or changed to settleable solids. *659*

second law of thermodynamics Physical and chemical processes proceed in such a way that the entropy of the universe (the system and its surroundings) increases to the maximum possible. *156*

secretory IgA (sIgA) The primary immunoglobulin of the secretory immune system. *See* IgA. *738*

secretory vacuole In protists and some animals, these organelles usually contain specific enzymes that perform various functions such as excystation. Their contents are released to the cell exterior during exocytosis. *585*

segmented genome A virus genome that is divided into several parts or fragments, each probably coding for the synthesis of a single polypeptide; segmented genomes are very common among the RNA viruses. *374*

selectins (sə-lek′tins) A family of cell adhesion molecules that are displayed on activated endothelial cells; examples include P-selectin and E-selectin. Selectins mediate leukocyte binding to the vascular endothelium. *712*

selective media Culture media that favor the growth of specific microorganisms; this may be accomplished by inhibiting the growth of undesired microorganisms. *105*

selective toxicity The ability of a chemotherapeutic agent to kill or inhibit a microbial pathogen while damaging the host as little as possible. *807*

self-assembly The spontaneous formation of a complex structure from its component molecules without the aid of special enzymes or factors. *65, 207*

sepsis (sep′sis) Systemic response to infection. This systemic response is manifested by two or more of the following conditions as a result of infection: temperature $>$38 or $<$36°C; heart rate $>$90 beats per min; respiratory rate $>$20 breaths per min, or pCO_2 $<$32 mm Hg; leukocyte count $>$12,000 cells per ml^3 or $>$10% immature (band) forms. Sepsis also has been defined as the presence of pathogens or their toxins in blood and other tissues. *933*

septate (sep′tāt) Divided by a septum or cross wall; also with more or less regular occurring cross walls. *556*

septic shock (sep′tik) Sepsis associated with severe hypotension despite adequate fluid resuscitation, along with the presence of perfusion abnormalities that may include, but are not limited to, lactic acidosis, oliguria, or an acute alteration in mental status. Gram-positive bacteria, fungi, and endotoxin-containing gram-negative bacteria can initiate the pathogenic cascade of sepsis leading to septic shock. *933*

septicemia (sep″tĭ-se′me-ah) A disease associated with the presence in the blood of pathogens or bacterial toxins. *514, 793*

septic tank (sep′tik) A tank used to process small quantities of domestic sewage. Solid material settles out and is partially degraded by anaerobic bacteria as sewage slowly flows through the tank. The outflow is further treated or dispersed in aerobic soil. *663*

septum (sep′tum; pl., **septa**) A partition or cross-wall that occurs between two cells in a bacterial (e.g., actinomycete) or fungal filament, or which partitions off fungal structures such as spores. Septa also divide parent cells into two daughter cells during bacterial binary fission. *286, 556*

serology (se-rol′o-je) The branch of immunology that is concerned with in vitro reactions involving one or more serum constituents (e.g., antibodies and complement). *774*

serotyping A technique or serological procedure that is used to differentiate between strains (serovars or serotypes) of microorganisms that have differences in the antigenic composition of a structure or product. *784*

serum (se′rum; pl., **serums** or **sera**) The clear, fluid portion of blood lacking both blood cells and fibrinogen. It is the fluid remaining after coagulation of plasma, the noncellular liquid faction of blood. *742*

serum resistance The type of resistance that occurs with bacteria such as *Neisseria gonorrhoeae* because the pathogen interferes with membrane attack complex formation during the complement cascade. *801*

settling basin A basin used during water purification to chemically precipitate out fine particles, microorganisms, and organic material by coagulation or flocculation. *652*

sex pilus (pi′lus) A thin protein appendage required for bacterial mating or conjugation. The cell with sex pili donates DNA to recipient cells. *63, 303*

sheath (shēth) A hollow tubelike structure surrounding a chain of cells and present in several genera of bacteria. *496*

shigellosis (shĭ′gəl-o′sis) The diarrheal disease that arises from an infection with a member of the genus *Shigella.* Often called bacillary dysentery. *931*

SHIME system (simulated human intestinal microbial ecosystem) A set of connected chemostat-like reactors that provide a sequence of environments similar to the human digestive system. *987*

Shine-Dalgarno sequence A segment in the leader of procaryotic mRNA that binds to a special sequence on the 16S rRNA of the small ribosomal subunit. This helps properly orient the mRNA on the ribosome. *244*

shingles (zoster; shing′g′lz) A reactivated form of chickenpox caused by the varicella-zoster virus. *872*

siderophore (sid′er-o-for″) A small molecule that complexes with ferric iron and supplies it to a cell by aiding in its transport across the plasma membrane. *104*

sigma factor A protein that helps the RNA polymerase core enzyme recognize the promoter at the start of a gene. *262*

sign An objective change in a diseased body that can be directly observed (e.g., a fever or rash). *850*

silage Fermented plant material with increased palatability and nutritional value for animals, which can be stored for extended periods. *986*

silent mutation A mutation that does not result in a change in the organism's proteins or phenotype even though the DNA base sequence has been changed. *249*

simple matching coefficient (S_{SM}) An association coefficient used in numerical taxonomy; the proportion of characters that match regardless of whether or not the attribute is present. *426*

single radial immunodiffusion (RID) assay An immunodiffusion technique that quantitates antigens by following their diffusion through a gel containing antibodies directed against the test antigens. *779*

site-specific recombination Recombination of nonhomologous genetic material with a chromosome at a specific site. *292*

skin-associated lymphoid tissue (SALT) The lymphoid tissue in the skin that forms a first-line defense as a part of nonspecific immunity. *709*

slash-and-burn agriculture The cutting down and burning of tropical vegetation to make mineral nutrients available for use by introduced agricultural crops. *672*

S-layer A regularly structured layer composed of protein or glycoprotein that lies on the surface of many bacteria. It may protect the bacterium and help give it shape and rigidity. *62*

slime The viscous extracellular glycoproteins or glycolipids produced by staphylococci and *Pseudomonas aeruginosa* bacteria that allows them to adhere to smooth surfaces such as prosthetic medical devices and catheters. More generally, the term often refers to an easily removed, diffuse, unorganized layer of extracellular material that surrounds a bacterial cell. *61, 919*

slime layer A layer of diffuse, unorganized, easily removed material lying outside the bacterial cell wall. *61*

slime mold A common term for members of the divisions *Acrasiomycota* and *Myxomycota.* *564*

slow sand filter A bed of sand through which water slowly flows; the gelatinous microbial layer on the sand grain surface removes waterborne microorganisms, particularly *Giardia,* by adhesion to the gel. This type of filter is used in some water purification plants. *653*

slow virus disease A progressive, pathological process caused by a transmissible agent (virus or prion) that remains clinically silent during a prolonged incubation period of months to years after which progressive clinical disease becomes apparent. *410, 893*

sludge (sluj) A general term for the precipitated solid matter produced during water and sewage treatment; solid particles composed of organic matter and microorganisms that are involved in aerobic sewage treatment (activated sludge). *658*

smallpox (variola; smawl′poks) Once a highly contagious, often fatal disease caused by a poxvirus. Its most noticeable symptom was the appearance of blisters and pustules on the skin. Vaccination has eradicated smallpox throughout the world. *876*

snapping division A distinctive type of binary fission resulting in an angular or a palisade arrangement of cells, which is characteristic of the genera *Arthrobacter* and *Corynebacterium.* *542*

sorocarp The fruiting structure of the Acrasiomycetes. *565*

sorus A type of fruiting structure composed of a mass of spores or sporangia. *565*

SOS repair A complex, inducible repair process that is used to repair DNA when extensive damage has occurred. *255*

source The location or object from which a pathogen is immediately transmitted to the host, either directly or through an intermediate agent. *854*

Southern blotting technique The procedure used to isolate and identify DNA fragments from a complex mixture. The isolated, denatured fragments are transferred from an agarose electrophoretic gel to a nitrocellulose filter and identified by hybridization with probes. *322*

specialized transduction *See* restricted transduction. *309*

species (spe′shēz) Species of higher organisms are groups of interbreeding or potentially interbreeding natural populations that are reproductively isolated. Bacterial species are collections of strains that have many stable properties in common and differ significantly from other groups of strains. *425*

specific immune response (acquired or specific immunity) *See* acquired immunity. *705*

spermosphere The region surrounding a germinating seed where released organic matter stimulates microbial growth. *974*

spheroplast (sfēr′o-plast) A relatively spherical cell formed by the weakening or partial removal of the rigid cell wall component (e.g., by penicillin treatment of gram-negative bacteria). Spheroplasts are usually osmotically sensitive. *61*

spike *See* peplomer. *374*

spirillum (spi-ril′um) A rigid, spiral-shaped bacterium. *44*

spirochete (spi′ro-kēt) A flexible, spiral-shaped bacterium with periplasmic flagella. *44, 479*

spleen (splēn) A secondary lymphoid organ where old erythrocytes are destroyed and blood-borne antigens are trapped and presented to lymphocytes. *708*

split or interrupted gene A structural gene with DNA sequences that code for the final RNA product (expressed sequences or exons) separated by regions coding for RNA absent from the mature RNA (intervening sequences or introns). *263*

spongiform encephalopathies Degenerative central nervous system diseases in which the brain has a spongy appearance; they appear due to prions. *893*

spontaneous generation (spon-ta′ne-us) The hypothesis that living organisms can arise from nonliving matter. *2*

sporadic disease (spo-rad′ik) A disease that occurs occasionally and at random intervals in a population. *849*

sporangiospore (spo-ran′je-o-spōr) A spore born within a sporangium. *539, 557*

sporangium (spo-ran′je-um; pl., **sporangia**) A saclike structure or cell, the contents of which are converted into an indefinite number of spores. It is borne on a special hypha called a sporangiophore. *68, 557*

spore (spōr) A differentiated, specialized form that can be used for dissemination, for survival of adverse conditions because of its heat and dessication resistance, and/or for reproduction. Spores are usually unicellular and may develop into vegetative organisms or gametes. They may be produced asexually or sexually and are of many types. *573*

sporogenesis (spor′o-jen′ĕ-sis) *See* sporulation. *69*

sporotrichosis (spo″ro-tri-ko′sis) A subcutaneous fungal infection caused by the dimorphic fungus *Sporothrix schenckii. 945*

sporulation (spor″u-la′shun) The process of spore formation. *69*

spread plate A petri dish of solid culture medium with isolated microbial colonies growing on its surface, which has been prepared by spreading a dilute microbial suspension evenly over the agar surface. *106*

sputum (spu′tum) The mucous secretion from the lungs, bronchi, and trachea that is ejected (expectorated) through the mouth. *829*

stalk (stawk) A nonliving bacterial appendage produced by the cell and extending from it. *490*

standard free energy change The free energy change of a reaction at 1 atmosphere pressure when all reactants and products are present in their standard states; usually the temperature is 25°C. *156*

standard reduction potential A measure of the tendency of a reductant to lose electrons in an oxidation-reduction (redox) reaction. The more negative the reduction potential of a compound, the better electron donor it is. *157*

staphylococcal food poisoning (staf″i-lo-kok′al) A type of food poisoning caused by ingestion of improperly stored or cooked food in which *Staphylococcus aureus* has grown. The bacteria produce exotoxins that accumulate in the food. *932*

staphylococcal scalded skin syndrome (SSSS) A disease caused by staphylococci that produce an exfoliative toxin. The skin becomes red (erythema) and sheets of epidermis may separate from the underlying tissue. *922*

starter culture An inoculum, consisting of a mixture of carefully selected microorganisms, used to start a commercial fermentation. *978*

stationary phase (sta′shun-er″e) The phase of microbial growth in a batch culture when population growth ceases and the growth curve levels off. *114*

statistics (stah-tis′tiks) The mathematics of the collection, organization, and interpretation of numerical data. *849*

stem-nodulating rhizobia Rhizobia (members of the genera *Rhizobium, Bradyrhizobium,* and *Azorhizobium*) that produce nitrogen-fixing structures above the soil surface on plant stems. These most often are observed in tropical plants and produced by *Azorhizobium. 682*

sterilization (ster″ĭ-lĭ-za′shun) The process by which all living cells, viable spores, viruses, and viroids are either destroyed or removed from an object or habitat. *137*

stigma (stig′mah) A light-sensitive eyespot, which is found in some algae and photosynthetic protozoa; it is believed to be involved in phototaxis, at least in some cases. *575*

stoneworts A group of approximately 250 species of algae that have a complex growth pattern, with nodal regions from which whorls of branches arise; they are abundant in fresh to brackish waters. *576*

strain A population of organisms that descends from a single organism or pure culture isolate. *425*

streak plate A petri dish of solid culture medium with isolated microbial colonies growing on its surface, which has been prepared by spreading a microbial mixture over the agar surface, using an inoculating loop. *107*

streptococcal pneumonia An endogenous infection of the lungs caused by *Streptococcus pneumoniae* that occurs in predisposed individuals. *905*

streptococcal sore throat (strep″to-kok′al) One of the most common bacterial infections of humans. It is commonly referred to as "strep throat." The disease is spread by droplets of saliva or nasal secretions and is caused by *Streptococcus* spp. (particularly group A streptococci). *905*

streptolysin-O (SLO) (strep-tol′ĭ-sin) A specific hemolysin produced by *Streptococcus pyogenes* that is inactivated by oxygen (hence the "O" in its name). SLO causes beta-hemolysis of blood cells on agar plates incubated anaerobically. *797*

streptolysin-S (SLS) A product produced by *Streptococcus pyogenes* that is bound to the bacterial cell but may sometimes be released. SLS causes beta hemolysis on aerobically incubated blood-agar plates and can act as a leukocidin by killing leukocytes that phagocytose the bacterial cell to which it is bound. *797*

streptomycin (strep′to-mi″sin) A bactericidal aminoglycoside antibiotic produced by *Streptomyces griseus. 816*

strict anaerobes *See* obligate anaerobes. *127*

stroma (stro′mah) The chloroplast matrix that is the location of the photosynthetic carbon dioxide fixation reactions. *85*

stromatolite (stro″mah-to′līt) Dome-like microbial mat communities consisting of filamentous photosynthetic bacteria and occluded sediments (often calcareous or siliceous). They usually have a laminar structure. Many are fossilized, but some modern forms occur. *423*

structural gene A gene that codes for the synthesis of a polypeptide or polynucleotide with a nonregulatory function. *277*

subacute sclerosing panencephalitis Diffuse inflammation of the brain resulting from virus and prion infections. *874*

subgingival plaque (sub-jin′jĭ-val) The plaque that forms at the dentogingival margin and extends down into the gingival tissue. *936*

substrate-level phosphorylation The synthesis of ATP from ADP by phosphorylation coupled with the exergonic breakdown of a high-energy organic substrate molecule. *177*

subsurface biosphere The region below the plant root zone where microbial populations can grow and function. *691*

sulfate reduction (sul′fāt) The process of sulfate use as an oxidizing agent, which results in the accumulation of reduced forms of sulfur such as sulfide, or incorporation of sulfur into organic molecules, usually as sulfhydryl groups. *614*

sulfonamide (sul-fon′ah-mīd) A chemotherapeutic agent that has the SO_2-NH_2 group and is a derivative of sulfanilamide. *812*

superantigen Superantigens are bacterial proteins that stimulate the immune system much more extensively than do normal antigens. They stimulate T cells to proliferate nonspecifically through simultaneous interaction with class II MHC proteins on antigen-presenting cells and variable regions on the β chain of the T-cell receptor complex. Examples include streptococcal scarlet fever toxins, staphylococcal toxic shock syndrome toxin-1, and streptococcal M protein. *732*

superinfection (soo″per-in-fek′shun) A new bacterial or fungal infection of a patient that is resistant to the drug(s) being used for treatment. *819*

superoxide dismutase (SOD; soo″per-ok′sīd dis-mu′tas) An enzyme that protects many microorganisms by catalyzing the destruction of the toxic superoxide radical. *128*

suppressor mutation A mutation that overcomes the effect of another mutation and produces the normal phenotype. *248*

Svedberg unit (sfed′berg) The unit used in expressing the sedimentation coefficient; the greater a particle's Svedberg value, the faster it travels in a centrifuge. *52*

swab (swahb) A wad of absorbent material usually wound around one end of a small stick and used for applying medication or for removing material from an area; also, a dacron-tipped polystyrene applicator. *827*

swarm cell A flagellated cell; the term is usually applied to the motile cells of the *Myxomycota.* *565*

symbiosis (sim″bi-o′sis) The living together or close association of two dissimilar organisms, each of these organisms being known as a symbiont. *596*

symbiosome The final nitrogen-fixing form of *Rhizobium* that is active within root nodule cells. *676*

symptom (simp′təm) A change during a disease that a person subjectively experiences (e.g., pain, bodily discomfort, fatigue, or loss of appetite). Sometimes the term symptom is used more broadly to include any observed signs. *850*

syndrome *See* disease syndrome. *850*

synthetic medium *See* defined medium. *105*

syntrophism (sin′trōf-izəm) The association in which the growth of one organism either depends on, or is improved by, the provision of one or more growth factors or nutrients by a neighboring organism. Sometimes both organisms benefit. This type of mutualism is also known as cross-feeding or the satellite phenomenon. *604*

syphilis (sif′ĭ-lis) *See* venereal syphilis. *923*

systematic epidemiology The field of epidemiology that focuses on the ecological and social factors that influence the development of emerging and reemerging infectious diseases. *859*

systematics (sis″te-mat′iks) The scientific study of organisms with the ultimate objective being to characterize and arrange them in an orderly manner; often considered synonymous with taxonomy. *422*

systemic lupus erythematosus (loo′pus er″ĭ-them-ah-to′sus) An autoimmune, inflammatory disease that may affect every tissue of the body. *770*

T

taxon (tak′son) A group into which related organisms are classified. *422*

taxonomy (tak-son′o-me) The science of biological classification; it consists of three parts: classification, nomenclature, and identification. *422*

TB skin test Tuberculin hypersensitivity test for a previous or current infection with *Mycobacterium tuberculosis.* *771*

T cell or **T lymphocyte** A type of lymphocyte derived from bone marrow stem cells that matures into an immunologically competent cell under the influence of the thymus. T cells are involved in a variety of cell-mediated immune reactions. *705, 745*

T-cell antigen receptor (TCR) The receptor on the T cell surface consisting of two antigen-binding peptide chains; it is associated with a large number of other glycoproteins. Binding of antigen to the TCR, usually in association with MHC, activates the T cell. *745*

T-dependent antigen An antigen that effectively stimulates B-cell response only with the aid of T-helper cells that produce interleukin-2 and B-cell growth factor. *753*

teichoic acids (ti-ko′ik) Polymers of glycerol or ribitol joined by phosphates; they are found in the cell walls of gram-positive bacteria. *56*

temperate phages Bacteriophages that can infect bacteria and establish a lysogenic relationship rather than immediately lysing their hosts. *308, 390*

template strand (tem′plat) A strand of DNA or RNA that specifies the base sequence of a newly synthesized complementary strand of DNA or RNA. *242*

terminator A sequence that marks the end of a gene and stops transcription. *244, 263*

tertiary treatment (ter′she-er-e) The removal from sewage of inorganic nutrients, heavy metals, viruses, etc., by chemical and biological means after microorganisms have degraded dissolved organic material during secondary sewage treatment. *661*

test A loose-fitting shell of an amoeba. *590*

tetanolysin (tet″ah-nol′ĭ-sin) A hemolysin that aids in tissue destruction and is produced by *Clostridium tetani.* *925*

tetanospasmin (tet″ah-no-spaz′min) The neurotoxic component of the tetanus toxin, which causes the muscle spasms of tetanus. Tetanospasmin production is under the control of a plasmid gene. *924*

tetanus (tet′ah-nus) An often fatal disease caused by the anaerobic, spore-forming bacillus *Clostridium tetani,* and characterized by muscle spasms and convulsions. *924*

tetracyclines (tet″rah-si′klēns) A family of antibiotics with a common four-ring structure, which are isolated from the genus *Streptomyces* or produced semisynthetically; all are related to chlortetracycline or oxytetracycline. *815*

tetrapartite associations (tet″rah-par′tīt) A mutualistic association of the same plant with three different types of microorganisms. *685*

T_H1 cell *See* regulator T cell. *751*

T_H2 cell *See* regulator T cell. *751*

T_H0 cell *See* regulator T cell. *751*

thallus (thal′us) A type of body that is devoid of root, stem, or leaf; characteristic of some algae, many fungi, and lichens. *537, 554, 573*

T-helper (T_H) cell A cell that is needed for T-cell–dependent antigens to be effectively presented to B cells. It also promotes cell-mediated immune responses. *751*

theory A set of principles and concepts that have survived rigorous testing and that provide a systematic account of some aspect of nature. *8*

thermal death time (TDT) The shortest period of time needed to kill all the organisms in a microbial population at a specified temperature and under defined conditions. *140*

thymus (thi′məs) A primary lymphoid organ in the chest that is necessary in early life for the development of immunological functions. T-cell maturation takes place here. *708*

thermoacidophiles A group of bacteria that grow best at acid pHs and high temperatures; they are members of the *Archaea.* *457*

thermophile (ther′mo-fīl) A microorganism that can grow at temperatures of 55°C or higher; the minimum is usually around 45°C. *126*

thrush (thrush) Infection of the oral mucous membrane by the fungus *Candida albicans;* also known as oral candidiasis. *949*

thylakoid (thi′lah-koid) A flattened sac in the chloroplast stroma that contains photosynthetic pigments and the photosynthetic electron transport chain; light energy is trapped and used to form ATP and NAD(P)H in the thylakoid membrane. *85*

thymine (thi′min) The pyrimidine 5-methyluracil that is found in nucleosides, nucleotides, and DNA. *217*

T-independent antigen An antigen that triggers a B cell into immunoglobulin production without T-cell cooperation. *754*

Ti or Ri plasmid A plasmid obtained from *Agrobacterium tumefaciens* that is used to insert genes into plant cells. *339, 684*

tinea (tin′e-ah) A name applied to many different kinds of superficial fungal infections of the skin, nails, and hair, the specific type (depending on characteristic appearance, etiologic agent, and site) usually designated by a modifying term. *943*

tinea capitis A fungal infection of scalp hair caused by species of *Trichophyton* or *Microsporum.* *943*

tinea corporis A fungal infection of the smooth parts of the skin caused by either *Trichophyton rubrum, T. mentagrophytes,* or *Microsporum canis.* *943*

tinea cruris A fungal infection of the groin caused by either *Epidermophyton floccosum, Trichophyton mentagrophytes,* or *T. rubrum;* also known as jock itch. *944*

tinea manuum A fungal infection of the hand caused by *Trichophyton rubrum, T. mentagrophytes,* or *E. floccosum.* *944*

tinea pedis A fungal infection of the foot caused by *Trichophyton rubrum, T. mentagrophytes,* or *E. floccosum;* also known as athlete's foot. *944*

tinea unguium A fungal infection of the nail bed caused by either *Trichophyton rubrum* or *T. mentagrophytes.* *944*

tinea versicolor A fungal infection caused by the yeast, *Malassezia furfur,* that forms brownish-red scales on the skin of the trunk, neck, face, and arms. *943*

titer (ti′ter) Reciprocal of the highest dilution of an antiserum that gives a positive reaction in the test being used. *742*

T lymphocyte *See* T cell. *705*

tonsillitis (ton′si-li′tis) Inflammation of the tonsils, especially the palatine tonsils often due to *S. pyogenes* infection. *905*

toxemia (tok-se′me-ah) The condition caused by toxins in the blood of the host. *794*

toxic shock-like syndrome (TSLS) A disease caused by an invasive group A streptococcus infection that is characterized by a rapid drop in blood pressure, failure of many organs, and a very high fever. It probably results from the release of one or more streptococcal pyrogenic exotoxins. *904*

toxic shock syndrome (tok′sik) A staphylococcal disease that most commonly affects females who use certain types of tampons during menstruation. It is associated with the production of toxic shock syndrome toxin by certain strains of *Staphylococcus aureus*. *922*

toxigenicity (tok″sĭ-jĕ-nis′ĭ-tē) The capacity of an organism to produce a toxin. *790*

toxin (tok′sin) A microbial product or component that can injure another cell or organism at low concentrations. Often the term refers to a poisonous protein, but toxins may be lipids and other substances. *794*

toxin neutralization The inactivation of toxins by specific antibodies, called antitoxins, that react with them. *756*

toxoid (tok′soid) A bacterial exotoxin that has been modified so that it is no longer toxic but will still stimulate antitoxin formation when injected into a person or animal. *767, 796*

toxoplasmosis (tok″so-plaz-mo′sis) A disease of animals and humans caused by the parasitic protozoan, *Toxoplasma gondii*. *957*

trachoma (trah-ko′mah) A chronic infectious disease of the conjunctiva and cornea, producing pain, inflammation and sometimes blindness. It is caused by *Chlamydia trachomatis* serotypes A–C. *925*

transamination (trans″am-i-na′shun) The removal of amino acid's amino group by transferring it to an α-keto acid acceptor. *192*

transcriptase (trans-krip′tās) An enzyme that catalyzes transcription; in viruses with RNA genomes, this enzyme is an RNA-dependent RNA polymerase that is used to make RNA copies of the RNA genomes. *406*

transcription (trans-krip′shun) The process in which single-stranded RNA with a base sequence complementary to the template strand of DNA or RNA is synthesized. *230*

transduction (trans-duk′shun) The transfer of genes between bacteria by bacteriophages. *308*

transfer host (trans′fer) A host that is not necessary for the completion of a parasite's life cycle, but is used as a vehicle for reaching a final host. *789*

transfer RNA (tRNA) A small RNA that binds an amino acid and delivers it to the ribosome for incorporation into a polypeptide chain during protein synthesis. *261*

transformation (trans″for-ma′shun) A mode of gene transfer in bacteria in which a piece of free DNA is taken up by a bacterial cell and integrated into the recipient genome. *228, 305*

transgenic animal or **plant** An animal or plant that has gained new genetic information from the insertion of foreign DNA. It may be produced by such techniques as injecting DNA into animal eggs, electroporation of mammalian cells and plant cell protoplasts, or shooting DNA into plant cells with a gene gun. *335*

transient carrier *See* casual carrier. *854*

transition mutations (tran-zish′un) Mutations that involve the substitution of a different purine base for the purine present at the site of the mutation or the substitution of a different pyrimidine for the normal pyrimidine. *246*

translation (trans-la′shun) Protein synthesis; the process by which the genetic message carried by mRNA directs the synthesis of polypeptides with the aid of ribosomes and other cell constituents. *230*

transmission electron microscope (trans-mish′un) A microscope in which an image is formed by passing an electron beam through a specimen and focusing the scattered electrons with magnetic lenses. *30*

transovarian passage (trans″o-va′re-an) The passage of a microorganism such as a rickettsia from generation to generation of hosts through tick eggs. (No humans or other mammals are needed as reservoirs for continued propagation of the rickettsias.) *913*

transpeptidation 1. The reaction that forms the peptide cross-links during peptidoglycan synthesis. 2. The reaction that forms a peptide bond during the elongation cycle of protein synthesis. *223, 270*

transposable elements *See* transposon. *298*

transposition (trans″po-zish′un) The movement of a piece of DNA around the chromosome. *298*

transposon (tranz-po′zon) A DNA segment that carries the genes required for transposition and moves about the chromosome; if it contains genes other than those required for transposition, it may be called a composite transposon. Often the name is reserved only for transposable elements that also contain genes unrelated to transposition. *298*

transversion mutations (trans-ver′zhun) Mutations that result from the substitution of a purine base for the normal pyrimidine or a pyrimidine for the normal purine. *246*

traveler's diarrhea A type of diarrhea resulting from ingestion of certain viruses, bacteria, or protozoa normally absent from the traveler's environment. One of the major pathogens is enterotoxigenic *Escherichia coli*. *932*

tricarboxylic acid cycle (TCA) The cycle that oxidizes acetyl coenzyme A to CO_2 and generates NADH and $FADH_2$ for oxidation in the electron transport chain; the cycle also supplies carbon skeletons for biosynthesis. *183, A–16*

trichome (tri′kōm) A row or filament of bacterial cells that are in close contact with one another over a large area. *472*

trichomoniasis (trik″o-mo-ni′ah-sis) A sexually transmitted disease caused by the parasitic protozoan *Trichomonas vaginalis*. *958*

trickling filter A bed of rocks covered with a microbial film that aerobically degrades organic waste during secondary sewage treatment. *659*

trihalomethanes (THMs) Halogenated one-carbon compounds formed during water disinfection; many of these compounds are potential carcinogens. *653*

tripartite associations (tri-par′tīt) A mutualistic association of the same plant with two types of microorganisms. *685*

trophozoite (trof″ o-zo′īt) The active, motile feeding stage of a protozoan organism; in the malarial parasite, the stage of schizogony between the ring stage and the schizont. *586*

tropism (tro′piz-əm) The movement of living organisms toward or away from a focus of heat, light, or other stimulus. *791*

trypanosome (tri-pan′o-sōm) A protozoan of the genus *Trypanosoma*. Trypanosomes are parasitic flagellate protozoa that often live in the blood of humans and other vertebrates and are transmitted by insect bites. *589, 957*

trypanosomiasis (tri-pan″o-so-mi′ah-sis) An infection with trypanosomes that live in the blood and lymph of the infected host. *957*

tubercle (too′ber-k′l) A small, rounded nodular lesion produced by *Mycobacterium tuberculosis*. *908*

tuberculoid (neural) leprosy (too-ber′ku-loid) A mild, nonprogressive form of leprosy that is associated with delayed-type hypersensitivity to antigens on the surface of *Mycobacterium leprae*. It is characterized by early nerve damage and regions of the skin that have lost sensation and are surrounded by a border of nodules. *916*

tuberculosis (too-ber″ku-lo′sis) An infectious disease of humans and other animals resulting from an infection by a species of *Mycobacterium* and characterized by the formation of tubercles and tissue necrosis, primarily as a result of host hypersensitivity and inflammation. Infection is usually by inhalation, and the disease commonly affects the lungs (pulmonary tuberculosis), although it may occur in any part of the body. *906*

tuberculous cavity (too-ber′ku-lus) An air-filled cavity that results from a tubercle lesion caused by *M. tuberculosis*. *908*

tularemia (too″lah-re′me-ah) A plaguelike disease of animals caused by the bacterium *Francisella tularensis* subsp. *tularensis* (Jellison type A), which may be transmitted to humans. *926*

tumble Random turning or tumbling movements made by bacteria when they stop moving in a straight line. *67*

tumor (too′mor) A growth of tissue resulting from abnormal new cell growth and reproduction (neoplasia). *411*

turbidostat A continuous culture system equipped with a photocell that adjusts the flow of medium through the culture vessel so as to maintain a constant cell density or turbidity. *121*

twiddle *See* tumble. *67*

two-component phosphorelay system A signal transduction regulatory system that uses the transfer of phosphoryl groups to control gene transcription and protein activity. It has two major components: a sensor kinase and a response regulator. *283*

type I hypersensitivity A form of immediate hypersensitivity arising from the binding of antigen to IgE attached to mast cells, which then release anaphylaxis mediators such as histamine. Examples: hay fever, asthma, and food allergies. *768*

type II hypersensitivity A form of immediate hypersensitivity involving the binding of antibodies to antigens on cell surfaces followed by destruction of the target cells (e.g., through complement attack, phagocytosis, or agglutination). *769*

type III hypersensitivity A form of immediate hypersensitivity resulting from the exposure to excessive amounts of antigens in which antibodies bind to the antigens and produce antibody-antigen complexes. These activate complement and trigger an acute inflammatory response with subsequent tissue damage. Examples: poststreptococcal glomerulonephritis, serum sickness, and farmer's lung disease. *770*

type IV hypersensitivity A delayed hypersensitivity response (it appears 24 to 48 hours after antigen exposure). It results from the binding of antigen to activated T lymphocytes, which then release cytokines and trigger inflammation and macrophage attacks that damage tissue. Type IV hypersensitivity is seen in contact dermititis from poison ivy, leprosy, and tertiary syphilis. *771*

type III secretion system *See* pathogenicity island. *794*

typhoid fever (ti-foid) A bacterial infection transmitted by contaminated food, water, milk, or shellfish. The causative organism is *Salmonella typhi,* which is present in human feces. *933*

U

ultramicrobacteria Bacteria that can exist normally in a miniaturized form or which are capable of miniaturization under low-nutrient conditions. They may be 0.2 µm or smaller in diameter. *640*

ultraviolet (UV) radiation (ul″trah-vi′o-let) Radiation of fairly short wavelength, about 10 to 400 nm, and high energy. *130, 144*

uracil (u′rah-sil) The pyrimidine 2,4-dioxypyrimidine, which is found in nucleosides, nucleotides, and RNA. *217*

V

vaccine (vak′sēn) A preparation of either killed microorganisms; living, weakened (attenuated) microorganisms; or inactivated bacterial toxins (toxoids). It is administered to induce development of the immune response and protect the individual against a pathogen or a toxin. *764*

vaccinomics The application of genomics and bioinformatics to vaccine development. *766*

valence (va′lens) The number of antigenic determinant sites on the surface of an antigen or the number of antigen-binding sites possessed by an antibody molecule. *731*

variable region (V_L and V_H) The region at the N-terminal end of immunoglobulin heavy and light chains whose amino acid sequence varies between antibodies of different specificity. Variable regions form the antigen binding site. *734*

vasculitis (vas″ku-li′tis) Inflammation of a blood vessel. *909*

vector (vek′tor) 1. In genetic engineering, another name for a cloning vector. A DNA molecule that can replicate (a replicon) and is used to transport a piece of inserted foreign DNA, such as a gene, into a recipient cell. It may be a plasmid, phage, cosmid or artificial chromosome. 2. In epidemiology, it is a living organism, usually an arthropod or other animal, that transfers an infective agent from one host to another. *322, 791, 854*

vector-borne transmission The transmission of an infectious pathogen between hosts by means of a vector. *857*

vehicle (ve′ĭ-k′l) An inanimate substance or medium involved in the transmission of a pathogen. *857*

venereal syphilis (ve-ne′re-al sif′ĭ-lis) A contagious, sexually transmitted disease caused by the spirochete *Treponema pallidum.* *923*

venereal warts *See* anogenital condylomata. *894*

verrucae vulgaris (vĕ-roo′se vul-ga′ris; s. **verruca vulgaris**) The common wart; a raised, epidermal lesion with horny surface caused by an infection with a human papillomavirus. *894*

vibrio (vib′re-o) A rod-shaped bacterial cell that is curved to form a comma or an incomplete spiral. *43*

viral hemagglutination (vi′ral hem″ah-gloo″tĭ-na′shun) The clumping or agglutination of red blood cells caused by some viruses. *776*

viral neutralization An antibody-mediated process in which IgA, IgM, and IgA antibodies bind to some viruses during their extracellular phase and inactivate or neutralize them. *756*

viremia (vi-re′me-ə) The presence of viruses in the blood stream. *791*

viricide (vir′i-sīd) An agent that inactivates viruses so that they cannot reproduce within host cells. *138*

virion (vi′re-on) A complete virus particle that represents the extracellular phase of the virus life cycle; at the simplest, it consists of a protein capsid surrounding a single nucleic acid molecule. *363*

virioplankton Viruses that occur in waters; high levels are found in marine and freshwater environments. *643*

viroid (vi′roid) An infectious agent of plants that is a single-stranded RNA not associated with any protein; the RNA does not code for any proteins and is not translated. *416*

virology (vi-rol′o-je) The branch of microbiology that is concerned with viruses and viral diseases. *362*

virulence (vir′u-lens) The degree or intensity of pathogenicity of an organism as indicated by case fatality rates and/or ability to invade host tissues and cause disease. *790*

virulence factor A bacterial product, usually a protein or carbohydrate, that contributes to virulence or pathogenicity. *792*

virulent bacteriophages (vir′u-lent bak-te′re-o-fājs″) Bacteriophages that lyse their host cells during the reproductive cycle. *390*

virus (vi′rus) An infectious agent having a simple acellular organization with a protein coat and a single type of nucleic acid, lacking independent metabolism, and reproducing only within living host cells. *363*

vitamin (vi′tah-min) An organic compound required by organisms in minute quantities for growth and reproduction because it cannot be synthesized by the organism; vitamins often serve as enzyme cofactors or parts of cofactors. *99*

volutin granules (vo-lu′tin) *See* metachromatic granules. *52*

W

wart (wort) An epidermal tumor of viral origin. *894*

wastewater treatment The use of physical and biological processes to remove particulate and dissolved material from sewage and to control pathogens. *658*

water activity (a_w) A quantitative measure of water availability in the habitat; the water activity of a solution is one-hundredth its relative humidity. *122*

water mold A common term for a member of the division *Oomycota.* *564*

Weil-Felix reaction A test for the diagnosis of typhus and certain other rickettsial diseases. In this test, the blood serum of a patient with suspected rickettsial disease is tested against certain strains of *Proteus vulgaris* (OX-2, OX-19, OX-K). The agglutination reactions, based on antigens common to both organisms, determine the presence and type of rickettsial infection. *910*

white piedra A fungal infection caused by the yeast *Trichosporon beigelii* that forms light-colored nodules on the beard and mustache. *943*

whole-genome shotgun sequencing An approach to genome sequencing in which the complete genome is broken into random fragments, which are then individually sequenced. Finally the fragments are placed in the proper order using sophisticated computer programs. *346*

whole-organism vaccine A vaccine made from complete pathogens, which can be of four types: inactivated viruses; attenuated viruses; killed microorganisms; and live, attenuated microorganisms. *766*

Widal test (ve-dahl′) A test involving agglutination of typhoid bacilli when they are mixed with serum containing typhoid antibodies from an individual having typhoid fever; used to detect the presence of *Salmonella typhi* and *S. paratyphi.* *775*

Winogradsky column A glass column with an anaerobic lower zone and an aerobic upper zone, which allows growth of microorganisms under conditions similar to those found in a nutrient-rich lake. *637*

woolsorter's disease *See* anthrax. *913*

wort The filtrate of malted grains used as the substrate for the production of beer and ale by fermentation. *982*

X

xenograft (zen′o-graft) A tissue graft between animals of different species. *773*

xerophilic microorganisms (ze″ro-fil′ik) Microorganisms that grow best under low a_w conditions, and may not be able to grow at high a_w values. *965*

Y

yeast (yēst) A unicellular fungus that has a single nucleus and reproduces either asexually by budding or fission, or sexually through spore formation. *554*

yeast artificial chromosome (YAC) A stretch of DNA that contains all the elements required to propagate a chromosome in yeast and which is used to clone foreign DNA fragments in yeast cells. *335*

yellow fever An acute infectious disease caused by a flavivirus, which is transmitted to humans by mosquitoes. The liver is affected and the skin turns yellow in this disease. *878*

YM shift The change in shape by dimorphic fungi when they shift from the yeast (Y) form in the animal body to the mold or mycelial form (M) in the environment. *556*

Z

zooflagellates (zo″o-flăj′-e-lāts) Flagellate protozoa that do not have chlorophyll and are either holozoic, saprozoic, or symbiotic. *588*

zoonosis (zo″o-no′sis; pl. *zoonoses*) A disease of animals that can be transmitted to humans. *849*

zooplankton (zo″o-plank′ton) A community of floating, aquatic, minute animals and nonphotosynthetic protists. *571*

zoospore (zo′o-spōr) A motile, flagellated spore. *573*

zooxanthella (zo″o-zan-thel′ah) A dinoflagellate found living symbiotically within cnidarians and other invertebrates. *579, 599*

zoster *See* shingles. *872*

***z* value** The increase in temperature required to reduce the decimal reduction time to one-tenth of its initial value. *140*

zygomycetes (zi″go-mi-se′tez) A division of fungi that usually has a coenocytic mycelium with chitinous cell walls. Sexual reproduction normally involves the formation of zygospores. The group lacks motile spores. *560*

zygospore (zi′go-spōr) A thick-walled, sexual, resting spore characteristic of the zygomycetous fungi. *558*

zygote (zi′gōt) The diploid (2n) cell resulting from the fusion of male and female gametes. *574*

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